9

TRANSFORMING LINGUISTICS

Theoretical linguistics has been enormously important in the history of cognitive science *as a whole*. But it might not have been. After all, why should psychologists, or philosophers, or anthropologists, or neurophysiologists, or computer scientists be interested in linguistics? If they happen to be concentrating on language, then perhaps it's relevant. If not, why should they bother with it?

As late as 1950, most of them didn't. And, in fact, most of them don't today. But in the late 1950s to 1970s all cognitive scientists had to pay some attention to it. This chapter explains why that was. (Why the interest then dwindled to near-zero is explained in Chapter 7.iii.a, and in Section ix.g below.)

The cyberneticists gave no thought to linguistics—and very little to language. The nearest they got was Warren McCulloch's 1920s search for a "logic" of verbs (4.iii.c), a pastime foreign to his cybernetic fellows. Even Gregory Bateson and Gordon Pask, who both championed language-based psychotherapy, weren't interested in the structure of language. When Alan Turing and others in the UK wrote the first computer programs involving language, they ignored linguistics (see x.a, below). And the (few) psychologists who focused on language paid scant attention to linguistic theory, and—crucially—didn't see their work as relevant for the study of mind *in general*. Nor was linguistics needed to spark off the cognitive revolution: that originated in psychology and AI, not linguistics (4.iii.e, 5.iii–iv, and 6.i–iii).

It turned out, however, that theoretical linguistics—specifically, the study of *syntax*—stoked the fire sparked off by other disciplines. (The contemporary work on phonetics was less relevant in this regard, and I shan't discuss it.) By the early 1960s no one involved in cognitive science, or in the philosophy of mind, could ignore it. The reason, in just two words: Noam Chomsky (1928–).

Linguistics turned computational with Chomsky. He tried to formulate a mathematical definition of language *as such*: a "computational" account, in the abstract sense later stressed by David Marr (7.iii.b). His work, including a pioneering paper on computer languages, encouraged others to persevere with, or to embark on, the computer modelling of language. But he himself didn't join them. He was highly sceptical about what's normally called computational linguistics, or natural language processing (NLP). He dipped his toes into the water only once (G. A. Miller and Chomsky 1963: 464–82)—by which time many others were already swimming in the shallows (see Section ix).

Chomsky's early publications—up to the mid-1960s—had a huge, and lasting, effect on cognitive science overall. This needn't have happened, even if his *linguistics* was, as one disciple later said, "*the* existence proof for the possibility of a cognitive science" (Fodor 1981*b*: 258). But Chomsky generalized his philosophical and methodological claims. He even defined linguistics as "part of psychology", focused on "one specific cognitive domain and one faculty of mind" (Chomsky 1980*b*: 1). Granted, he cared little about the psycholinguistic processing details—especially when they turned out to be less consilient with his theory than had at first appeared (Chapter 7.ii.a). But he saw his linguistics, in effect, as a theory of *mind*.

Specifically, he revived the then hugely unfashionable doctrine of *nativism*. This is the view that the mind/brain of the newborn baby is already equipped with knowledge of, or dispositions towards, language—and various other things, too. Modern proponents of nativism, such as his ex-pupil Steven Pinker (1954–), cite many biological data that were unknown to Chomsky (Pinker 1994, 2002). But he remains their key inspiration. Indeed, a very recent book by a leading anti-nativist has acknowledged that his "arguments [for linguistic nativism] are still widely read, *and carry as much weight today as any newly-produced publications*" (Sampson 2005: 7; italics added). In brief, he's still a major force.

Chomsky's influence on cognitive science was beneficial in many ways. In particular, he deepened the nascent questioning of behaviourism, and encouraged "mentalist" theories couched in terms of internal rules and/or representations (Chapter 6.i.e). He revivified research on nativism (7.vi), and indirectly encouraged the growth of evolutionary psychology (8.ii.d–e and iv–v). He offered a vision of theoretical rigour which inspired linguists and non-linguists alike. And, despite his own scepticism, his work encouraged others to attempt the computer modelling of mind. In short, for the field as a whole, he was a Good Thing.

Nevertheless, this chapter starts, unusually, with a health warning: *beware of the passions that swirl under any discussion of Chomsky* (Section i).

Next, Sections ii–v locate Chomsky in relation to the three centuries of language scholarship—including linguistically inspired philosophies of mind—that preceded him. The ancient writings are discussed in some detail here because he himself made a point of comparing his own work to them in a provocative way. I begin by distinguishing predecessors from precursors (in Section ii), and then consider the various writers named by Chomsky as his "rationalist" precursors (see Sections iii–iv). Section v sketches the state of theoretical linguistics during Chomsky's youth.

In Section vi, I describe how he first sought to change it. Section vii deals with his nativism, and the associated attack on behaviourism, while Section viii says a little about the subsequent changes in Chomskian theory. Various fundamental critiques of his account of language are outlined in Section ix, which also shows the need for the health warning.

Finally, the last two sections describe the early days, and the eventual maturation, of work in the computer modelling of language. These matters could have been discussed in Chapter 10, for NLP was central to GOFAI research. I decided to place them here instead, largely because the problems experienced in NLP recalled theoretical questions highlighted by the scholars featured in Sections iv–v, below.

A word on demarcation: this chapter deals with linguistics rather than psycholinguistics. The focus is on the nature of language *as such*. Psychological questions about how we understand, communicate, and remember linguistic meanings do crop up here, of course. But they're addressed more directly in many other parts of the book (e.g. Chapters 7.i.c, ii, iii.d, and iv.d–e; 8.i.a–b and vi; 10.iii.a and e; 12.x; and 16.i.c and iv.c–d). Indeed, Chomsky's influence led many linguists to ignore such questions: see Section ix.g, below.

9.i. Chomsky as Guru

Many people, including youngsters, take Chomsky as their political guru. Indeed, it's mostly because of his courageous and uncompromising political writings—which far outnumber his publications on linguistics—that he's the most-quoted living author (Barsky 1997: 3). I'll say a little about his politics in Section vii.a, below. But it's his role as a *scientific* guru that's relevant here.

I said, above, that Chomsky was a Good Thing for the development of cognitive science. So he was. But to acknowledge this isn't to accept the tenfold Chomsky myth.

a. The tenfold Chomsky myth

Those who uncritically take Chomsky as their scientific guru share ten beliefs:

* They think (1) that besides giving a *vision* of mathematical rigour, he always achieved it in his own work.
* They believe (2) that his linguistic theory was, or is now, indisputably correct in all essentials.
* They hold (3) that the (nativist) psychological implications he drew from it were convincingly argued at the time, and
* (4) that they are now empirically beyond doubt.
* On matters of history, they suggest (5) that his work of the 1950s was wholly original.
* And they suppose (6) that his writings of the 1960s were—as he himself claimed—the culmination of an august tradition of rationalism dating back 300 years.
* Some also assume (7) that without Chomsky's grammar there would have been no computer modelling of language.
* Many believe (8) that Chomsky was responsible for the demise of behaviourism.
* Most take for granted (9) that he reawakened and strengthened the discipline of linguistics.
* And many assume (10) that linguistics is as prominent in cognitive science today as it was, thanks to Chomsky, in the late 1950s to 1970s.

Each of these beliefs, strictly interpreted, is false. But all, suitably qualified, approach the truth. (The least defensible are nos. 9 and 10: in some ways, as we'll see, he *set back* the field of linguistics, and *undermined* its relevance for cognitive science in general.)

Moreover, many people in Chomsky's heyday accepted all of them (except no. 10, of course)—and a significant number still do, with no. 10 now added. That's why

Chomsky is not only a pivotal thinker in theoretical linguistics, but also a hugely important figure in the wider story of cognitive science.

However, "important" means *influential*, *provocative*, *fruitful* . . . perhaps even *largely beneficial*. It doesn't necessarily mean *right*.

b. A non-pacific ocean

The tenfold myth attracts passions of surprising depth. Surprising, that is, to anyone who believes the Legend: that science is a disinterested enterprise (see 1.iii.b).

While doing the research for this chapter, I was astonished to discover the lack of mutual respect between the two camps in linguistics, Chomskyan and non-Chomskyan. I'd known there were unpleasant tensions of course, but I hadn't realized their degree. They range from unscholarly ignorance of the very existence of important competing theories (as I discovered in a recent conversation with a young MIT linguist) to vitriolic hostility and public abuse.

As one illustration, a leading American linguist who (unusually) has an open mind on the issue of nativism has recently bemoaned "what is, I am afraid, an often disturbingly unserious, indeed irresponsible approach to the innateness position on the part of its major advocate" (Postal 2005, p. viii). Paul Postal (1936–) then quotes some "revealing", "outrageous", and "grotesquely untrue" remarks by Chomsky, the burden of which is that all the arguments against innateness are such confused nonsense that nobody ever bothers to answer them. And his verdict is that "These statements [by Chomsky] are not scientific comments but rather the analog of awful, partisan political discourse" (2005, p. ix).

As another illustration, a non-Chomskyan complained to me about Chomsky's theorizing about syntax:

> Anything that any human being actually says is at once defined by Chomsky out of the discipline, leaving him and his followers free to pursue their strange little version of navel-gazing, free from any contamination from the real world. *And I can't forgive Chomsky for perverting my discipline in this sick manner*. (Larry Trask, personal communication)

Larry Trask told me, a few months before his tragically early death in 2004, that he was happy for his remark to be quoted, since he'd said much the same in print. (And I'm confident that he'd have been happy with the italics, which are mine.)

Third, the computational linguist (and AI–vision researcher) Shimon Edelman (1957–) has recently referred to "the Bolshevik manner of [Chomsky's] takeover of linguistics", and the "Trotskyist ('permanent revolution') flavor of the subsequent development of [his] doctrine" (S. Edelman 2003). To be fair, he allows that there were cultural reasons for the "takeover" in early 1960s America (Koerner 1989). These were the widespread counter-cultural rejection of the views of the older generation (see 1.iii.c), and the post-Sputnik increase of research funding—especially at MIT (see 11.i). Thanks to that money, many new departments of linguistics were set up in the USA at that time, and it was "inevitable" that they'd be staffed by young lecturers trained at MIT (James McCawley, quoted in Koerner 1994). So Edelman isn't suggesting that Chomsky was an all-powerful tyrant, solely responsible for what happened. By the same token, the Chomskyan "revolution" wasn't caused by purely *intellectual* factors (S. Edelman 2003: 675).

And last, an anonymous linguist who reviewed this chapter accused me of pursuing "a personal vendetta" against Chomsky instead of "doing intellectual history". He, or possibly she, couldn't have been more wrong. I'm not a professional linguist, so have no axe to grind either way. (And like Postal, I have an open mind about linguistic nativism: see Chapter 7.vi.) Moreover, it's precisely because I'd done the intellectual history carefully that I'd realized just how questionable, and in certain respects just how damaging, some of Chomsky's claims were.

It's a telling sign that my account—which isn't written in a polemical spirit (though it does quote polemics, from both sides), and which gives chapter and verse for every criticism—should have been misinterpreted in that way. So be warned: these are turbulent waters. When it comes to Chomsky, few linguists, if any, are emotionally neutral.

(The same reviewer wanted me to change the chapter because "it will upset people". Well, I've no doubt that it will upset some people—though it will probably hearten others. But I haven't changed it: it's an honest account of what I found in the literature, and of what I was told in conversations with several leading linguists.)

Far from having mellowed with time, the passions have risen over the years. In Sections viii and ix, we'll see why this is so. But before then, we'll consider Chomsky's early work (up to the mid-1960s), its immediate reception, and—first of all—its intellectual background.

9.ii. Predecessors and Precursors

Let's start with some banalities:

A predecessor is a person who discussed the same topic as some later scholar.

A precursor is a predecessor who said similar things about it.

A precursor needn't have influenced the scholar concerned: intellectual anticipation isn't the same as intellectual ancestry.

Identifying precursors requires one to decide just what one means by "similar".

Similarity isn't merely a question of degree, but of nature. Two œuvres may have more or less of a certain feature, and/or they may have partly different features.

Finally, someone who interprets "similarity" too broadly not only obscures important differences, but casts all the scholars concerned in intellectual roles for which they aren't best suited. So reflected glory may be more distorting than illuminating.

a. Why Chomsky's 'history' matters

The points just listed, which relate to the caveats given in Chapter 1.iii.f, are pertinent in any discussion of the history of ideas. But they're especially relevant here. They concern not only Chomsky's intellectual pedigree, but also his rhetorical efforts to share in the glory associated with great names of the past.

Those efforts weren't mere self-seeking. Rather, they were a form of philosophical defensiveness. Chomsky had deliberately suppressed his (nativist) ideas on language-and-mind when writing his first book (1957), judging them to be "too audacious" for publication (see Section vi.e). When he did finally announce them, he defended them partly by locating them in the context of respected thinkers of the past (1965, 1966). In so doing, however, he misrepresented those thinkers in various ways.

"So what?" you may ask. "Why bother to rap Chomsky over his historical knuckles? He wasn't a historian, after all." Well, no. But historical accuracy is important here for two reasons.

He himself, in defending nativism, gave his Cartesian–rationalist labelling a high profile (and many Chomsky fans uncritically took him at his word). Today, his long-time champion Jerry Fodor (1998*c*) speaks of "the New Rationalists", of whom he counts himself one (7.vi.d). We need to know just how literally this tag should be taken.

Even more to the point, we must try to grasp the shifting subtleties of the long-standing debate about "innate ideas" if we're to understand nativism *in general*. In this context, we need to know what the philosophers he referred to actually said—which often differed, more or less, from what he implied that they said.

On the one hand, Chomsky himself repeatedly named as precursors people—dating back to the early seventeenth century—whose views differed significantly from his own, and implied that they were more similar to each other than is the case. (In so far as they are indeed his precursors, they're anticipatory rather than ancestral: he studied most of them only after developing his own ideas.)

On the other hand, the fact that linguists today situate themselves—in agreement or opposition—with respect to Chomsky might suggest that his work was a complete break with what went before. In other words, it suggests that his teachers were mere predecessors, not precursors.

In discussing these matters, we must distinguish his views on language-and-mind from his theory of language (primarily, of syntax) as such. The "one hand" (above) concerns the former. The "other hand" concerns the latter.

With regard to his grammatical theory, one might say that Chomsky had no precursors. More accurately, the sense in which others before him had "said similar things" was an attenuated one. He himself claimed that the transformational model of language was "not at all new", having been anticipated by the Port-Royal logicians of the seventeenth century (Chomsky 1964: 15–16). But that was true only in the most general sense. Even those of his immediate predecessors who developed formal theories of syntax did so in much less detailed ways. Chomsky's grammatical work, although clearly dependent on contemporary advances in formal linguistics and information theory, was largely novel (see Sections v–vi).

As for his views on the relation between language and mind, Chomsky was the first to say that he had many precursors (Chomsky 1964: 8–25). But most, he says, were long dead and largely forgotten:

> These early modern contributions were scarcely known, even to scholarship, until they were rediscovered during the second cognitive revolution, after somewhat similar ideas had been independently developed. (Chomsky 1996*a*: 7)

That remark apparently applies to Wilhelm von Humboldt (1767–1835), whose work one historian sees as having suffered "a pattern of recurrent, or successive, amnesia" in American linguistics (Hymes and Faught 1981: 98). And it applies also to the Port-Royal *Grammar* (see Section iii.c), which another historian of linguistics has described as "forgotten, [and] often misunderstood" by the end of the nineteenth century (Koerner and Asher 1995: 174). But it doesn't apply so clearly to the Port-Royal *Logic*, which was being translated into English for the fourth time just as Chomsky's first book appeared. And I know from my own experience that Herbert of Cherbury (on innate ideas) was routinely recommended to Cambridge philosophy undergraduates in the 1950s. Perhaps professional linguists weren't reading these people, but others certainly were.

These examples suggest that some of Chomsky's remarks about his precursors may need to be taken with a pinch of salt. Perhaps a sackful: a third specialist historian has gone so far as to say that Chomsky's version of the history of linguistics is "fundamentally false from beginning to end" (Aarsleff 1970: 583).

Even if that judgement is too harsh, there are systematic problems with regard to Chomsky's account. I noted, above, that talk of precursors depends on what one counts as similar. Chomsky interpreted the similarities so broadly that his descriptions of his forerunners were often misleading. In particular, his frequent uses of "rationalist" and "Cartesian" were inaccurate. And it's not at all clear that what his forebears had meant by the "inner form" of language, or by "creativity", was what he meant by these words.

The central doctrine of "Cartesian linguistics", said Chomsky, is that "the general features of grammatical structure are common to all languages and reflect certain fundamental properties of the human mind" (Chomsky 1966: 59). In other words, "Cartesian" linguistics holds that there is some inborn grammatical core of all natural languages.

Chomsky did attach a caveat to his inclusive term ("Cartesian"), pointing out that the rationalist/nativist writers he had in mind fell into more than one philosophical school. Nevertheless, he implied that they comprised a coherent tradition of humanist rationalism, one that had been forgotten but had now been revived by his own research (Chomsky 1966, 1968).

If we're to situate that research in its historical context, we need to understand how his nominated precursors differed—from him and from each other—as well as how they were alike. In this section, we'll consider their general philosophical background; specific examples are discussed in Sections iii and iv.

b. The rationalist background

In broad outline, the writers Chomsky named as his "rationalist" precursors agreed on two things. They believed that language is the origin of human thought, and that it's innate in every child.

Some—rationalists in the strict sense—interpreted "thought", here, to mean access to necessary truth. Some, but not all, remarked that the second (nativist) belief implies the existence not only of some specifically human propensity for language, but also of some common core of language as such. Some believed in, or hoped for, a past (or future) universal language, intelligible to everyone because it was (or would be) wholly or largely natural, not purely conventional. Some saw language as the essential core of humanity. And some stressed the role of language in creativity and culture.

As we'll see in Section vii.d, Chomsky wasn't a rationalist in the strong sense. He didn't posit innate access to necessary linguistic principles, still less to necessary truth. Nor was he interested in projects aimed at designing a universal language, or at locating the origins of language in prehistory. But he did believe that language is what makes us human, that language and thought are intimately connected, that language is essentially creative, that all languages share a common grammatical core, and that linguistic ability is innate in (only) the human species.

In Chomsky's own judgement, his most important precursor was René Descartes. As we saw in Chapter 2.iii.c, Descartes taught that language is uniquely human. We don't share it with any animal, nor could it be provided to a machine:

> If you teach a magpie to say good-day to its mistress, when it sees her approach, this can only be by making the utterance of the word the expression of one of its passions. For instance, it will be the expression of the hope of eating, if it has always been given a tidbit when it says the word. Similarly, all the things which dogs, horses and monkeys are taught to perform are only the expression of their fear, their hope or their joy [all of which are entirely mechanical: see Chapter 2.ii.d] . . . But the use of words, so defined, is peculiar to human beings. (Descartes 1646: 207)

No machine, according to Descartes, could "arrange words variously in response to the meaning of what is said in its presence, as even the dullest men can do". The human ability to arrange words variously was to be heavily stressed by Chomsky, who both sought to explain it and used it as a stick to beat behaviourism. However, what Chomsky meant by this ability wasn't what Descartes meant by it (see Section iv.f).

As for where language ability comes from, Descartes believed that it's provided to all human minds by God. His reason was that, because of its variability, it can't be explained in material terms. He pointed out that deaf-mutes—fellow human beings—spontaneously use gestures and make up signs that other people can learn to understand (1637, ch. 5).

It's not at all clear, however, that Descartes was a "Cartesian" according to Chomsky's definition (above). He didn't specifically say that language has a core structure, of which we have innate knowledge. Indeed, he described languages as unpredictably diverse, to be learnt from the environment much as other social customs are. And he attributed the variability of language to the innate power of reason (the ability to come up with new and appropriate thoughts), not to any property of language as such. He did believe, however, that some human abilities rest on innate knowledge.

c. The puzzle of innate ideas

In general, for Descartes, our knowledge depends heavily on innate ideas. These exist in the mind prior to any experience.

The idea of God must be innate, he said, since the concept of infinity could be neither derived nor—as Thomas Hobbes had argued (see 2.iii.b)—extrapolated from experience. The true proposition "God exists" is also innate (said Descartes), because it is self-contradictory to suppose its falsity. Similarly, he saw the necessary principles of logic and mathematics as present in the mind a priori.

For Descartes, no true *propositions* about the empirical world are innate: only experience can teach us that roses are red and violets blue. Here, he was less nativist than Chomsky, for whom knowledge of certain empirical facts about language—facts which might have been different—is inborn (see Section vii.c–d). But Descartes sometimes argued that the *sensory concepts* themselves must be innate, since no genuine causal relation between brain and mind is possible:

> [Besides the ideas of movements and figures] so much the more must the ideas of pain, colour, sound and the like be innate, that our mind may on occasion [*sic*] of certain corporeal movements, envisage these ideas, for they have no likenesses to the corporeal movements. (trans. Haldane and Ross 1911: i. 42)

Descartes wasn't the first to posit innate ideas, whether concepts or propositions (McRae 1972; Chomsky 1966: 59–72). They'd been spoken of for centuries. Plato's doctrine of *anamnesis*, introduced in *The Meno* to explain the slave boy's intuitive grasp of geometry, is a well-known example. Most rationalist writers followed Plato in attributing necessity to innate ideas. Descartes's occasional suggestion that even sensations are innate was unusual.

But familiarity didn't bring clarity. There was much controversy about just what this doctrine amounted to, in Descartes's writings as in other people's. This was so, irrespective of whether it concerned only concepts or also propositions.

Did it mean that we have *actual* ideas—of number, extension, God, or even colour—prior to any experience? Or did it mean that our *potential* to do arithmetic and geometry, to think about God, or to perceive red and blue, is a natural capacity, or disposition, possessed from birth by all human minds—as opposed to a learnt skill? And if the latter, *just what* is a "potential", "capacity", or "disposition" to have certain ideas? Are these expressions anything more than philosophically misleading ways of saying that we can, in fact, do so?

Many of Descartes's correspondents, including the arch-empiricist Hobbes, raised these questions. Comparing innate mental dispositions to a family's inborn propensity for a certain disease, Descartes replied:

> I never wrote or concluded that the mind required innate ideas which were in some sort different from its faculty of thinking; but when I observed the existence in me of certain thoughts which proceeded, not from extraneous objects nor from the determination of my will, but solely from the faculty of thinking which is within me, then . . . I termed them INNATE. (trans. Haldane and Ross 1911: i. 442)

But his reply didn't end the debate. On the contrary, it raged on long after he died.

John Locke (1690) interpreted the doctrine literally, and rejected it. Or rather: "he formulated every clear version he could think of and showed each to be either trivially true or obviously false" (Goodman 1969: 138). He likened the newborn mind to a blank wax tablet, or tabula rasa (which, trivially, has certain potentialities rather than others). Locke's view was very widely accepted at the time—not least, for its 'enlightened' political implications. Indeed, the entry on "Idea" in Ephraim Chambers's *Cyclopaedia* of 1728 stated that "our great Mr. Locke seems to have put this Matter out of dispute".

But Chambers was over-optimistic. Gottfried Leibniz (1690), specifically targeting Locke's discussion, interpreted the doctrine dispositionally, and defended it. He compared the mind to a veined block of marble, which both enables and constrains the figures that can be sculpted from it.

Empiricists accepted Locke's interpretation. They saw knowledge in general as built from passively received sensations, and language as the learnt ability to use certain words to stand for certain ideas.

In justification, they pointed out that languages are very different, and that children learn to speak as the people surrounding them do. They remarked that feral children, much discussed in the seventeenth and eighteenth centuries (Itard 1801), don't use language spontaneously, but will learn it if rescued while still young (for a modern discussion, see Curtiss 1977). They argued that dispositional accounts of innate ideas were merely pretentious ways of stating a commonplace: that people can learn language and mathematics. And they complained that positing innate ideas prevents one from asking more searching questions about the origins of knowledge. These empiricist arguments were still prominent in the twentieth century, and were marshalled by some of Chomsky's critics (see Section vii.c–d).

Those of a rationalist cast of mind favoured Leibniz's interpretation. But it was unclear just what cognitive dispositions might be involved and/or just how they contribute to knowledge. Rationalists claimed, for instance, that necessary truths are innate, but they couldn't explain why geometry and arithmetic seem to provide genuinely new knowledge.

That changed after the late eighteenth century, thanks to Immanuel Kant's *Critique of Pure Reason* (1781). The problem of innate ideas—at least as regards our knowledge of the physical world—seemed, to many, to have been solved by Kant's distinction (sketched in Chapter 2.vi.a) between empirical intuitions and categories on the one hand and sensory experience on the other. The former are (he said) innate, but are merely structuring principles. Only the latter can give actual content to our knowledge. In Kant's words: "Thoughts without content are empty, intuitions without concepts are blind" (Kant 1781: 61).

It was more difficult, however, to gloss language as innate in this sense. For there seem to be no universal structuring principles. Whereas we all acknowledge causality and Euclidean space, our languages appear chaotic. Not only do we use different words for the same thing (*children*, *enfants*, *tamariki* . . .), a fact linguists call "the arbitrariness of the sign", but we put the words together in very different ways. In short, languages differ not only in vocabulary, but also in grammar.

This seemingly incontrovertible truth, coupled with the growth of empiricism outlined in Chapter 2, led—in scientific circles—to the demise of the rationalist view of language. By the end of the 1940s, when Chomsky was a student, the predominant school of scientific linguistics favoured environmentalist accounts of language learning (see Section v.b).

That situation was partly due to the dominance of behaviourism in American psychology. But even anti-behaviourists fought shy of nativist theories of language. Thus the Gestalt psychologists, who developed neo-Kantian explanations of the perception of causation and identity, didn't do so for language. And Jean Piaget, who recognized universal features of language such as temporal order and hierarchy, saw them as based in the universals of logic. These in turn, he said, are constructed by the infant's sensori-motor activity in the physical world (Piaget 1952*b*; Piatelli-Palmarini 1980). In short, he saw language as neither unique nor innate.

In humanist circles, by contrast, the nativist viewpoint on language and mind survived. In the three centuries separating Descartes's birth from Chomsky's, it was elaborated in various ways. Literally Cartesian versions of innate ideas gave way to neo-Kantian and Romantic ones. And, in place of Descartes's mix of pure mentalism and mechanistic biology, there arose a holistic view of mind as grounded in self-organization, similar to the autonomous development of the embryo (see 2.vi). The diversity of languages wasn't denied: quite the contrary (see Section iv.b). But the basic humanist commitment to species-specific linguistic nativism remained.

9.iii. Not-Really-Cartesian Linguists

When is a door not a door? And when is a Cartesian not a Cartesian? The first question is more swiftly answered than the second.

The people named by Chomsky as Cartesians (and undeniably influenced by Descartes) differed between themselves about matters of interest to us—including some of the questions listed in Chapter 1.i.a. Many of those questions are still (largely) unanswered. So, quite apart from evaluating Chomsky's accuracy as a historian, there's good reason to consider just what they did say and just what they didn't say.

a. Descartes's disciple

Géraud de Cordemoy (*c.*1620–84), sometime Director of the Académie française, is the only person named as a precursor by Chomsky (e.g. 1996*a*: 3) who was an orthodox Cartesian. Born four years after Descartes, he was one of his leading disciples. Indeed, he was a member of the party who travelled to Stockholm in 1666 to bring Descartes's exhumed remains back to France—minus the writing forefinger, which the French ambassador was allowed to cut off, and the skull, which had been stolen. (After being sold several times, the skull ended up in the Musée de l'homme—Lindeboom 1979: 13–14.)

Descartes had died of pneumonia sixteen years earlier, having risen daily at four o'clock in the Swedish winter to teach Queen Christina (aka Greta Garbo) philosophy in her freezing castle. Cordemoy outlived his master by over thirty years. For much of that time, he was spreading, and elaborating on, Descartes's views.

In promulgating the Cartesian philosophy, Cordemoy devoted an entire essay to language, the *Discours physique sur la parole* (Cordemoy 1668: 196–256). As the title implies, he had much to say about the physical aspects of language (the lungs, speech organs, ears, and nerves), pointing out that the human bodily machine is naturally suited for producing and hearing it. He even discussed the bodily aspects of writing. Nevertheless, he argued that language is possible only for beings with souls (minds), which is to say, human beings. It can't be explained in purely physical terms, nor mimicked by artificial machines—and it isn't acquired by the learning methods used by animals.

Cordemoy marvelled at the child's acquisition of language, and remarked that it doesn't seem to depend on reinforcement:

> It's true that one usually tries to excite some emotion (such as joy) in infants, by exclaiming when one shows them something while speaking its name, so that they are more attentive and, being more affected by this method of teaching, retain the words better.
>
> But, no matter what trouble one takes to teach them certain things, one often finds that they know the names of thousands of other things, which one has not attempted to show them. And the most surprising thing about this is that, by the time they are two or three years old, and by the power of their attention alone, they are able to pick out the name of something from all of the constructions that one uses to talk about it.
>
> Next, and with the same concentration and discernment, they learn the words that signify the properties of the things whose names they know.
>
> Eventually, extending their knowledge further, they notice certain actions or movements of these things; and simultaneously observing people talking about them, they distinguish—by their ability to pay attention to the movements, and to hear people repeat certain words in combination with the names of the things or of their properties—those words which signify actions. (Cordemoy 1668: 213–14; my trans.)

Adverbs, he said, are acquired much later, because modifications of actions are less important to the infant than the actions themselves. And grammarians teaching adults a foreign language concentrate first on nouns, then adjectives, and then verbs—"in imitation of the precepts given by nature to children" (p. 215).

Words are partly physical, in so far as they are sounds produced by our vocal apparatus. In that, they resemble the cries of animals. But animal 'signs' are communications without meaning. Their communicative effect results from the harmony between beast and beast, and between animal and environment, established by God as a system of interlocking reflexes. Words, by contrast, have meaning as well as sound, for (as Descartes had taught) they are the voluntary expression of conscious thoughts guided by reason. This is evident from the unpredictability and aptness of the words produced by human beings.

Moreover, said Cordemoy, souls are universally the same. Differences in linguistic ability, and in intelligence in general, are due to imperfections in the brain, not the mind. Freed of the body, all human souls would be equal, because equally rational. (To deal with the philosophical mind–body problem lurking here, Cordemoy developed a neuropsychological version of occasionalism: Chapter 2.iii.b.)

Whether Cordemoy was a "Cartesian" as well as a Cartesian is questionable. (We've already seen that Descartes himself wasn't.) The quotations given above might be read as implying that some specifically grammatical capacity leads the infant to concentrate first on nouns, then on adjectives, and finally on verbs. And perhaps Cordemoy did believe that. However, his remarks about the modifications of actions being less important to the child than the actions themselves suggest that it is the child's general interests, not its specifically grammatical preparedness, that guide the order of language acquisition.

In sum, Cordemoy may not have believed that grammar as such is foreshadowed in the infant's mind. But he certainly believed that humans have a natural capacity for language, which has no parallel in either animals or automata.

b. Arnauld and the abbey

The belief in some universal basis of language was promoted also by Antoine Arnauld. He was a highly valued correspondent of Descartes (2.ii.g)—and of Leibniz, too (Arnauld 1662, p. xxxv).

Although he was influenced by Descartes's philosophy, Arnauld wasn't a Cartesian but a (highly controversial) Jansenist. He was known primarily for holding unorthodox views on grace and the Eucharist. More to the point, here, he rejected Descartes's claim that we have innate knowledge of God. He didn't think that God's existence can be proved by the light of reason, but taught that Christian theism is—literally—our best bet (Arnauld 1662: 357). That is, he accepted the famous "wager" argument of his friend Blaise Pascal—who spent his last years at Arnauld's home base, the Abbey of Port-Royal.

Port-Royal was a Cistercian abbey for women, allowed by papal decree to accept male seculars in retreat. Arnauld was associated with it from early childhood: one sister was appointed Abbess in 1602, and four more—as well as his widowed mother—were nuns. In maturity, he was a key member of the influential 'Port-Royal logicians', and the main author of their most famous publications: the *Grammaire générale et raisonnée* and its successor treatise on reasoning, *La Logique; ou, L'Art de penser* (Lancelot and Arnauld 1660; Arnauld 1662). These texts openly adopted some of Descartes's ideas. But, as many of their pietistic examples showed, they weren't promoting pure Cartesianism. In particular:

> Port-Royal, thanks to the excellent philosophical tools employed by Arnauld, in general grammar developed a branch of Cartesianism which Descartes himself hadn't emphasized: namely, the study and analysis of language in general, assumed to be founded in reason alone. This Cartesian branch, planted and nurtured at Port-Royal, to some extent advanced the ideas familiar in the seventeenth century, and anticipated the labours of the eighteenth century, in which it would be taken up directly by [among others] du Marsais [and] Condillac . . . (Sainte-Beuve (1867), quoted in Cornelius 1965: 121; my trans.).

The scholastic logic of the medieval period had leant strongly on grammar when it was first developed. Indeed, medieval treatises contain many statements broadly consonant with Arnauld's (and Chomsky's) views. Roger Bacon (1214–94), for example, taught that "Grammar is substantially the same in all languages, even though it may vary accidentally" (cited in Lyons 1991: 172).

But most medieval logicians—as opposed to grammarians—weren't much interested in the grammar of real sentences. Rather, they used natural language to construct a semi-formal language with its own grammar, in order to demonstrate certain formal inference patterns.

By the opening of the seventeenth century, then, the links between logic and language were primarily with rhetoric (I. Thomas 1967). The Port-Royalists changed that.

c. The Port-Royal *Grammar*

In their fresh approach to logic, Arnauld and his colleagues revived its relation to grammar—which characterizes human speech, they said, but not the mimicking of word sounds by parrots (Lancelot and Arnauld 1660: 21). They aimed to state a "rational grammar". By this they meant one that is universally shared, and based on necessary principles of reason rather than contingent (and diverse) facts of linguistic usage.

The Port-Royal group distinguished a sentence's underlying propositional structure from its final expression. In this, they were following the medieval and Renaissance logicians. They claimed that words have a "natural order", in the sense that they provide the natural expression of our thoughts. Or rather, words considered as parts of speech (Subject, Verb, Object, Copula) have a natural order, although words as constituents of phrases or idioms may not. The natural logic underlying the various parts of speech may differ from the common-sense view: all verbs, for instance, were analysed as the Copula and an attribute (so that *Peter lives* became *Peter is alive*). Language and mind (rationality) were thus seen as intimately connected.

Moreover, just as reason is universal, so are some grammatical features. These weren't formal syntactic rules, as they were for Chomsky. Rather, they were features which—because of their natural relation to thought—aid the function of language: communication. (Significantly, the two subtitles of the *Grammar* advertised 'The Fundamental Principles of the Art of Speaking' *before* 'The Reasons of the General Agreement, and the Particular Differences of Languages'.)

For instance (the group argued), all languages that distinguish singular and plural must *naturally* ensure that nouns and adjectives, and verbs and nouns/pronouns, agree in their number (Lancelot and Arnauld 1660: 148–9). For to do otherwise would be to confuse the hearer, who hopes to discover the speaker's thought.

The Port-Royalists were well aware that languages don't always behave 'naturally'. English, for instance, distinguishes singular and plural in nouns, but not in adjectives: both *man* and *men* can be *learned*. Greek is puzzling, too, since plural neuter nouns take singular verbs. Arnauld dealt with such counter-examples by saying that a "figure of discourse" or idiom (which may lend a superficial elegance to the language) will have "some word [implicitly] understood", and that the hearer copes "by considering the thoughts more than the words themselves" (Lancelot and Arnauld 1660: 149). The clear implication was that some natural languages might be more natural (closer to rational thought) than others. No prizes need be offered for guessing which one was favoured by the denizens of Port-Royal:

> [There] is scarce any language, which uses these [superficially unnatural] figures less than the French: because it particularly delights in perspicuity, and in expressing things as much as possible, in the most natural and least intricate order; tho' at the same time it yields to none in elegance and beauty. (p. 154)

Arnauld distinguished between agreement, which is universal (except in a figure of discourse), and government, which is "almost entirely arbitrary" (Lancelot and Arnauld 1660: 148). By government, he meant the fact that using one word can cause some alteration in the form of another. An example discussed at some length is the phenomenon of case: genitive, dative, accusative, and so on. The case, which is a logical relation in thought, may or may not be apparent in the form of the noun (compare the Latin *terra*/*terram* with the English *ground*/*ground*). There may or may not be a special particle for a certain case (such as the French *de*). And a verb may or may not always take the same case (consider *servir quelqu'un* and *servir à quelque chose*). Moreover, since "cases and prepositions were invented for the same use", many prepositions govern the noun they are used with—thus *ad terram*, but *ab terra* (chs. 6 and 11, and pp. 149–52).

The various languages differ greatly in whether, and if so how, they mark this or that case by changes in the form of words and/or by the insertion of extra words. Occasionally, different governments alter the sense of the governed word. So in Latin (Arnauld pointed out), *cavere alicui* and *cavere aliquem* mean *to watch over a person's safety* and *to beware of him*, respectively. Such meaning shifts are grounded not in logic, but in arbitrary linguistic convention (p. 152). (As these examples show, the Port-Royalists did consider several European languages. But their main focus was on French, and some of their universalist claims were inconsistent even with English or Greek.)

With respect to those grammatical constructions where they believed that logic is involved, the Port-Royalists gave many detailed examples of the natural order they had in mind. In discussing relative clauses, for instance, Arnauld related specific grammatical forms to their "subordinate propositions" in thought (Arnauld 1662: 118–21). He distinguished "explicative" and "restrictive" uses of words like *which* and *who*, and argued that a particular grammatical conversion (ellipsis) can produce the underlying thought in one case but not the other.

For example, the subordinate proposition expressed by the words "men, who were created to know and love God" can be attained by substituting the antecedent for the pronoun itself, thus: "men were created to know and to love God". The reason is that the relative clause here is (for Arnauld) explicative: according to his theology, the idea of "man" includes the idea of a being created to know and love God. By contrast, the restrictive clause in the sentence "men who are pious are charitable" doesn't express the subordinate proposition (arrived at by a similar pronominal substitution) that "man as man is pious". Rather, it states that men may be pious ("the idea of piety is not incompatible with the idea of man"), and that the idea "charitable" can be affirmed of the complex idea "pious man".

Chomsky described the Port-Royalists' work as an anticipation of his own contrast between "deep" and "surface" structure. Further, he endorsed some of their specific hypotheses about how these are correlated (Chomsky 1966: 33–51, 97).

He claimed, for instance, that they employed a type of phrase-structure grammar (see Section vi.c). They held that the deep structure, or thought, consists of one or more elementary (subject–predicate) propositions, whereas the surface structure may be syntactically very complex: recursively nested, for example. Moreover, expressing the thought as an actual sentence may require more than just expressing each elementary proposition in words. It may also involve rearranging, replacing, or deleting parts of the first-stage (unspoken) sentences—which is to say, transforming them (Chomsky 1966: 40).

Broad similarities certainly exist. But these concern aspects of Chomsky's thought that weren't especially original or controversial. Moreover, some of Chomsky's comparisons are questionable in detail.

For instance, he says that Arnauld's treatment of relative clauses anticipated various "modern" logical–semantic insights, including the distinction between meaning and reference "in pretty much their contemporary sense" (Chomsky 1996*a*: 7).

It's true that the explicative–restrictive contrast rests on a distinction between the meaning of the term concerned (in this case, "man") and those objects that are picked out by it (actual men—who may or may not be pious). However, some distinction between meaning and reference had already been made by medieval logicians. Moreover, the Port-Royal contrast between the "comprehension" and "extension" of a general term wasn't equivalent to any distinction made in modern logic. It muddled singular and general terms, and also gave a definition of meaning—according to which it's part of the meaning of "triangle" that the internal angles equal two right angles—which is now regarded as over-inclusive (Kneale and Kneale 1962: 318 ff.).

Indeed, modern logicians see Arnauld's definition of meaning as an example of psychologism (2.ix.b). They therefore criticize the Port-Royal group as "the source of a bad fashion of confusing logic with epistemology" (Kneale and Kneale 1962: 316). That's why, by the 1950s, the Port-Royalists were only a minority taste in philosophical logic.

Nonetheless, the Port-Royal *Logic* was a creative advance at the time, and dominated logic for the next 200 years. And their *Grammar* was regularly reprinted until 1846. Consequently, their view that grammar reflects thought was widespread throughout this period, and still respected in the late nineteenth century.

d. Deaf-mutes and Diderot

One of the thinkers who accepted this Port-Royalist view was Denis Diderot (1713–84). He was editor-in-chief of the *Encyclopédie*, the 'bible' of the French Enlightenment—a project that had started out as a scheme to translate Chambers's two-volume *Cyclopaedia* (see Chapter 2.ii.c), but which ended up as twenty-eight newly written volumes (Porter 2001: 8). And he, too, was someone named by Chomsky as a precursor.

Descartes, as we've seen, had referred briefly to deaf-mutes, in making a point about the universality of reason and communication. But Diderot wrote a 17,000 word essay about them, exploring the relations between gesture, language, and thought (Diderot 1751). Part of his aim was to discuss the origins of language, and to ask whether there is some "natural" universal language, intelligible to all human beings.

Both questions were prominent topics of debate in the eighteenth century. They featured in several novelists' tales of imaginary voyages—some of which described "universal" artificial languages and symbol systems (Cornelius 1965).

Some people argued that the spontaneous signing of deaf-mutes represented the universal gestural language from which conventional languages had originally developed, and which might now be deliberately extended to form a worldwide communication system (Knowlson 1975; Large 1985). This 'handmade' system would be naturally intelligible to all, because it would be logically transparent (much as the Russell–Whitehead logical notation was later intended to be: 4.iii.c). Others had suggested new sign languages and/or alphabets, designed for universal use (Large 1985: 19–63).

The most interesting of these projects was that of John Wilkins (1614–72), Bishop of Chester and a founder member—and first Secretary—of the Royal Society. More interested in phonetics than in grammar, he proposed a universal alphabet, systematically designed to encompass "all such kinds of simple sound, which can be framed by the mouths of men" (J. Wilkins 1668: 357–62).

His reference to "the mouths of men" was no mere rhetorical flourish. For his notation was comprised of iconic symbols depicting various positions of the speech organs: root/tip of tongue, one/two lips, top/root of teeth, back/front of palate, parts of mouth, and nose. It also marked whether or not the sound involved breathing. In addition, Wilkins designed a more easily usable alphabet. This had some highly stylized iconic features, and some coherence in using similar symbols to represent similar sounds.

The supposed intelligibility of this language wasn't grounded only in the universal ability to make speech sounds and mouth movements. For the systematic coherence spread into semantics, too. Wilkins gave every letter a significance, based on the forty classes into which, according to him, *everything* could be fitted. Each class was subdivided into differences, and these into species. To each class, he gave a two-letter syllable, and a consonant and a vowel were assigned to each difference and species, respectively. So flame was called *deba*: the *de* meant 'element', the *b* meant 'fire' (the first of the four elements), and the *a* meant a small part of the element: namely, a 'flame'. In short, semantic hierarchies were explicitly reflected in phonetics, and in spelling too.

Another aim of Diderot's essay, regarded by some as "one of the outstanding examples of literary criticism in the eighteenth century" (A. M. Wilson 1972: 123), was to distinguish various rhetorical genres. So he compared prose, poetry, drama, scientific writing, and so on.

Here, Descartes was lurking in the background. His clear prose, which reflected his views on the best "method" of thinking, had caused a revolution in scientific and philosophical writing. This is evident, for example, in the striking stylistic differences between the three versions of Joseph Glanville's *Vanity of Dogmatizing* (Glanville 1661–76). These arose because, between the first and second drafts, Glanville adopted the Royal Society's recommendation of Descartes's prose over the more florid writing of the early Renaissance. (One aspect of this was the rejection of interpretative accounts of Nature, considered as God's text, in favour of empirical ones: see Chapter 2.iii.b.)

Diderot went further, arguing that authors should choose not only the most appropriate genre for the type of thought they wished to express, but also the most fitting language. He had in mind not just choice of vocabulary (*le mot juste*), but selection among natural languages themselves. And here, Chomsky has remarked (1965: 7), he illustrated the lasting influence of Port-Royal. For he declared, as Arnauld had done before him, that French surpasses all other languages in the degree to which it matches the order of our thoughts. Everyone, in so far as they are rational, must think their thoughts in the same order. Because French matches this rational order best, he argued, it is the natural language for philosophy and science. Greek, Latin, Italian, and English are more appropriate for less reasoned enterprises, such as the theatre—and in general for persuasion, emotionalism, and deceit.

Only his French readers, no doubt, found that particular claim compelling. But Diderot's view that grammar has some principled basis in thought was widely shared.

9.iv. Humboldt's Humanism

Ten years after the final publication of the *Encyclopédie*, the Enlightenment criticisms of traditional institutions and beliefs were followed by Kant's radical critique of human knowledge itself. Accordingly, some late eighteenth-century philosophers, described as "Cartesians" by Chomsky, conceptualized language in neo-Kantian terms.

That is, they posited—however vaguely—some linguistic equivalent of the empirical intuitions and categories (2.vi.a). This was supposed to be a formative principle shared by all humankind, which moulds all conceivable languages much as the Kantian intuitions inform our ideas about physical reality.

Typically, this 'grammar' wasn't thought of as a static set of given principles—like the Port-Royal logic, or even Kant's space and time. Rather, it was a developing, self-organizing, inner form. As such, it was broadly comparable to the organic forms then being posited in *Naturphilosophie* and Johann von Goethe's biological morphology (see 2.vi.e).

One of these neo-Kantian writers was Johann Herder (1744–1803), who in a prizewinning essay on the origins of language argued that language wasn't given to us in its full form by God, but isn't wholly invented either (Herder 1772). It's part of our innermost nature, and very different from the narrowly focused and unvarying instincts of animals. Rather, it's the ground of mankind's freedom and self-development. Indeed, Herder compared the genesis of language to the mature embryo pressing to be born. He also described a nation's mother tongue as an aspect of the national "soul", or "spirit".

All these notions were to be taken up by the Romantics in general, and by Humboldt in particular. Humboldt, in turn, would have a strong influence on nineteenth-century Romanticism. (In the early 1800s, he spent four years in Paris at a time when William Wordsworth, for instance, was living there too.)

a. Language as humanity

Herder's essay was read, about ten years later, by the young Humboldt—in his teens at the time. Eventually, Humboldt became a renowned humanist scholar (and sometime Prussian ambassador to England). But he wasn't just a fine scholar: he added some enormously influential ideas of his own.

Humboldt's philosophy of language, like Kant's epistemology, has been described as a Copernican revolution (Hansen-Love 1972). For it presents the human mind as the creative origin of knowledge and culture, instead of being dependent on them for its ideas. Accordingly, his position (following the *verum factum* tradition of Giambattista Vico: see 1.i.b) promised a deep knowledge of language and culture—much deeper than scientific knowledge of material things could ever be.

Humboldt saw human beings as *set apart* from nature (including animals) by the mental faculty of language, through which the human mind or spirit is both developed and expressed. Like Herder, he thought of language in terms of organic development, describing it as a "living seed" and "an internally connected organism" (Humboldt 1836: 61, 21). He was thinking not only of the development of language in infancy but, even more, of the historical development of language from its earliest roots:

Language, regarded in its real nature, is an enduring thing, and at every moment a *transitory* one . . . In itself it is no product (*Ergon*), but an activity (*Energeia*). Its true definition can therefore only be a genetic one. For it is the ever-repeated *mental labour* of making the *articulated* sound capable of expressing *thought* . . . To describe languages as a *work of the spirit* is a perfectly correct and adequate terminology, if only because the existence of spirit as such can be thought of only in and as activity. (Humboldt 1836: 49).

Humanity as such was continually celebrated by Humboldt. He often did so in spiritual terms, as when he said that "thought at its most human is a yearning from darkness into light, from confinement into the infinite" (1836: 55). This causes a recurring difficulty when one tries to relate his work to modern linguistics and cognitive science (and when one translates his term *Geist* as "mind" rather than "spirit").

For instance, he pointed out, as Cordemoy had done before him, that human anatomy seems to be especially apt for the production of speech (a fact to be explored in fascinating detail over 100 years later: Lenneberg 1967). But he *didn't* explain these anatomical phenomena as due to the benevolent design of God. Instead, he saw them as aspects of the activity of the creative human spirit. He even linked bipedalism to—or explained it in terms of?—our language and humanity:

And suited, finally, to vocalization is the upright posture of man, denied to animals; man is thereby summoned, as it were, to his feet. For speech does not aim at hollow extinction in the ground, but demands to pour freely from the lips towards the person addressed, to be accompanied by facial expression and demeanour and by gestures of the hand, and thereby to surround itself at once with everything that proclaims man human. (1836: 56)

In short, although Humboldt's observations were often highly astute, many of his scientific questions were fundamentally different from those posed in the empiricist tradition. His work therefore sets one of the "traps" mentioned in Chapter 1.iii.a. To interpret Humboldt's questions and/or answers in terms of the explanatory categories of modern science—which Chomsky frequently did—is often to distort them.

It would be misleading, for instance, to praise Humboldt's naturalistic (i.e. non-theological) explanation of bipedalism as being about as close as a pre-Darwinian scientist could have got. For Humboldt was not only pre-Darwin: he was also pro-Goethe. In other words, his conception of "science" was very different from the empiricist's (see 2.vi).

Like Goethe (a close friend, and a neighbour), Humboldt favoured organic holism over analytic science:

The comprehension of *words* is a thing entirely different from the understanding of *unarticulated sounds* . . . [The word] is perceived as articulated, [which perception] presents the word directly through its form as part of an infinite whole, a language. (1836: 57–8)

It is thus self-evident that in the concept of linguistic form no detail may ever be accepted as an *isolated fact*, but only insofar as a method of language-making can be discovered therein. (p. 52)

Connected discourse . . . is all the more proof that language proper lies in the act of its real production. It alone must in general be thought of as the true and primary, in all investigations which are to penetrate into the living essentiality of language. The break-up into words and rules is only a dead makeshift of scientific analysis. (p. 49)

To be sure, language involves "a system of rules" (p. 62). But that system is an organic whole, which "grows, in the course of millennia, into an independent force", restricting how we can express (and think) our thoughts. Its generative principle is humanity itself ("the unity of human nature"):

Language belongs to me, because I bring it forth as I do; and since the ground of this lies at once in the speaking and having-spoken of every generation of men, so far as speech-communication may have prevailed unbroken among them, it is language itself which restrains me when I speak. But that in it which limits and determines me has arrived there from a human nature intimately allied to my own, and its alien element is therefore alien only for my transitory individual nature, not for my original and true one. (p. 63)

The similarity between this passage and Chomsky's nativism is evident. But the differences are arguably even greater (see subsections e–g, below).

b. Languages and cultures

Drawing partly on Diderot's *Encyclopédie*, and in particular on the three volumes of its summary (the *Encyclopédie méthodique*) that were devoted to grammar and literature, Humboldt studied all the European languages, ancient and modern. But he didn't stop there.

He also described several native American languages, relying on novel data brought back by his younger brother Alexander—a noted scientist much admired by Charles Babbage (Hyman 1982: 72–4), who spent five years exploring the Americas. In addition, he was familiar with many oriental examples, including Chinese, Sanskrit, and Hebrew. And, thanks to the interest in non-Indo-European languages that was aroused by European colonialism in the eighteenth century, he also had knowledge of Melanesian and Polynesian—and the various dialects of Kawi, the ancient literary and sacred language of Java.

Kawi was especially interesting to him largely because it seemed to be a mixed language: although its vocabulary was Sanskrit, its grammar was Malayan. Moreover, its geographical position suggested (to him) that it might be linked to virtually all the known languages of the world. Given Humboldt's passion for intellectual synthesis, and his view that "language" is the fundamental unifying force within all humankind, Kawi seemed to offer the hope of integrating "old world" and "new world" languages—and cultures (Humboldt 1836, p. xii).

His three volumes on Kawi, prepared in the closing years of his life, included a 350-page introduction that gives the clearest statement of his views on *language in general* (Humboldt 1836). As well as considering syntax, morphology, and (especially) phonetics, Humboldt studied the various literatures and orally transmitted poetry and prose. When he thought of language, then, he had in mind both linguistics and literature.

These wide-ranging studies weren't mere scholarly stamp collecting: Humboldt was no Dr Casaubon. They were driven by Humboldt's novel ambition to distinguish language groups in terms of their grammatical (and phonetic) structure, rather than their historical origins. They were intended to show, too, that all languages are essentially alike:

My aim is [not to gather the external details of individual languages, but] a study that treats the faculty of speech in its inward aspect, as a human faculty, and which uses its effects, languages, only as sources of knowledge and examples in developing the argument. I wish to show that what makes any particular language what it is, is its grammatical structure and to explain how the grammatical structure in all its diversities still can only follow certain methods that will be listed one by one . . . (Humboldt 1836, p. xiv)

His studies also underpinned Humboldt's distinctive position on the relation between language and culture, and on the individuality of cultures.

If thoughts can be expressed only in language, difference in language implies difference in culture. Humboldt argued that each natural language is unique. *Perfect* translation is impossible, since apparently equivalent words will have different associations in the minds of speakers of the two languages (and even in individual speakers of "one" language). Nevertheless, *good* translation—on which he had many interesting things to say (Novak 1972)—can expand the language and mentality of the reader.

Although he held that everything can in principle be expressed in every language, he pointed out that in practice languages (cultures) differ in what they choose to express. Translations of literary masterpieces can thus awaken latent potential in the receiving nations. (Scientific thought, he said, varies less across languages.)

Each culture, for Humboldt, has its own *Weltansicht*, its unique manner of regarding the world. Human individuals are deeply informed by their culture, as well as by their unique life experience. And their cultural identity must be thought of holistically. For a language, like an organism, is a self-organizing whole, not a collection of independently definable parts.

c. Humboldt lives!

Humboldt's stress on the diversity, and the holism, of cultures informed some early twentieth-century work on topics relevant to cognitive science. So, too, did his views on the near-identity of language and thought.

The holistic view of language was to be taken up, for example, by the linguists Ferdinand de Saussure (1857–1913) and Roman Jakobson (1896–1982). Although his birth-date locates Saussure firmly in the nineteenth century, his general intellectual presence dates from well into the twentieth. For his influence blossomed only after his death, with the posthumous publication of his lecture notes on "general" linguistics (Saussure 1916).

After a lifetime of detailed research in historical philology, Saussure had finally considered language as such. (It was he who coined the phrase "the arbitrariness of the sign"—see ii.c, above.) He described language as a holistic system, in which the meaning of one word, or sign, can be defined only in relation to others. To include a word, such as *cat*, in a language is not to provide an inert pointer referring to some pre-existing class, but to identify some class by distinguishing it from others picked out by different words (*kitten*, *dog* . . .).

Jakobson was a neo-Humboldtian whom Chomsky has acknowledged as a major influence in the origination (not just the *post hoc* justification) of his own thinking. Based at Harvard while Chomsky was a Junior Fellow there, Jakobson concentrated on phonology, not meaning, and combined holism with nativism (compare Humboldt's "inner form", discussed below). He claimed—as it turned out, wrongly (Sampson 1974)—that the seemingly chaotic speech sounds heard around the world can be defined in terms of twelve oppositional "distinctive features" (Jakobson 1942/3; Jakobson *et al.*

1952). These, he said, comprise a hierarchical system of phonological capacities shared by all human beings. Each language selects only some of the possibilities available.

As for linguistic diversity, Humboldt's ideas were echoed by the anthropological linguist Franz Boas (1858–1942), who set in hand the systematic description of the fast-disappearing native American tongues (Boas 1911). (It was he who'd translated the story used in Frederic Bartlett's experiments on memory: see Chapter 5.ii.b.)

Whereas previous accounts of these languages had focused largely on vocabulary, Boas focused on grammar. He argued that each natural language has a distinct grammatical structure. Moreover, he said, the structural differences can't always be expressed by using shared grammatical categories, as when one says that adjectives are usually put after the noun in French but before it in English. Rather, they may need grammatical categories specifically designed to describe the particular syntax concerned.

Edward Sapir (1884–1939) and Benjamin Whorf (1897–1941) went further, claiming that virtually none of the concepts of one language can be understood (thought) by people brought up to speak another. Whorf even held that each language conceals a specific metaphysics, or ontology, unintelligible to people with different mother tongues (Carroll 1956). On this view, there is no 'true' ontology: languages differ arbitrarily from each other, and views of reality differ likewise (cf. 1.iii.b).

This claim—that a person's thought can't escape their language—was hotly disputed through the 1960s and beyond (Hoijer 1954; R. Brown 1958; Hymes 1964: 149–53; Rosch 1977; Lucy 1992). Empiricist philosophers complained that the Sapir–Whorf 'hypothesis' is slippery and vague, and if clarified becomes either obviously true or obviously false (M. Black 1959). But many social scientists, doubting the obviousness, garnered evidence to test it.

Some of these people, influenced by Chomsky's nativism (see Section vii.c–d), claimed that apparently arbitrary differences in colour vocabulary are in fact underlain by universally shared colour categories (O. B. Berlin and Kay 1969; Rosch/Heider 1972; Rosch 1973; see Chapter 8.iia–b). This early work had glaring methodological faults (Sampson 1980: 95–102). But Chomskyans still reject the Sapir–Whorf hypothesis, saying "there is no scientific evidence" for it and attributing it to "a collective suspension of disbelief" (Pinker 1994: 58).

Others followed Sapir and Whorf, suggesting that cognitive differences may be associated with, or even grounded in, syntactic variations as well as vocabulary. For instance, the Navaho language uses a different form of the verbs for handling things (such as *throw*), depending on the shape of the object handled. This grammatical distinction isn't found in Indo-European languages, and suggests that speakers of Navaho may have a relatively keen awareness of shape (Carroll and Casagrande 1966: 26–31; Lucy 1992: 198–208). Only "suggests": independent tests need to be done to confirm any specific instance of the Whorfian hypothesis.

d. A fivefold list

There's a puzzle here, however.

To concentrate on the syntactic differences between languages seems decidedly un-Chomskyan, given that Chomsky posited a universal grammar. Indeed, Humboldt's aim of describing the grammatical structure of as many languages as possible was defined as the major goal of linguistics by the very people against whom Chomsky reacted in the 1950s (see Section v.b, below). And Chomskyans, as remarked above, reject neo-Humboldtian views on the boundless diversity of languages and concepts.

Yet Humboldt was repeatedly named by Chomsky as an important precursor, who had essentially similar aims. In his first act of homage to the "Cartesian" linguists, Chomsky devoted more space to Humboldt than to anyone else (Chomsky 1964: 17–25; 1966: 19–28).—Why is this?

The answer is fivefold:

* First, Humboldt insisted on the universality of language in the human species.
* Second, he saw it as an innate mental faculty, or force, distinct from other psychological abilities and from the instincts of animals.
* Third, he saw language as inexplicable in terms of its ultimate origins (although he explained many features of specific languages by reference to earlier ones).
* Fourth, he contrasted the finite number of objects in the physical world, including the fixed number of speech sounds, with the infinite creativity of language, which can always come up with some new thought. Language, therefore, implies freedom.
* And fifth, he saw language as having an organic "inner form" that unites humanity, irrespective of culture, in a fundamental way.

Stated briefly, as I've just done, the similarities between Humboldt and Chomsky seem to be clear—indeed, overwhelming. In truth, however, they're neither.

We needn't waste time cavilling over the first two items. These (nativist) beliefs are evident in Humboldt's reply to the familiar empiricist argument that children will learn quite different languages, depending on who brings them up:

> Man is everywhere one with man, and development of the ability to use language can therefore go on with the aid of every given individual [i.e. caretaker]. It occurs no less, on that account, from within one's own self; only because it always needs an outer stimulus as well, must it prove analogous to what it actually experiences, and can do so in virtue of the congruence of all human tongues. (Humboldt 1836: 59)

He also argued (like Chomsky) that the infant's linguistic capacity must be shared with everyone else—otherwise, how could they learn so quickly (pp. 58–9)?

So far, so similar. But the last three items need fuller discussion.

e. Origins

The third item is included in the list because Chomsky claimed that there could be no evolutionary explanation of language—or at least, that we could never find one. Language, he argued, is a holistic system with an all-pervasive structure, not a collection of independently acquired habits or tricks. Humboldt had said much the same thing. But Humboldt, born almost 100 years before *On the Origin of Species*, couldn't consider the possibility of a Darwinist evolutionary account. Chomsky could, and did.

Chomsky was no less pessimistic than Humboldt about our ever understanding the origin of language as such:

> There seems to be no substance to the view that human language is simply a more complex instance of something to be found elsewhere in the animal world. This poses a problem for the

biologist, since, if true, it is an example of true "emergence"—the appearance of a qualitatively different phenomenon at a specific stage of complexity of organization. (Chomsky 1986: 62)

In fact, the processes by which the human mind achieved its present stage of complexity and its particular form of innate organization are a total mystery . . . . It is perfectly safe to attribute this development to "natural selection", so long as we realize that there is no substance to this assertion, that it amounts to nothing more than a belief that there is some naturalistic explanation for these phenomena. (p. 83)

Admittedly, "evolution" was listed as one of only five keywords for Chomsky's official précis of one of his books (1980*b*: 1). However, this was due not to the author but to the editor, on the grounds that talk of universal grammar predictably raises questions about origins (A. N. Chomsky, S. Harnad, personal communications).

Chomsky's pessimistic remarks of the 1980s, cited above, implied that "emergence" is inexplicable. They implied, also, that the naturalistic explanation of language would refer to some single genetic mutation, or possibly some small set of mutations, that occurred early in our species' history, whose effect on the "emergent" phenotype is opaque. If that were so, then we couldn't hope to find it/them, nor to understand its/their significance if we did.

Similar anti-evolutionary scepticism has often been associated with the complex structure of the eye. Darwin himself admitted that to explain the eye in terms of natural selection "seems, I freely confess, absurd in the highest degree" (quoted in Cronly-Dillon 1991: 15). Nevertheless, he rebutted this scepticism in general terms, and evolutionary biology can now do so in detail. It turns out that the eye is neither a miracle nor even a singularity: eyes have evolved independently a number of times. (Or perhaps not: evidence that all types of eye have evolved from the same root is given in Gehring and Ikeo 1999.) Possibly, then, we shall one day understand the evolution of language much as we now understand the evolution of the eye.

Some Chomskyans, indeed, have speculated at length about the evolution of "the language instinct" (Pinker 1994). As for Chomsky himself, he now allows that we're beginning to understand how a qualitatively different phenomenon can arise spontaneously from a simpler base (see Chapters 14.x.a–b and 15). By the new millennium, he had even suggested that language might be based in spontaneous physical self-organization:

There is little doubt that the human language faculty is a core element of specific human nature . . . .

Recent work suggests more far-reaching possibilities. The minimal conditions on usability of language are that languages provide the means to express the thoughts we have with the available sensorimotor apparatus. *One far-reaching possibility* is that, in non-trivial respects, the language faculty approaches an optimal solution to these minimum design specifications (with "optimality" characterised in natural computational terms). If true, that would suggest interesting directions for the study of neural realization, and perhaps for the further investigation of the critical role of physical law and mathematical properties of complex systems in constraining the "channel" within which natural selection proceeds.

What the work on optimal design of language seems to suggest is that language *might* be more like the appearance of familiar mathematical structures in nature such as shells of viruses or snowflakes than like prey/predator becoming faster to escape/catch one another [see Chapter 15, below]. When the brain reached a certain state, some small change *might* have led to a reorganisation of structure that included a (reasonably well-designed) language faculty. *Maybe.* (Chomsky 1999: 31; italics added)

However, if natural selection must somehow underlie language, as it underlies every other biological phenomenon, that's not to say that Chomsky expects a detailed evolutionary explanation. On being charged (on the evidence of the passage just quoted) with changing his views on the role of natural selection, he replied:

[I presuppose] that natural selection is operative in this case (as in others). [And I have raised some tentative] questions about the role of "the 'channel' within which natural selection proceeds." That natural selection proceeds within such a "channel" is too obvious to have been questioned by anyone. Only the most extreme dogmatist would produce a priori declarations about its role in any particular case, whether it is virus shells, slime molds, bones of the middle ear, infrared vision, or whatever.

As before, I take no particular stand on the matter, for the simple and sufficient reason that virtually nothing significant is known—even about vastly simpler questions, such as the evolutionary origins of the waggle dance of honeybees. (Chomsky, email to "evolutionary psychology" list, 19 Oct. 1999; italics added)

Chomsky's agnosticism, here, is well judged. It's not clear that one can go much further than speculation on this topic. Languages don't fossilize. Nor do soft structures such as vocal cords (although oral bones and skulls do). Moreover, animal communication systems seem to be very unlike ours, having only finite productive power—as both Humboldt and Chomsky were keen to point out.

On the other hand, mammalian movements are highly variable—and recent work suggests that gestural language is grounded in neuroscientific mechanisms (such as mirror neurones; Arbib 2005) originally evolved for controlling other bodily actions (14.vii.c). Even the logic of *subject–predicate* has been attributed to specific neural circuits in the brain, embodying mechanisms necessary for many tasks performed by non-human primates (Hurford 2003).

In sum, if scepticism about the very possibility of linguistic evolution is out of place, pessimism about our ever being able to detail it may not be (Maynard Smith and Szathmáry 1995, ch. 17; 1999, ch. 13).

f. Creativity of language

The fourth item is included on the list (in subsection d, above) because Chomsky sees Humboldt as anticipating his own notion of the creativity of language. He first argued for this in 1962, at the International Congress of Linguists in Cambridge, Massachusetts; his paper was revised and expanded before publication (Chomsky 1964: 17–22).

Certainly, Humboldt constantly described language—alias the human spirit—as creative, indeed as self-creative. And he spoke of language making infinite use of finite means:

But just as the matter of thinking, and the infinity of its combinations, can never be exhausted, so it is equally impossible to do this with the mass of what calls for designation and connection in language. In addition to its already formed elements, language also consists, before all else, of methods for carrying forward the work of the mind, to which it prescribes the path and the form.

> The elements, once firmly fashioned, constitute, indeed, a relatively dead mass, but one which bears within itself the living seed of a never-ending determinability. (Humboldt 1836: 61)

Whether Chomsky is right to see Humboldt, and the other "Cartesians", as a precursor in this respect is another matter.

Chomsky's definition of the creativity of language referred to syntax, not thought (see Section vi.b). Admittedly, when (in the early 1960s) he included semantics within his grammar, novel meaning would automatically accompany novel form. But even then, he still prioritized syntax (see Section viii.c).

Descartes, by contrast, had focused on the power of (language-informed) reason or imagination to generate new thoughts, expressible in language. And Humboldt, too, had spoken of the power of language to define new thoughts. Chomsky stressed the purely formal fact that his grammar can generate infinitely many new sentences and deep structures, saying that "this 'creative' aspect of language is its essential characteristic" (Chomsky 1964: 8), and that "recursive rules . . . provide the basis for the creative aspect of language use" (Chomsky 1967: 7). His later complaint that "A number of professional linguists have repeatedly confused what I refer to here as 'the creative aspect of language use' with the recursive property of generative grammars, a very different matter" was thus misdirected (Chomsky 1972*a*, p. viii). In short, his notion of linguistic creativity was not that of his august predecessors.

Moreover, Chomsky's concept of creativity fails to capture all the cases ordinarily covered by the term, including those which Humboldt seems to have in mind here:

> At every single point and period, therefore, language, like nature itself, appears to man—in contrast to all else that he has already known and thought of—as an inexhaustible storehouse, in which the mind can always discover something new to it, and feeling perceive what it has not yet felt in this way. (Humboldt 1836: 61)

What Chomsky calls creativity is the exploration of an unchanging conceptual space, or thinking style (namely, generative grammar). Some human creativity is like this: run-of-the-mill jazz improvisation, for instance, or mundane examples of what Thomas Kuhn (1962) called normal science. But the most interesting cases are not.

These cases involve either unfamiliar combinations of familiar ideas, or the transformation of an existing conceptual space by altering one or more of its defining dimensions (Boden 1990*a*/2004, 1994*b*; cf. 13.iv). Such transformations make it possible to generate structures that were previously impossible (Humboldt's "things not felt this way before"?). Successive transformations over days, years, or even centuries (Humboldt, again?) can evolve the thinking style in complex and unpredictable ways: the development of post-Renaissance tonal music, for example (C. Rosen 1976; Boden 1990*a*: 59–62).

Both these types of creativity go beyond what Chomsky means by the term. And both, arguably, were what the Romantics were interested in. The poet Samuel Taylor Coleridge (1772–1834), who popularized neo-Kantianism in England, highlighted the mind's propensity for combinational creativity (Livingston Lowes 1930; Boden 1990*a*, ch. 6; see also the preamble to Chapter 12, below). And transformational creativity is perhaps comparable to the more structured, developmental, creativity posited by Humboldt.

I say "arguably" and "perhaps", here, because this point depends on how we understand the fifth item on our list. Just what did Humboldt mean by the "inner form" of language?

g. The inner form

In his Kawi-introduction, Humboldt described this inner form as a creative human force that is unfurled and developed in life, a vital linguistic instinct whose expression can be shaped, favoured, and hindered by external circumstances (Humboldt 1836, esp. 48–53). The environment doesn't impose itself on a passive tabula rasa, but elicits an active response from a mind inherently suited to make linguistically relevant distinctions—between speech sounds and other sounds, for example. And, he said, even (indeed, especially) the languages of "so-called savages" show a typically human diversity of expression.

The broad similarity to Chomsky's approach is, again, evident. And Chomsky declared that Humboldt's notion of form as generative process was "his most original and fruitful contribution to linguistic theory" (Chomsky 1964: 17). But the similarity between the two linguists becomes more fuzzy on closer inspection.

Humboldt's definitions of the inner form of language were both various and vague:

> The constant and uniform element in this mental labour of elevating articulated sound to an expression of thought, when viewed in its fullest possible comprehension and systematically presented, constitutes the *form* of language. (1836: 50)

So far, so (apparently) Chomskyan. However, Humboldt immediately added a gloss that doesn't sound Chomskyan at all:

> In this definition, form appears as an *abstraction* fashioned by science. But it would be quite wrong to see it also in itself as a mere non-existent thought-entity of this kind. In actuality, rather, it is the quite individual *urge* whereby a nation gives validity to thought and feeling in language. (p. 50)

Two pages later, having remarked that "the characteristic form of languages depends on every *single* one of their smallest *elements*", and that "language, in whatever shape we may receive it, is always the spiritual exhalation of a nationally individual life", he passed from nation states to the ultimate nature of language:

> From the foregoing remarks it is already self-evident that by the form of language we are by no means alluding merely to the so-called *grammatical form* . . . The concept of form in languages extends far beyond the rules of *word order* and even beyond those of *word formation*, insofar as we mean by these the application of certain general logical categories, of active and passive, substance, attribute, etc. to the roots and basic words. It is quite peculiarly applicable to the formation of the *basic words* themselves, and must in fact be applied to them as much as possible, if the nature of the language is to be truly recognizable. (p. 51)

In that quotation, Humboldt seemed to have historical philology in mind. The crucial questions would then concern (for instance) how the Indo-European languages developed, and how they differ from the Malayan group, rather than (for example) the relations between active and passive, or the hidden depths of *John is easy/eager to please*. But Chomsky interprets inner form as embracing "the rules of syntax and word formation as well as the sound system and the rules that determine the system of concepts that constitute the lexicon" (Chomsky 1966: 26–7). And Chomsky's interpretation finds support in the following remark of Humboldt:

> [We must engage in] a laborious examining of fundamentals, which often extends to minutiae; but there are also details, plainly quite paltry in themselves, on which the total effect of languages is dependent, and nothing is so inconsistent with their study as to seek out in them only what is great, inspired, and pre-eminent. Exact investigation of every grammatical subtlety, every division of words into their elements, is necessary throughout, if we are not to be exposed to errors in all our judgments about them. It is thus self-evident that in the concept of linguistic form no detail may ever be accepted as an *isolated fact*, but only insofar as a method of language-making can be discovered therein. (Humboldt 1836: 52)

Because Humboldt defined "inner form" in so many different (and imprecise) ways, it's not clear whether he really is Chomsky's precursor in this fifth sense. Indeed, one might say that he never *defined* it at all: "what it means is never revealed either by way of explanation or example, let alone definition which is a device he seems to have spurned" (Aarsleff, in Humboldt 1836, p. xvi). It's hardly surprising, then, that "for a hundred years all discussion has failed to converge on any accepted meaning" (p. xvi). ("*All* discussion" includes scholars who can read Humboldt in the original German, as I myself cannot; so I don't think the translation can be to blame, here.)

When Humboldt was more specific about just what these formal relations might be, he was often less plausible. Thus he remarked that certain stories have developed independently in different cultures, explaining this not as the result of shared human experiences but as the unfolding of innate ideas (Novak 1972: 128). He even claimed that similar sounds are used by numerous languages to express similar ideas—where he wasn't thinking of onomatopoeia, as in *buzz*, *miaow*, or perhaps *crack* (Jespersen 1922: 57).

In short, Humboldt's "inner form" of language was specified only a little more clearly than Leibniz's veins of marble. Even his friends criticized him for his irremediable vagueness (Humboldt 1836, p. xv). And one of his twentieth-century admirers, the Danish linguist Otto Jespersen (1860–1943), put it in a nutshell: "Humboldt, as it were, lifts us to a higher plane, where the air may be purer, but where it is also thinner and not seldom cloudier as well" (Jespersen 1922: 57). If such a remark could be made even by an admirer, less sympathetic critics would clearly need some persuading.

A twentieth-century linguist wishing to revive the unfashionable humanist/nativist ("Cartesian") approach would face several challenges:

* They'd have to clarify the relation between linguistic structure and meaning.
* They'd have to explain just how linguistic variability is possible, and how it differs from the changes (learning) found in other types of behaviour.
* They'd have to specify what the universal innate ideas of language might be, and reconcile them with the diversity of actual tongues.
* And they'd have to show that nativist theories of language aren't mere optional alternatives to empiricist accounts, but inescapably more appropriate.

In Sections vi–viii, we'll see how Chomsky tried to do all those things. And in Section ix we'll see that by the 1980s, despite his enormous influence over the preceding quarter-century, some of his fundamental arguments were being rejected.

First, however, we must understand why it was that "Cartesian" linguistics became unfashionable in the first place.

9.v. The *Status Quo Ante*

Chomsky's immediate predecessors had no interest in the tasks just listed. The professional linguists he encountered as a student dismissed questions about inner forms of grammar, although some did engage with Jakobson's inner forms of phonology. They assumed that languages are arbitrarily diverse systems, wholly learnt from the environment.

If Humboldt's influence was still strong (whether acknowledged or not), it was in his wish to describe the many distinct grammars exemplified by natural languages, and to avoid historical bias in doing so. These two aims suffused the tradition in which Chomsky was trained: American "descriptive", or "structuralist", linguistics.

Following Boas's lead (see Section iv.c), the structuralists produced detailed descriptions, or taxonomies, of the grammatical categories of many little-known languages. They saw linguistics as an autonomous enterprise based firmly in the present, an example of what Saussure (1916) had called "synchronic" linguistics. That is, they aimed to describe speech, morphology, and grammars by a scientific method not tainted by history—nor, as we'll see, by introspection.

Like any intellectual movement, this one included a variety of positions. With respect to structuralism as a whole, it's been said that Chomsky's linguistics "continued some fundamental traits of its predecessor, recovered others, and unwittingly rediscovered still others" (Hymes and Faught 1981: 1). What's most relevant for our purposes is that the dominant school in the late 1930s–1940s—Bloomfieldian linguistics—adopted a broadly behaviourist methodology, and that it included theorists who tried to analyse linguistic structure in formal terms. Chomsky's work would undermine the former and emphasize the latter.

a. Two anti-rationalist 'isms'

The Bloomfieldians' rejection of introspection, and their striving for formalism, were inherited from two closely related movements in early twentieth-century thought. These were behaviourism in psychology and positivism in the philosophy of science.

Introspection was ousted from (American) psychology long before being banished from linguistics. Boas's follower Leonard Bloomfield (1887–1949) started out as an admirer of Wilhelm Wundt, whose mentalistic folk psychology informed his first book (Bloomfield 1914). As a professor of German, he could read Wundt in the original: the first five (of ten) volumes of the *Volkpsychologie*, had been published by the time Bloomfield wrote his text. (A brief summary appeared in English two years later: Wundt 1916.) His book also paid homage to Humboldt, as the founder of general linguistics (1914, ch. 10).

With hindsight, it's intriguing that Bloomfield recounted Wundt's theory of "inner forms". A meaning, or inner form, may be expressed in various grammatical sequences, as in *Mary ate the apple* and *The apple was eaten by Mary*—plus, of course, equivalent sentences in other languages. Wundt had argued that linguistics should focus on *sentences*, where a 'sentence' can't be defined in terms of words or phrases, but is recognized intuitively by native speakers. He'd also held that processes of transformation (*Umwandlung*) produce grammatical sequences from inner forms.

But Bloomfield didn't follow up these ideas. In the very year in which he praised Wundt's mentalistic work in a leading psychological journal (Bloomfield 1913), another such journal published John Watson's seminal paper on 'Psychology as the Behaviorist Views It' (J. B. Watson 1913). That paper swept the board in psychology: Watson was elected President of the American Psychological Association only two years later. Eventually, Watson would defeat Wundt in linguistics too.

Bloomfield soon abandoned Wundt, and his pre-Watsonian talk of inner forms and hidden transformations. Indeed, thanks to a "shocking" book very different from his first one, he became the leader of "scientific" structuralist linguistics, which downplayed meanings and had no place for inner forms (see below). But this didn't happen overnight. Two decades would pass before the behaviourist approach conquered in linguistics.

Watson's iconoclasm had been fuelled by frustration with the experimental impasse in psychology, *not* by independent philosophical argument. But the later behaviourists justified their approach by appeal to logical positivism, which swept the philosophy of science in the 1920s and 1930s.

Several leading positivists emigrated from Europe to America shortly before the Second World War. They included Hans Reichenbach (1891–1953) and Rudolf Carnap—one of whose protégés was Walter Pitts (see 4.iii–iv).

Besides stressing observational evidence and operational definition, these philosophers tried to axiomatize physics (an aim which intrigued the young Herbert Simon: Chapter 6.iii.a). In this, they were still following Descartes's formalist dream, which had inspired modernism—in science, art, and politics—since its origin in the seventeenth century (Toulmin 1999: 152–60). Believing that other sciences could then be defined in terms of physics, they hoped to "unify" science within a single deductive system.

Some behaviourists tried to do much the same thing, expressing their theories in terms of psychological definitions and quasi-axioms. The most detailed examples were developed by Clark Hull in the late 1930s and early 1940s (C. L. Hull 1943) (see 5.iii.b). But perhaps the most ambitious appeared much earlier, when logical positivism was relatively new: Albert Weiss's 'One Set of Postulates for a Behavioristic Psychology' (1925).

Weiss listed ten "postulates", or axioms, of psychology, outlining its potential unification with physics. These ranged from the motions of electrons and protons, through behaviour (including language, as "the characteristic factor in human behavior"), to civilization. And, strange as it may seem, civilization was defined by Weiss partly by reference to "electron–proton movements" (Postulate 9). The underlying thread supposedly linking electrons to civilization was Weiss's assumption that "human behavior and social achievement are . . . forms of motion", as opposed to "psychical or mental phenomena".

As for language, this was defined as a system of "sensorimotor and contractile effects" linking stimulus and response with environmental changes.

If that were so, then sentences about propositional attitudes, using (intentional) verbs such as *know*, *believe*, *want*, and *hope*, could be analysed in purely truth-conditional (extensional) terms. But it's not clear that this is possible. The truth of *Mary believed that the bracelet had been lost*, for instance, is independent of the truth of the embedded sentence about the bracelet.

Philosophers were aware of this difficulty. So much so, that many saw intentionality as an insuperable logical–metaphysical boundary between psychology and natural science (Chisholm 1967; cf. Boden 1970). (What's more, many still do: see Chapter 16.) But the behaviourists ignored it. They either avoided psychological terms entirely or assumed—with Weiss—that they could be interpreted as forms of motion (stimulus and response).

Not all behaviourists went so far as trying to axiomatize psychology, but they did accept the positivists' general viewpoint. Indeed, they shamelessly took it for granted. Psychology textbooks before the 1970s typically opened with a chapter first exulting that psychology had at last 'escaped' from philosophy, and then uncritically offering a positivist account of science.

b. The shock of structuralism

Bloomfield was influenced by both behaviourism and positivism. He was converted to the former by Weiss (Bloomfield 1931), and invited by the philosopher Otto Neurath to contribute one of the first essays in a positivist encyclopedia of "unified science" (Bloomfield 1939). (Not that one can put too much weight on that: Thomas Kuhn's (1962) devastating attack on positivism—and Popperianism—would later appear as an annexe to the same encyclopedic review.)

At one stage, Bloomfield followed Weiss's example by formulating a set of fifty postulates for scientific linguistics and, for comparison, twenty-seven postulates for historical linguistics (Bloomfield 1926). More to the point, he referred repeatedly to "stimulus" and "reaction" in his most influential book, *Language* (Bloomfield 1933, e.g. 29–30).

This was the volume which tipped the linguistic profession—at least, in America—into the behaviourist mould. To be sure, the neo-Humboldtians Boas, Sapir, and Whorf weren't converted. They continued to study language (and meaning) in much the same way as before. But the mainstream shifted.

Bloomfield's new account was seen on its publication as "a shocking book: so far in advance of current theory and practice that many readers, even among the well-disposed, were outraged by what they thought a needless flouting of tradition" (B. Bloch 1949: 92). Nevertheless, it soon became "almost orthodoxy" (p. 92).

Part of this new orthodoxy was the *rejection* of speculation on the relation between language and mind:

> [Many linguists accompany their] statements about language with a paraphrase in terms of mental processes which the speakers are supposed to have undergone. The only evidence for these mental processes is the linguistic process; they add nothing to the discussion, but only obscure it. (Bloomfield 1933: 17)

Another part was the rejection of *meaning* as a theme for linguistics. Meaning, Bloomfield argued, can't be studied scientifically. Meanings are in practice defined (even by mentalists) "in terms of the speaker's situation and, whenever this seems to add anything, of the hearer's response" (p. 144). Rigorously defining the speaker's situation is impossible, however, for we don't know what associations the speaker has learnt (a point with which Humboldt would have agreed).

Accordingly, the Bloomfieldians focused on phonology and syntax, not semantics. They identified their theoretical categories without reference to meaning, by attending to physical measurements and statistical distribution patterns. They all tried to define their terms clearly. But many wished to replace verbal definitions by formal–symbolic ones. If that could be done, they thought, they might even identify scientific (meaning-free) methods by which phonological, morphological, and grammatical categories could be reliably—perhaps automatically—discovered.

In this journey towards formalism, they were running in parallel with empiricist philosophy of language—though with less delicate tread. During most of Bloomfield's life, the dominant movements were logical atomism and logical positivism, both of which drew heavily on Bertrand Russell's logic (Urmson 1956).

Some philosophers focused on meaning. For instance, they tried to define class concepts by lists of necessary and sufficient conditions; and they discussed the semantics of definite descriptions (noun phrases starting with *the*), and of the English words *and*, *if–then*, and *some*. (There's more to the English *and* than meets the logician's eye: according to logic, *London is the capital of England and fish have gills* is unproblematically true, but these two conjuncts would never be linked by *and* in normal conversation.)

It was this general approach which led the young McCulloch to try to formulate a logic of verbs: distinguishing past, present, and future while ignoring grammatical questions, such as where to place the past participle (4.iii.c). But it didn't affect mainstream linguistics, which had outlawed studies of *meaning*. The philosophy of language that was potentially relevant for structuralism was concerned rather with *grammar*.

Some positivists discussed logical/linguistic syntax at length (Carnap 1934; Reichenbach 1947, ch. 7). Indeed, Carnap explicitly argued that the best way to understand natural grammar is to compare it with the syntax of an artificially constructed language. But this positivist work attracted only the bravest structuralist souls. It was highly abstract (Carnap's formal notation was notoriously obscure, as remarked in Chapter 4.iii.f), and said very little about the specifics of natural languages. Not until the late 1940s was there a structuralist grammar that could compare in sophistication with logical studies of syntax.

Bloomfield himself, in his "shocking" book, offered a very simple formalism to explain the appearance of new word forms. His "proportional formulae" represented the selection of new forms by analogy with previously experienced ones: see Figure 9.1. (Bloomfield 1933: 406).

Bloomfield used these formulae to explain both correct uses of regular forms, such as the plural *cows*, and overgeneralizations of irregulars, such as the past-tense *dreamed* instead of *dreamt* (cf. Chapters 7.vi.a and 12.vi.e). He applied proportional formulae also to syntactic innovations in the history of a given language: for instance, the change in sixteenth-century English when people started to introduce subordinate clauses by the word *like* (as in *to do like Judith did*) (Bloomfield 1933: 407).

sow : *sows* = *cow* : *x*

scream : *screams* : *screaming* : *screamer* : *screamed*
= *dream* : *dreams* : *dreaming* : : *x*

Fig. 9.1. Proportional formulae for creating new word forms. Reprinted with permission from Bloomfield (1933: 406)

However, to call these expressions "formulae" was to claim too much. One can't reasonably complain that they had to be interpreted intuitively, for in the early 1930s the notion of automatic computation had yet to be defined (see 4.i.c), still less implemented. Even so, they were highly informal. Apart from the proportionality sign borrowed from mathematics, there was no symbolization here, nor even an explicit separation of the words into constituent morphemes (such as *scream* and *-ed*). This wasn't a new notation, nor a grammatical calculus, merely a mnemonic for summarizing certain morphemic patterns.

Bloomfield's gloss on Figure 9.1 showed that it wasn't a description of conscious (verbally reportable) processes, either:

> Psychologists sometimes object to this formula, on the ground that the speaker is not capable of the reasoning which the proportional pattern implies. If this objection held good, linguists would be debarred from making almost any grammatical statement . . . Our proportional formula of analogy and analogic change, like all other statements in linguistics, describes the action of the speaker and does not imply that the speaker himself could give a similar description. (Bloomfield 1933: 406)

Unconscious (physiological) mechanisms were doubtless responsible for the speaker's behaviour: Bloomfield didn't believe in magic. But the linguist's job was to describe the abstract structure of the linguistic behaviour, not to speculate on the processes by which it was achieved.

c. The formalist Dane

A more ambitious linguistic formalism was published a few years later by Jespersen. Situated as he was in Europe, Jespersen wasn't a card-carrying structuralist. Indeed, Chomsky (1996*a*) names him as a "Cartesian" precursor. But his early attempts at formalism attracted praise from Bloomfield (1933: 86), and were cited by Reichenbach in his positivist analysis of "conversational" language (Reichenbach 1947, ch. 7).

Initially, Jespersen used his notation to describe only phonemes, which he analysed in considerable detail. His "analphabetic" phonetic symbolism, a twentieth-century version of Wilkins's project of the 1660s, represented many distinct states of the vocal organs. In his *Analytic Syntax* (1937), Jespersen extended his notation to cover grammatical structure too. His work was not only ambitious, but original:

> So far as I know, this is the first complete attempt at a systematic symbolization of the chief elements of sentence-structure, though we meet here and there with partial symbolizations . . . (Jespersen 1937: 97)

He compared his notation with the formulae of chemistry and logic (p. 13). And (without digressing to explain the individual symbols) we can see that it was an analytic tool far in advance of Bloomfield's proportionalities. The phrases *burning hot soup* and *wide open windows* were both rendered as "2/321", and *curious little living creatures* was "21(21(21))"—(see p. 19). The sentence *I saw the soldiers, some of them very young indeed* was represented as "S V O [1^qp1 323]"—(see p. 47), and *I've found the key that you spoke of* as "S V O (12(3^c/1^c*S_2 V p*))"—(see p. 76). And the two superficially similar sentences *The story is too long to be read at one sitting* and *The story is too long to read at one sitting* were represented respectively as "S V P (32p1(I^b3))" and "S* V P(32p1(IO^o*3))"—(see p. 64). As for more complicated examples, the sentence *Brother Juniper, forgetting everything except the brother's wishes, hastened to the kitchen, where he seized a knife, and thence directed his steps straightway to the wood where he knew the pigs to be feeding* became this: "S(1 1 2(Y O p1($1^2$1)) Vp1 2(3^c S V O & 3 V O($S^2$1) 3p1 2(3^c S V O(S_2 I)"—(see p. 93). As Jespersen laid it out, it was as follows:

Brother Juniper, forgetting everything except the brother's wishes,
S(1 1 2(Y O p1(1^2 1))
hastened to the kitchen, where he seized a knife, and thence directed
V p 1 2(3^c S V O & 3 V
his steps straightway to the wood where he knew the pigs to be
O(S^2 1) 3 p 1 2(3^c S V O(S_2 I)
feeding. [NB Opening/closing brackets are unequal in number.]

Jespersen stated his general policy as "to follow the sentence or word combination that is to be analyzed word for word". (Even without the detailed explanations, one can see this strategy at work in the examples given above.) However, he allowed many exceptions to this rule. Specifically, "such combinations as *the man*, *a man*, *has taken*, *will take*, is *taking*, etc. (generally also *to take*), even *can take*, are reckoned as one unit" (Jespersen 1937: 15).

Some of Jespersen's remarks about language recall Humboldt, whom (as remarked in Section iv.g) he admired greatly—and, with hindsight, Chomsky too. For example:

> What is certain is that no race of mankind is without a language which in everything essential is identical in character with our own . . . (Jespersen 1922: 413)

> The complexity of human language and thought is clearly brought before one when one tries to get behind the more or less accidental linguistic forms in order to penetrate to their notional kernel. Much that we are apt to take for granted in everyday speech and consider as simple or unavoidable discloses itself on being translated into symbols as a rather involved logical process, a fact that is shown, for instance, by the number of parentheses necessary in some of the examples. (Jespersen 1937: 15)

The reference to the "notional kernel" betrays Jespersen's belief, previously argued in his *Philosophy of Grammar* (1924), that behind the syntactic categories which describe the partly "accidental" form of an actual language there are deeper categories that are independent of existing languages. He saw the linguist's task as to investigate the relations between notional and syntactic categories, and to discover in what way all natural languages are "essentially identical".

Despite these Chomskyan overtones, Jespersen's work was significantly different from Chomsky's. Most importantly, his grammatical formalism wasn't generative. It described the syntactic structure of a sentence as given, not how it could in principle be built. (The term "generative" is here intended in its timeless mathematical sense, indicating the class of structures defined by some set of derivational rules.) As a corollary, Jespersen didn't explicitly ask what makes word strings grammatical or ungrammatical, since his theory was applied only to sentences actually present in the corpus.

(A "corpus" is a representative sample of sentences, or phrases, gleaned from some natural source: books, conversations, interviews . . . In Jesperson's time it might have held only a few score items. Today, it may hold hundreds of thousands. The CHILDES corpus, for instance, is a database containing over 150 megabytes of speech-exchanges between parents and children of various ages, some using a second language and some the mother tongue; sub-corpora can be extracted which hold only utterances of parents/carers, or only utterances of 4-year-olds . . . and so on.)

Jespersen failed to consider sentence generation partly because this typically requires syntactic procedures to be performed recursively. For instance, one noun phrase can be nested inside another, and that in another . . . and so on. But the mathematics of recursion wasn't yet understood.

Even fifteen years later, when functioning machines using recursion were available, the scientific potential of recursion wasn't obvious. The philosopher Carl Hempel (1905–97) observed in 1952 that "recursive definitions which play an important role in logic and mathematics . . . are not used in empirical science" (Hempel 1952: 11 n. 11). As Chomsky put it in the 1960s:

> The fundamental reason for the inadequacy of traditional grammars is a . . . technical one. Although it was well understood [by traditional grammarians] that linguistic processes are in some sense "creative", the technical devices for expressing a system of recursive processes were simply not available until much more recently. In fact, a real understanding of how a language can (in Humboldt's words) "make infinite use of finite means" has developed only within the last thirty years, in the course of studies on the foundations of mathematics. (Chomsky 1965: 8)

d. Tutor to Chomsky

The first linguist to take recursion seriously was Zellig Harris (1909–92) at the University of Pennsylvania—where the ENIAC was built in the mid-1940s (3.v.b). Harris taught Chomsky as an undergraduate and as a graduate student, and they kept in close contact after the younger man moved to Massachusetts.

Chomsky later acknowledged Harris as his main inspiration, and a frequent discussant of his own work (Chomsky 1975: 4). Harris was no more interested than Bloomfield in the humanist/nativist ('language-and-mind') questions listed at the close of Section iv.g. But if anyone was a precursor of Chomskyan grammatical analysis, it was he.

In 1951 Harris published a book widely received as the most important contribution to descriptive linguistics since Bloomfield's *Language*. His *Methods in Structural Linguistics* was shown to Chomsky in proof, on joining Harris's seminar in 1947. It contained a detailed discussion of how to use distribution patterns to identify phonemes, morphemes, morpheme sequences (phrases), and other linguistic classes. In addition, it offered a formalism for representing such classes. This formalism—which Chomsky adopted and modified—was used by Harris to state a finite set of grammatical rules that could describe an infinite set of sentences.

When electronic computers appeared on the horizon, Harris hoped also to define a formal method which, if expressed as a computer program, would automatically identify the grammatical classes and patterns within a set of sentences drawn from an unknown language with an alien grammar (Z. S. Harris 1951, 1968). In this, a computer-age version of a long-standing structuralist goal, he didn't succeed.

That should have been no surprise, for the task is in principle impossible. It assumes that scientific categories and hypotheses can be derived from the data by theory-free induction—a view that was already being questioned by Popperian philosophers (K. R. Popper 1935; Goodman 1951). One can't even identify phonemes and morphemes without some guiding grammatical theory (Chomsky 1962: 125). However, Harris's system was "the most ambitious and the most rigorous attempt that has yet been made to establish what Chomsky was later to describe as a set of 'discovery procedures' for grammatical description" (Lyons 1991: 34).

It turned out, many years later, that the discovery procedures needed may be less specifically linguistic than some Popperians might expect. Recent work in the machine learning of language shows that relatively "light" theoretical assumptions can go a long way, if combined with powerful statistical techniques (see Section xi.a, below). These assumptions aren't language-specific in the sense that interested Chomsky, for they aren't grammatical.

Nevertheless, the general Popperian point still stands. In the 'pure' statistical analysis of the linguistic data, words are given for free by the spaces between the input letter strings. Moreover, phonetic categories—in the form of the alphabet—are taken for granted. From the point of view of psycholinguistics, the latter is entirely defensible. Although nativism with respect to syntax is highly questionable (see Section vii.c–d), nativism with respect to phonetics isn't: there's a finite set of human speech sounds, and newborn babies pay more attention to them than they do to sounds in general. Refutation of the Popperian claim would require successful induction over purely acoustic features—and even then, the acoustically structured anatomy of the cochlea could be seen as providing an implicit theory. (Although whether, and if so in what sense, acoustic feature detectors are really "innate" is debatable: cf. Chapter 14.vi.b and x.a–b.)

Harris introduced Chomsky not only to the project of rigorous formalism, but also to the notion of transformations (Z. S. Harris 1952, 1957, 1968). This idea wasn't mentioned in *Methods*, but was already prominent in Harris's mind. Indeed, by 1947 (when the book was completed) he was discussing it in his seminars, and in conversations with other linguists (Bar-Hillel 1970: 292; Z. S. Harris 1952: 18–25). The young Chomsky was a research assistant for some of the necessary analyses (Z. S. Harris 1952, n. 1). By the time of his own first publications, Harris's work on transformations was well known.

"Discourse analysis" was, in effect, a scientific study of the phenomenon of literary genre that had so fascinated Diderot. It showed how to give a structuralist description of entire connected texts, including spoken conversation. Its first public presentation (in 1949) was to an audience of anthropologists, for Harris's motivation in defining transformations was to apply linguistics to culture and personality. One might almost say that Humboldt's assimilation of culture and linguistics was being revived—but from a very different philosophical base, and with a very different methodology.

Instead of asking a priori what type of language should be used for a certain type of task, as Diderot had done, Harris asked what sentence structures actually occur within discourses devoted to that task. He answered this question in terms of inter-sentence distribution patterns, defined in terms of grammatical "transformations" specifying relations between entire sentences or morpheme sequences. Two syntactically complex sentences might be transforms of one, simpler, kernel sentence. And he found that "often it is possible to show consistent differences of structure between the discourses of different persons or in different styles or about different subject matters" (Z. S. Harris 1952: 1).

Rhetoricians and literary critics, Diderot included, had known for centuries that some such differences exist. And they had identified various genres and authorial signatures accordingly. But discourse analysis could describe these structural differences more precisely, as well as discovering many that hadn't been noticed before.

Harris's Presidential Address to the Linguistic Society of America in 1955 summarized his approach. Two sentence structures, related (for instance) as *active/passive*, or even as *question/answer*, were classed as transforms of each other on the basis of their observed distribution patterns. Specifically, they must share the same set of individual co-occurrences of morphemes. For example, *The kids broke the window* and *The window was broken by the kids* share the same verb and two nouns, although in reverse order. (The sentence *The young people destroyed the pane of glass* wouldn't count as a transform: Harris approached meaning only indirectly, through word distributions.) Problems arise with the morpheme pair -*ed/will*, since the first co-occurs with *yesterday* whereas the second does not (*The cliff crumbled yesterday*, *The cliff will crumble tomorrow*). Accordingly, just what is to count as the relevant "sentence environment" is not a straightforward matter. Harris (1957) gave a detailed discussion of how this problem can be addressed. (I can't resist remarking that the cliff actually crumbled on 11 January 1999, a few weeks after this passage was written, when Beachy Head—a famous landmark, and suicide point, near my home—sent tens of thousands of tons of chalk crashing into the English Channel.)

e. Not quite there yet . . .

If Harris's work was a crucial inspiration for Chomsky, it was nonetheless different from his in two—maybe three—important ways.

First, Harris defined his transformations by reference to distributional (statistical), and purely surface, features of sentences, as we've seen. Chomsky, by contrast, defined transformational grammar in terms of deep and surface structures. Or rather, he did so in his publications in the 1950s and 1960s, which are the crucial ones for our purposes. (More recently, he has dropped the deep/surface distinction: see Section viii.b.)

Another often remarked difference is that Chomsky's theory was generative whereas Harris's wasn't. Chomsky certainly saw his grammar as being original in this way, and Harris apparently agreed: "Noam Chomsky has combined transformational analysis [which features also in my own system] with a generative theory of sentence structure [which does not]" (Harris 1968: 4).

This is puzzling, however. It's true that Harris's aim was to describe sentences, not to show how they could be generated. But, as noted above, his formal theory potentially covered an infinite set of sentences. In that sense, it was a generative system—even though Harris didn't describe his rules as "derivational", nor use them to discuss sentence generation (cf. Hymes and Faught 1981: 166).

The third difference is uncontestable, and the most significant of all. As Harris put it: "Chomsky [unlike myself] has produced a mathematical specification of context-free languagelike systems within a spectrum of languagelike systems" (Z. S. Harris 1968: 4). In other words, Harris didn't locate his formalism within *the class of all grammars*.

That idea was developed by Chomsky. In so doing, he brought theoretical linguistics into the ambit of information theory, computer science, AI, and computational psychology.

9.vi. Major Transformations

The major transformation in twentieth-century linguistics dates back to 1957, with the publication of Chomsky's *Syntactic Structures*. It wasn't a high-profile event. On the contrary, the volume appeared in an obscure specialist series, from a small publishing house in Holland. Chomsky himself said later that they accepted it only because it was recommended by Jakobson, and that it wouldn't have been noticed but for a long, and near-simultaneous, review in a leading linguistics journal by his student Robert Lees (1957).

Lees didn't merely recommend the book, he championed it. Chomsky's "first established convert", he "went around to linguistics conventions and got the ball rolling" (Weimer 1986: 300). As Howard Gardner put it, he was "playing Huxley to Chomsky's Darwin" (H. Gardner 1985: 189). This was to become an all-too-common role for Chomsky's students, as we'll see (Section viii.a).

Nevertheless, Lees and Jakobson weren't the only ones to be impressed. This little book, of only 118 pages (and by an author only 29 years old), revolutionized the subject.

a. Chomsky's first words

If linguists had had their eyes open, the revolution might have happened a few years earlier. For Chomsky's first book wasn't his first publication. A handful of technical papers had already appeared, most in journals not read by linguists, such as the *Journal of Symbolic Logic*.

One of these was Chomsky's paper on 'Three Models for the Description of Language' (1956). This was a hierarchical classification of grammars, defined in purely abstract (non-linguistic) terms. With hindsight, it's clear that this paper, not *Syntactic Structures*, was Chomsky's most original and most well-founded contribution—even if many of Chomsky's fans have never read it (cf. also Chomsky and Miller 1958; Chomsky 1959*a*).

'Three Models' did what Harris admitted he'd never thought of doing: it located the grammar of natural language within the class of all possible grammars. At the top of Chomsky's formal hierarchy were the "recursively enumerable" languages: namely, all languages having some finite definition. A subset of those were the "context-sensitive" grammars, which in turn contained the "context-free" grammars (see below). At the lowest level were the "regular" grammars, definable in very simple terms. (Twenty years later, other linguists would insert "indexed grammars" between the context-sensitive and context-free levels: see Section ix.d.)

Computer scientists were interested immediately. Similar research was already being done by some of them, but Chomsky's was a highly valued contribution.

His distinction between context-free and regular grammars (Chomsky 1956) helped them to codify, respectively, the lexical structure of artificial languages and the high-level structure of programs. Moreover, he proved in 1962 that different types of computational device—again, abstractly defined—would be needed to deal with the various classes of grammar. A universal Turing machine (see Chapter 4.i.c) can compute *any* definable language. Push-down stacks (see Chapter 10.v.b) can compute context-free languages (an insight that helped establish the theoretical basis of modern parsers, and of compiler design). And, since regular grammars are lower down in the hierarchy, they can compute those too. But push-down stacks aren't necessary for dealing with regular grammars. Such grammars can be computed by finite-state machines (described below).

The linguists, by contrast, didn't notice these early papers. For they weren't published in 'their' journals. (Chomsky's initial connection with MIT was the Research Laboratory of Electronics, from which the Linguistics Department later emerged: see 10.ii.a.) In other words, the scientific cooperation dreamt of by Descartes (2.ii.b–c) was—temporarily—prevented by disciplinary specialism.

Nor did they read the mimeograph/microfilm versions of the typescript of Chomsky's voluminous background research, which had been available to the cognoscenti since 1955. Even the cognoscenti had found it hard going, and very few (perhaps none?) devoured it from first page to last. This wasn't mere laziness on their part: the typescript was neither easily legible nor readily intelligible. Many long passages consisted of sequences of definitions expressed in algebraic symbolism. Indeed, in 1956 MIT Press had refused to publish it in its original form. (It was eventually published twenty years later: Section ix.d.)

In short, most professional linguists, if they'd encountered this work at all, weren't interested. They saw Chomsky's line of enquiry as "unpromising and exotic" (Chomsky 1975: 2). They weren't unaccustomed to formalism, as we saw in Section v.c–e. But whereas Harris (and Jespersen) had employed formalism in the service of meticulous empirical description, Chomsky's early research focused on formalism *as such*.

Most psychologists weren't interested either. Some mathematical psychologists realized the crucial implication of the 'Three Models' theme: namely, that human minds must have a level of computational power capable of dealing with whatever class of grammar is appropriate to natural languages. But, working within the information-theoretic paradigm, they assumed that regular grammars would suffice.

It was *Syntactic Structures*, described by Chomsky as a "sketchy and informal outline" of this background material, which set the linguistic world afire. Its impact for linguistics was immediately likened—in the review that Chomsky mentioned (above)—to the transition from alchemy to chemistry (Lees 1957: 375–6). And as we'll see, it had a revolutionary effect outside linguistics too.

b. The need for a generative grammar

The research summarized in *Syntactic Structures* was inspired by Harris's discourse analysis. It also reflected Chomsky's interdisciplinary education: at Harris's suggestion, he'd studied mathematics and philosophy as well as linguistics. Formalism had flourished in the philosophy of science for some years, as we've seen. But Chomsky went beyond mere formalism. His approach was fundamentally mathematical in a sense in which Harris's wasn't.

Chomsky asked two new questions about natural-language syntax. On the one hand, he sought to state a generative grammar—as opposed to a merely descriptive one—for a given language (such as English). On the other hand, he aimed to compare the computational power of various types of grammar.

Readers of *Syntactic Structures* weren't burdened with the bristling technicalities of his 1956 paper, but those ideas underlay the argument. Although he didn't locate his new grammar at a particular point within the grammar hierarchy, Chomsky's focus on differences in computational power largely explains the book's wide influence: across linguistics, computer science, psychology, and the philosophy of language.

The brief text of *Syntactic Structures* took a fresh, not to say shocking, approach. It denied two fundamental structuralist (Bloomfieldian) assumptions: that the goal of linguistics is to define a discovery procedure for grammars, and that language can be explained in probabilistic, behaviourist, terms. Moreover, it started out not from a rich description of actual usage but from dry definitions mentioning not a word of English, Sanskrit, or Kawi.

In the opening pages, Chomsky defined language, and the goal of linguistics, in abstract terms:

> I will consider a *language* to be a set (finite or infinite) of sentences, each finite in length and constructed out of a finite set of elements. All natural languages are languages in this sense . . . Similarly, the set of "sentences" of some formalized system of mathematics can be considered a language. The fundamental aim in the linguistic analysis of a language L is to separate the *grammatical* sequences which are sentences of L from the *ungrammatical* sequences which are not sentences of L and to study the structure of the grammatical sequences. The grammar of L will thus be a device that generates all of the grammatical sequences of L and none of the ungrammatical ones. (Chomsky 1957: 13)

A generative grammar is a set of (timeless) derivational rules. It doesn't tell us just how to build or parse a specific utterance. In other words, it isn't a computer program. However, it is a computational notion. The "device" mentioned here—and the "machine" mentioned elsewhere (e.g. p. 52)—was a purely mathematical one, as is a Turing machine (see 4.i.c–d).

But if Chomsky wasn't talking about computers he was, indirectly, talking about people:

> Any grammar of a language will *project* the finite and somewhat accidental corpus of observed utterances to a set (presumably infinite) of grammatical utterances. In this respect, a grammar mirrors the behavior of a speaker who, on the basis of a finite and accidental experience with language, can produce or understand an indefinite number of new sentences. Indeed, any explication of the notion "grammatical in L" . . . [offers] an explanation for this fundamental aspect of linguistic behavior. (p. 15)

Even so, his grammar wasn't a set of instructions for psychological processing. Rather, it represented the underlying grammatical competence of the speaker (see Chapter 7.iii.a).

In offering his new vision of what a theoretical linguistics should be like, Chomsky asked four interrelated questions:

* How can we discover a grammar for a given set of sentences?
* How can we know that a grammar generates all and only these sentences?
* How can we evaluate alternative grammars?
* And how can we express the differences between grammars, of either artificial or natural languages?

His answer to the first question wasn't what structuralists expected. On his view, no scientific ("practical and mechanical") discovery procedure can be found (1957, ch. 6). Even Harris, whose work was more careful than most, hadn't succeeded in defining one. The best we can hope for (and what Harris had actually provided), said Chomsky, is an evaluation procedure: a method for deciding, given a finite set of sentences, which of two candidate grammars is preferable. Even this is difficult to define formally, for grammars—like all scientific theories—are evaluated largely in terms of their simplicity, a notoriously elusive notion.

A candidate grammar for a natural language must generate all and only the grammatical sentences. But how do we know which these are? No search of an existing corpus can identify them. Statistics are inappropriate, since most sentences in any reasonable corpus will be one-offs: uttered on only one occasion. And if a sentence isn't there at all, so what? Any native speaker can, and frequently does, come up with new ones.

This is a matter not merely of observation, but of syntax. (At this stage, it was syntax, not semantics—still less, reason or imagination—which was given the credit for the creativity of language: see Section iv.f, above.) One can produce new sentences, for instance, by conjoining two or more old ones, or by recursively nesting one inside another. (Whether the result will always be intelligible was addressed in Chapter 7.ii.b.)

It follows, said Chomsky, that judgements of grammaticality must be left to the intuitions of native speakers. For they can reliably say whether a given word sequence is grammatical even if they can't explain their judgement.

Data-respecting critics soon complained—and still do—that by *native speakers* he meant *a single native speaker whose intuitions one trusts*—in Chomsky's case, Chomsky. Such a source hardly seems reliable: Chomsky himself changed his mind twice about his famous example "Colourless green ideas sleep furiously". And how were these armchair intuitions to be monitored? The structuralists, too, had used native informants to assess grammaticality, but had insisted that the conditions in which evidence is elicited from them be carefully controlled, to prevent experimental bias of various kinds (Z. S. Harris 1957, sect. 4). Disagreement arose, in psychology as well as linguistics, over the role of extensive sampling as against the consideration of individual cases: see Chapter 7.iii.d. Chomsky himself, however, remained content to rely on his natural, untutored, intuitions.

Assuming that we've agreed on a set of grammatical English sentences, we must ask "what sort of device can produce this set". This requires us, said Chomsky, to compare the power of different conceivable grammars. He addressed this task by starting from Claude Shannon's information theory (4.v.d).

c. Beyond information theory

Shannon had defined a number of artificial languages in the 1940s, suggesting that:

> These artificial languages are useful in constructing simple problems and examples to illustrate various possibilities. We can also approximate to a natural language by means of a series of simple artificial languages. (Shannon and Weaver 1949: 13)

Chomsky agreed that artificial languages are useful comparison points for natural language. But, with his grammar hierarchy lurking in the background, he saw the prospects for Shannon's serial "approximations" as hopeless.

The simplest example Chomsky considered, lying at the lowest ("regular") hierarchical level, was a *finite-state grammar* (*language*, *machine*), or FSG. This is a grammar—of which there are infinitely many—that describes a device with a finite number of states, including an initial state and a final state, and a finite set of rules for passing from one state (considered in isolation) to the next (cf. Shannon and Weaver 1949: 15–16). In a linguistic context, "passing from one state to the next" would involve either emitting or accepting a word, depending on whether the FSG was being used to produce word strings or to parse them, respectively.

Taking up Shannon's discussion, Chomsky pointed out that some FSGs could produce only a finite number of "sentences"—in some cases, only two (see the first state-diagram in Figure 9.2) (Chomsky 1957: 19). Clearly, such examples couldn't model the apparently infinite potential of natural languages. But a more powerful type of FSG could be defined by adding one or more closed loops, so that instead of passing directly from state n to state $n + 1$, the device may—optionally—return to state n again, and again . . . before finally leaving it. As a linguistic analogy, the sentence *The man comes* can be extended to *The old man comes*, *The nasty old man comes*, etc. by inserting one or more adjectives at the relevant point. This type of FSG is recursive, so can produce infinitely many sentences (compare using five adjectives, or fifty, or . . .).

Because of the potentially infinite number of sentences of English (French, Kawi . . .), Chomsky had the extended type of FSG primarily in mind (cf. Chomsky and Miller 1958). Of the two FSGs shown in Figure 9.2, then, he was less interested in the first than in the second, which contains a closed loop. Indeed, the simpler type is now sometimes forgotten, or anyway ignored (as an uninteresting special case). So FSGs *as such* are sometimes said to be capable of producing infinitely many word strings, or sentences.

Shannon had pointed out that a specific probability could be assigned to each possible state transition allowed by an FSG, and therefore that different artificial languages could be defined by assigning distinct sets of probabilities. Such probabilistic FSGs are called "Markovian". In natural languages, too, the successor probabilities for both letters and words vary. In English, though not in Polish, the state *z* is highly improbable as a successor to *c*. Similarly, the state *man* is fairly probable, given the state *old*, whereas the state *because* is highly improbable as a successor to *the*. If realistic probability measures were added for each state transition, Shannon said, something resembling natural language would result.

To illustrate this point, Shannon suggested a number of simple rules to produce various Markovian letter strings and word strings. These strings approximated English more and more closely as the number of previous letters or words considered (zero,

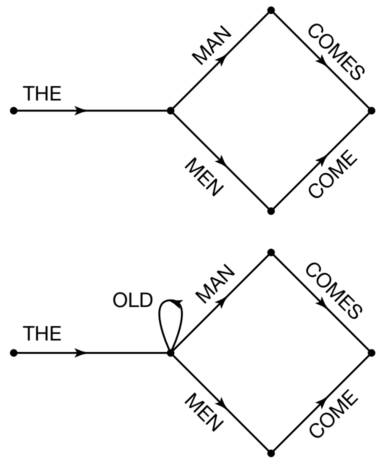

FIG. 9.2. State-diagrams for two FSGs. The first can produce only two sentences, whereas the second—because of the loop—can produce infinitely many. Redrawn with permission from Chomsky (1957: 19)

one, or two) increased. For instance, choosing words independently of each other (but guided by their individual frequencies) resulted in this:

REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NATURAL HERE HE THE A IN CAME THE TO IF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE. (Shannon and Weaver 1949: 14)

Not all of the two-word sequences included here would actually occur in English, and only one of the four-word sequences (AND SPEEDILY IS AN). By contrast, allowing the choice of each word to depend upon its immediate predecessor produced this:

THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED. (p. 14)

This passage contains many sequences of four or more words that could occur in real sentences. (Anyone but an information theorist would say ". . . that could make sense".) Indeed, the twelve-word sequence from FRONTAL to POINT could occur in a guidebook on country walks, in a sentence such as "It is because of the savage frontal attack on an English writer that the character of this point on the pathway has changed."

The method Shannon used to compose the latter word string was a simple one. Taking a book from his bookshelves, he opened it at random, and wrote down a word selected randomly from the open page. Next, he opened the book at another randomly chosen page, and read on until he encountered the previously recorded word. At that point, he wrote down the immediately following word. Then he repeated the process again and again, always using the most recently recorded word as his guide.

Clearly, he didn't pick *Alice's Adventures in Wonderland.* Indeed, he may have deliberately avoided picking a 'literary' text as his sample source. The reason is that this method would grind to a halt if any newly chosen word occurred only once in the entire book.

(i) *Sentence* $\rightarrow$ *NP* + *VP*
(ii) *NP* $\rightarrow$ *T* + *N*
(iii) *VP* $\rightarrow$ *Verb* + *NP*
(iv) *T* $\rightarrow$ *the*
(v) *N* $\rightarrow$ *man*, *ball*, etc.
(vi) *Verb* $\rightarrow$ *hit*, *took*, etc.

FIG. 9.3. Rewrite rules for a simple phrase structure grammar. Reprinted with permission from Chomsky (1957: 26)

Even this simple method was time-consuming. It would be interesting to go further, said Shannon, by considering more than one previous word. But the labour involved in hand-searching the sample text would become enormous even at the two-word stage. (George Miller, instead of searching texts, asked people to volunteer 'the next word' after having been given one, two, or . . . six words: the higher-order, more redundant, strings were remembered more easily: G. A. Miller and Selfridge 1950.)

If computers had been available at mid-century, to collect the necessary statistics, many structuralists would have been interested. Think of the textual studies that might have been done on the basis of Harris's discourse analysis, for instance. Today, such studies are possible. But in the 1950s, computing technology was still in its infancy. In practice, then, Shannon's approximations weren't followed up by those linguists whose theoretical sympathies were similar to his.

Chomsky, in any case, wasn't one of their number. He saw the statistical approach as a dead end in theory, not just in practice. He believed he could prove, as a matter of principle, that FSGs, although they can generate infinitely many new sentences, cannot produce all the examples of everyday English. Specifically, they cannot generate sentences involving nested expressions, where the selection (in Shannon's terms, the probability) of word $n + 1$ depends not on its immediate predecessor, word n (nor on the m immediate predecessors), but on some word indefinitely many places earlier. Consider this sentence, for instance: *The cat with black fur and long whiskers was sitting on the mat*. The ninth word here depends not on the eighth, but on the second: replace *cat* by *cats*, and you must change *was* to *were*. It's structure which is doing the work here, not statistics.

Such nested dependencies, Chomsky argued, can in principle occur on infinitely many levels. (We saw in Chapter 7.ii.b, however, that *in practice* they cannot.) They require a "phrase-structure grammar", in which sentences are derived by a series of ordered "rewrite rules". Rewrite rules *as such* were invented not by Chomsky, but by the logician Emil Post (1943: see 10.v.e). But Chomsky was the first to apply them to natural language. He used them to produce phrases, including terminal elements (words), on various hierarchical levels (see Figure 9.3).

Unlike Jespersen twenty years earlier, Chomsky aimed to account for every single word in the sentence. So he analysed nominal phrases like *the man* and *a man* into two parts—and when he came to discuss transformations (see below), he analysed auxiliary phrases such as *has taken*, *will take*, and *is taking*, and showed the underlying relations between them.

The complete grammatical structure of the sentence, though not the order in which the rewrite rules were applied, can thus be represented by a tree diagram in which every

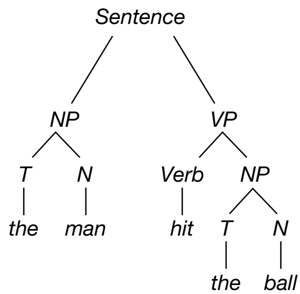

Fig. 9.4. Phrase-structure diagram showing the derivation of "The man hit the ball". Redrawn with permission from Chomsky (1957: 27)

word is separately considered (see Figure 9.4). This diagram shows how the grammatical categories are linked: from top to bottom, not—as in Markovian grammars—from left to right.

d. Transformational grammars

Phrase-structure grammars weren't new. They implicitly underlay immediate-constituent analysis, in which a sentence is recursively broken into contiguous constituents. This method had been used by many people, from the Port-Royal logicians to Harris, not only to parse unproblematic sentences but also to explain ambiguities. Linguists had long known, for instance, that the phrase structure of *The old men and women* could be either *The (old men) and women* or *The old (men and women)*, and that the meanings differed accordingly.

Similarly, the intuitive notion that different grammars might have different computational power wasn't new. Shannon, after all, had already contrasted FSGs having finite or infinite string sets.

The novelty in Chomsky's argument—as considered so far—was to define phrase-structure grammars as a general class, to prove that they're more powerful than Markovian (probabilistic) approaches, and to show that *natural* languages require grammars whose computational power is at least as great as that of a phrase-structure grammar. Significantly, he did the latter by relying, not, as Shannon had done, on some intuitive notion of computational power, but on his new classification of grammars. Each level in the hierarchy could generate all the strings generated by the level below, and more. FSGs lay at the lowest level, and phrase-structure grammars lay above them.

Chomsky was speaking here about *context-free* phrase-structure grammars. In his previous mathematical analysis, he'd distinguished between context-free (CF) and context-sensitive (CS) phrase-structure grammars.

In a CF grammar, each rewrite rule can be applied whenever the one-and-only symbol on the left-hand side is present. In a CS grammar, more than one symbol occurs on the left-hand side of the rewrite rules; a rule may rewrite one of them, and carry the others over unchanged. In the hierarchy of computational power, CS grammars had been located much higher than CF grammars. It was already clear that CS grammars included many "languages" of interest only to mathematicians, not to linguists. For linguists, the more interesting, more restrictive, and more plausible hypothesis was that natural languages have CF grammars. Indeed, the term "phrase-structure grammar" is often used to mean CF grammar.

At this point in the argument, Chomsky went further: he claimed that grammars of even greater computational power than phrase-structure grammars are needed for natural language. (My use of the word "claimed" here, instead of "proved", will be justified in Section ix.d–e.) In other words, natural languages involve a grammar located somewhere between phrase-structure grammars and universal Turing machines.

Chomsky admitted: "I do not know whether or not English is itself literally outside the range of [context-free phrase-structure] analysis." But he claimed that it could be so described (if at all) "only clumsily", by a complex, ad hoc, and unrevealing theory (p. 34). The reason he gave was that certain simple, and intuitively obvious, ways of describing sentences can't be expressed in terms of phrase structures, as then understood.

Everyday examples that require something more than a phrase-structure grammar, said Chomsky, include sentences involving auxiliary verbs (*The man has read the book*, *The man has been reading the book*); passive sentences (*The book is read by the man*); interrogatives (*Does the man read the book?*, *Will the man have been reading the book?*); *and* wh-questions (*Why/where/when/how does the man read the book? Who reads the book?*). It's intuitively obvious that there are close grammatical relations between the sentence *The man reads the book* and all these other sentences. But, Chomsky claimed, these relations can be simply expressed only by rules of the general form: *If the derivation of a grammatical sentence S-1 is such-and-such, then another grammatical sentence, S-2, can be produced by altering S-1 thus-and-so*. (*Sentence*, here, is a technical term, that covers any legal structure within a grammatical derivation. So *The man VP* is a sentence in this sense, for it is a structure derivable on the way to the terminal structure, *The man reads the book*.)

Intuitively, all these sentences seem to be related to the last one: *The man reads the book*. What's therefore required, said Chomsky, is a set of rules that can start with this sentence and derive all the others. Because rules like this define ways of transforming one sentence, or tree structure, into another (S-1 into S-2), grammars that contain them are called transformational grammars, or TGs.

A TG consists of a "base component"—a phrase-structure grammar that generates the initial trees—plus one or more transformational rules. This general definition covers many examples, including Harris's work and perhaps even Port-Royal grammar (see Sections iii.c and v.d–e). But the term TG is widely used to refer only to Chomsky's grammar, or variations of it—a fact that has aroused angry criticism, as we shall see (in Section ix.d).

TGs in general can't be located at any specific point in the grammar hierarchy, because the base component needn't be context-free (though it usually is), and because it's not clear how much—if any—computational power is added to it by the various transformations. The latter point makes it difficult to place even a particular TG in the hierarchy. The assumption implicit in *Syntactic Structures* was that a TG must lie somewhere between Turing machines and CS grammars—but just where was unclear.

In Chomsky's 1957 terminology, the simplest of the English sentences given above, and the point of reference for all of the others, is a "kernel" sentence. In other words, *The man reads the book* is derivable from its underlying grammatical form (which is derived by phrase-structure rewrite rules) by using only obligatory transformations (p. 45). These are the transformations that must be applied if the terminal sentence is to be grammatical at all. Non-obligatory transformations are applied either to the forms underlying kernel sentences, or to previous transforms thereof. In deciding which transformations are obligatory in English, and which optional, Chomsky relied on grammatical intuitions such as the one just remarked.

Chomsky's general claim, here, was that sentences of natural language must be represented on more than one theoretical level. In broad terms at least, many linguists were quick to agree. One later recalled Chomsky's argument as having had a "dramatic input" at the time:

> By God, native speakers DID associate active and passive counterparts . . . Yet the two were different and even unrelated in a phrase-structure analysis. His ideas seemed salient, dramatically salient, from the outset. (Hymes 1972)

As a result, "within a few years, quite a number of linguists were working on transformational grammar" (Chomsky 1975: 4).

Within a few years, also, the philosopher Hilary Putnam noticed that transformational grammars as defined above have the same power as a Turing machine (Putnam 1961: 101 ff.). If we allow just anything to count as S-1 and S-2, then this class of grammars has potentially infinite power. Any conceivable language, whether natural or artificial, recursive or non-recursive, could be derived by some TG (if it's computable at all). The set of prime numbers, for example, can be TG-generated.

This point was awkward for linguists claiming that TGs underlie natural languages. A scientific theory should not only explain what *does* happen, but also restrict what *could* happen (see 7.iii.d).

A grammar, for instance, should generate all *and only* the acceptable sentences of the language concerned. If any conceivable language, even including the (highly un-natural) set of prime numbers, can be 'explained' by some TG, then the class of such grammars can't exclude any possibilities. To be sure, a given TG might explain English and exclude French. (To someone seeking a TG to fit *all* natural languages, as Chomsky later did, this would be a weakness rather than a strength: see Section vii.d.) But Chomsky's supposedly surprising claim in 1957 was that each natural language requires *some TG or other*, and that *some TG or other* will suffice to explain it. Putnam's work suggested that this amounted merely to saying that natural languages are Turing-computable. That had been assumed for years by logicist philosophers (see Chapter 4.iii.c), so was hardly exciting.

The Turing-equivalence of transformational grammar was to be discussed in detail in the 1970s (Peters and Ritchie 1973). Partly because of this, by 1980 some linguists had rejected transformations altogether. They offered more restrictive scientific theories of natural language, relying on grammars low down in Chomsky's hierarchy.—But that was for the future (see Section ix.d–f, below).

Meanwhile, Putnam's theoretical point was shunted into the background for over twenty years, as linguists tried to discover *which transformations in particular*, and in which order, are needed to represent a given natural language—usually, English. The specific transformations that Chomsky hypothesized for English included, for instance, rules for converting a kernel sentence into a question either by moving the auxiliary verb to the beginning (so *The man will read* becomes *Will the man read?*), or by inserting the appropriate form of the auxiliary *do* at the beginning, and altering the form of the main verb (so *The man reads* becomes *Does the man read?*).

The details don't concern us (pp. 61–84). Nor need we discuss the rather different account of transformations that Chomsky was to give in his third book, in which kernel sentences, with their compulsory/optional transformations, were nowhere to be seen (Chomsky 1965). (Transformations as such became rarer in Chomsky's later work, as many functions previously effected by transformations were transferred to other rules of the grammar: see Section viii.b.)

More important, for our purposes, is the general form of the argument in *Syntactic Structures*. For this had implications far beyond linguistics. It alerted philosophers of language, psycholinguists, and the early computational psychologists. (One might say that it didn't alert psycholinguists so much as *create* them; the countless studies of "verbal behaviour" were focused on learning, not language—word strings, not sentences: see Chapter 6.i.e.) But it *didn't*, yet, alert psychologists in general.

e. So what?

Chomsky's analysis had radical—and to many, highly unwelcome—implications. For it followed, as he took pains to point out, that utterances (i.e. "sentences" in the everyday, non-technical sense) must be theoretically represented on several levels.

On the one hand, there is the surface structure: namely, the grammatical form of the sequence of words (morphemes, phonemes) that constitutes the sentence. The surface structure is interpreted phonetically, to produce a spoken utterance. On the other hand, there is the deep structure: the derivational history (including rewrite rules, kernel forms, and transformations), which will usually have many levels. The deep structure is interpreted semantically, to give the sentence's meaning.

(Although the technical terms "deep/surface" were introduced in Chomsky's second book, a related distinction was clearly made in the first. Similarly, the terms "competence/performance" didn't feature there, but the relevant distinction did.)

From the point of view of Bloomfieldian linguists (and behaviourist psychologists, too), this added insult to injury. Having already been told that statistical theories can't describe language, they were now being told that the observable aspects of language are underlain by a complex unobservable structure that must be represented in any adequate theory. Mentalism had returned, with a vengeance.—Mentalism, but not introspectionism: Chomsky agreed with Bloomfield that the speaker has no direct access to inner mechanisms.

One might argue that grammar is a mathematical exercise having nothing to do with mentalism. Indeed, theoretical computer scientists, who were much influenced by Chomsky's analysis of the computational power of different types of "language", gave no thought to psychology. And Chomsky himself said nothing, here, about the broader mentalistic implications that were to be so controversial in the 1960s (see Section vii.c–d).

In other words, *Syntactic Structures* contained no explicit comment on behaviourism, although Chomsky's anti-Markovian arguments were clearly relevant to it. Moreover, there was no mention of universal grammar, and therefore no hint that linguistics has nativist psychological implications, nor any relevance for the philosophy of mind. The theory of generative grammar wasn't yet being explicitly presented as "a particular sub-branch of cognitive psychology", still less as a study of "the human language faculty as such" (Chomsky 1975: 9).

The reason these "Cartesian" hypotheses weren't mentioned wasn't that Chomsky hadn't yet thought of them, but that he was deliberately suppressing them. They would have clashed with structuralist assumptions about the diversity of language and the emptiness of talk about language-and-mind—and the book was radical enough, even without them. Indeed, its ideas were so unconventional that, as he later recalled: "The one article I had submitted on this material to a linguistic journal had been rejected, virtually by return mail" (Chomsky 1975: 3). Understandably, then, he felt it to be "too audacious to raise the psychological issues which were in the background of my mind" (p. 35).

However, psychology wasn't ignored entirely. Linguistics, he said, is relevant to psycholinguistics—specifically, to the study of meaning. He was concerned not with vague claims like the Sapir–Whorf hypothesis, but with clearly specified theories about the psychological abilities (though not the processing details) involved in understanding:

> [My] formal study of the structure of language . . . may be expected to provide insight into the actual use of language, i.e. into the process of understanding sentences. (p. 103)

> [The] notion of "understanding a sentence" must be partially analyzed in grammatical terms. To understand a sentence it is necessary (though not of course sufficient) to reconstruct its representation on each level, including the transformational level where the kernel sentences underlying a given sentence can be thought of, in a sense, as the "elementary content elements" out of which this sentence is constructed. (p. 107–8)

The last remark occurred in the context of a discussion of the prospects for semantics. Here, the book was rather more acceptable to Chomsky's Bloomfieldian predecessors. For meaning was sidelined. Any theory of syntax, Chomsky argued, must be independent of semantics. Indeed, semantics is secondary to syntax, for understanding the meaning of a sentence rests on grammatical analysis.

Even so, structuralists were being challenged. Although Chomsky agreed with them that a science of meaning can't be based on vague appeals to introspection, or on unknown past associations, he did say that a (syntax-based) science of meaning is in principle possible. Indeed, semantics would play a fundamental role in his next book, in which he described a grammar as a set of rules relating the meaning of a sentence to its phonetic structure (Chomsky 1965).

These passages alerted psychologists, some of whom were aghast and others inspired. The repercussions in psycholinguistics, and the relevance of Chomsky's distinction between competence and performance, were discussed in Chapter 7.ii–iii.

In Chapters 5 and 6, we considered the growing revolt against behaviourism—to which Chomsky was an enthusiastic latecomer. Now, let's look more closely at what he had to say about the then prevailing psychological orthodoxy.

9.vii. A Battle with Behaviourism

Chomsky initiated a battle with behaviourism, but not the war itself. Others were fighting at the same time, and some had reached the battlefield earlier (see 5.iv and

6.i–iii). But it was Chomsky who flourished the sword most cuttingly and wounded his opponents most bloodily. And it was he who returned to the attack over and over again, in both his scientific and his political writing. Consequently, he's been awarded most of the victory medals, even though others merited them too.

(Not being one for half-measures, he also rejected Piaget's "epigenetic" compromise between nativism and behaviourism: see Chapters 7.vi.g and 14.x.c. Contributions to the Chomsky–Piaget debate are collected in Piatelli-Palmarini 1980.)

Chomsky's training in descriptive linguistics had been informed by behaviourist assumptions, as we've seen (in Section v.a). But in the early 1950s, he ignored this psychological movement. He looked at it in detail only after he'd finished his basic theoretical research on language, in 1956–7. Indeed, he said later that the critique of finite-state languages with which he'd opened *Syntactic Structures* was "an afterthought", not included in his larger manuscript (Chomsky 1975: 40).

From the late 1950s on, however, he pursued the behaviourists relentlessly. By the mid-1960s, he had rejected—or rather, expanded—the definition of linguistics given in *Syntactic Structures*. Now, he identified the goal of linguistics in explicitly nativist terms—with radical implications for psychology, and for the philosophy of mind.

a. Political agenda

In his pursuit of behaviourism, Chomsky was driven at least as much by political passion as by abstract argument. For he saw it in political terms, and didn't like what he saw.

He linked behaviourism with the social-scientific 'experts' whose advice to the US government, and especially to the military, he—and countless others—regarded as political anathema. AI scientists were suspect too, despite their non-behaviourist approach, for their work had made 'smart' bombs and aerial reconnaissance possible (Chapter 11.i).

Chomsky expressed this complaint on many public platforms at the height of the Vietnam war. Indeed, he spoke in the very first public demonstration against the war, held on the Boston Common in October 1965. He spoke, but he wasn't heard:

> the mobs were so hostile that none of us could say an audible word. The only reason we weren't killed, I suppose, was that there were hundreds of police—who didn't like what we were saying one bit, but didn't want to see anyone murdered on the Common. (quoted in Swain 1999: 29)

Besides braving mobs on the Boston Common, he was put in prison on more than one occasion (Swain 1999: 24). He was thus a powerful voice in the counter-culture (Chapter 1.iii.c).

He still is. For, unlike his MIT colleague Joseph Weizenbaum (11.ii.d), he continued—and intensified—his complaints long after the 1970s. He repeated them, with meticulous documentary evidence, all around the world in connection with various other events—most recently, those of 11 September 2001 in New York, and their grisly aftermath in Afghanistan and Iraq. He still berates "respected intellectuals" for justifying the actions of "violent and murderous states"—including the USA (Chomsky 2001: 73; 2005). And he still sees political engagement as a responsibility of the intellectual (1996*c*).

His speeches continue to draw a multitude of listeners: his Royal Institute of Philosophy Lecture in 2004, for instance, saw about 3,000 youngsters queuing along the London streets (Chomsky 2005). And his writings have attracted adulatory praise from political sympathizers (Barsky 1997). They doubtless drew some people to his scientific writings who wouldn't otherwise have been interested in them.

Given the combination of his political and linguistic work, he eventually became the most-quoted living writer, and the eighth most quoted in history (Barsky 1997: 3). (The other seven were Marx, Lenin, Shakespeare, Aristotle, Plato, Freud—and, in fourth place, the 'author' of the Bible.)

In making these politico-scientific links, Chomsky repeatedly criticized what he saw as the philosophical assumptions underlying the science. Specifically, he claimed that empiricist and rationalist views of humanity inherently tend to promote political oppression and freedom, respectively (e.g. Chomsky 1972*b*: 45–54 and *passim*; cf. N. V. Smith 1999).

The justice of this general claim isn't obvious; some have argued that the reverse is true (Sampson 1979*a*). For instance, Locke's political philosophy—which underlay the English, French, and American revolutions—hardly suggests that empiricism is incompatible with liberty. Indeed, Locke saw doctrines of innate ideas as politically dangerous, since they encourage "blind credulity" whereby the populace becomes "easily governed" by a "dictator of principles and teacher of unquestionable truths" (J. Locke 1690: I. iv. 25). One of his followers, Lord Chesterfield, even wrote to his son that "A drayman is probably born wyth as good organs as Milton, Locke, or Newton" (Strachey 1924–32, pp. ii–136). True to his time, however, when Locke framed the constitution of Carolina in 1669 he gave slave-owners full jurisdiction over their slaves (Porter 2001, pp. xxii–xxiii). Lockean liberty had its limits.

However that may be, this dual significance helps explain the virulence with which Chomsky attacked behaviourism in general, and Burrhus Skinner in particular. Skinner—who wrote the first book on "verbal behaviour"—not only theorized language in a manner Chomsky saw as scientifically vacuous, but also rejected traditional assumptions about the way to the good life (Skinner 1948) and humanist conceptions of freedom and dignity (Skinner 1971).

These views didn't endear him to the counter-cultural Chomsky (1.iii.c). He attacked them in typically uncompromising terms:

> Consider a well-run concentration camp with inmates spying on one another and the gas ovens [visibly] smoking in the distance, and perhaps an occasional verbal hint as a reminder of the meaning of this reinforcer. It would appear to be an almost perfect world . . . In the delightful culture we have just designed, there should be no aversive consequences . . . Unwanted behavior will be eliminated from the start by the threat of the crematoria and the all-seeing spies . . . Nevertheless, it would be improper to conclude that Skinner is advocating concentration camps and totalitarian rule (though he also offers no objection). Such a conclusion overlooks a fundamental property of Skinner's science, namely, its vacuity. (Chomsky 1973: 129–30)

The familiar vocabulary of freedom and dignity, said Chomsky, far surpasses Skinner's terminology in discussions of politics. Indeed, it's all we have: there's no scientific theory of politics, and perhaps there never can be. In refusing to use such humanistic expressions, therefore, Skinner was being not scientifically scrupulous but politically irresponsible.—Perhaps merely politically neutral? No, Chomsky insisted, for that comes to the same thing: to "offer no objection" is, in effect, to support the status quo.

Later, Chomsky widened this attack to include all quietist academics. He contrasted Albert Einstein's "very comfortable life" of research, interrupted by "a few moments for an occasional oracular statement", with Russell's passionate espousal of nuclear disarmament and vociferous critiques of the Vietnam war, which landed him in prison on more than one occasion (Barsky 1997: 32–3). (Russell was one of the first public critics of the H-bomb, and in 1958 became founding president of the Campaign for Nuclear Disarmament; his pacifist speeches and writings around the time of the First World War had also got him jailed—see Monk 1996: 525–40; 2000: 367–90, 405–13.) In general, Chomsky argued that intellectuals should actively, if perforce riskily, confront "privilege and authority"—both on public platforms and in their personal lives. Certainly, he himself has never stopped doing so.

For the rest of our discussion here, let's ignore the politics. If this brands us as irresponsible quietists, so be it. For Chomsky also gave non-political arguments against behaviourism.

b. That review!

Politics aside, Chomsky made three main claims in his ongoing critique of behaviourism:

* First, that behaviourist theories are in principle inadequate to represent grammatical structure, which exists on several theoretical levels.
* Second, that Skinner—the most influential behaviourist in this area—used covertly mentalistic terms to explain language, while having no theoretical right to do so.
* And third, that human babies, and *only* human babies, can acquire language only because they have innate knowledge of its fundamental structure.

One could coherently admit the first two claims while rejecting the (more controversial) third. Nevertheless, by the end of the 1960s, many cognitive scientists had accepted all three.

The first claim was the theoretical core of Chomsky's dispute with behaviourism, and rests on his argument in *Syntactic Structures* (presented there without explicit reference to behaviourism). The structure and "creativity" of language can't be described by Markov processes, but must be represented as a hierarchy of theoretical levels (see Section vi). The central idea here wasn't new: Karl Lashley had expressed it in the 1940s, in relation to speech and other motor skills (Chapter 5.iv.a). But Chomsky put it in mathematical terms.

He took this criticism to apply to behaviourism in general. As regards those behaviourist theories which (unlike Skinner's) posited internal Ss and Rs, he was justified: mathematically, it doesn't matter whether the Ss and Rs are observable or not. (Compare the objection that production systems are merely an internalist form of behaviourism: Chapter 7.iv.ii.)

Whether he was justified as regards all possible forms of behaviourism was less clear. Some empiricist philosophers argued that behaviourism, or (as they put it) scientific psychology, shouldn't be defined so restrictively as to include only Markovian theories based on conditioned response (e.g. Quine 1969: 96–7). (It turned out, later, that simple statistical methods can enable a connectionist network to learn the past tense, even generating the detailed errors made by children: see Chapter 12.vi.e and x.d.)

The second prong of Chomsky's attack was unsheathed in his notorious review of Skinner's book on the psychology of language (Skinner 1957; Chomsky 1959*b*). This book was published in the same year as *Syntactic Structures*, but had been circulating in draft for some time. It was based on Skinner's William James Lectures, given at Harvard in 1947, and in his own mind represented his crowning achievement. (He'd been working on it for twenty-three years—Skinner 1967: 402.)

Chomsky allowed that in his animal research Skinner had been scrupulous in defining his terms, and had made genuine scientific discoveries. On turning to language, however, he had abandoned both experiment and scruple. His central theoretical concepts of *stimulus*, *response*, and *reinforcement* were being applied so loosely as to be vacuous. And his new, specifically linguistic, concepts (such as *tact* and *mand*), though defined in putatively observational terms, were actually being understood mentalistically (*mand* as an indiscriminate amalgam of *command*, *request*, *question*, *prayer*, *advice*, and *warning*, for instance).

This was hardly surprising, Chomsky said. For if one struggled to interpret Skinner's vocabulary in a strictly behaviourist fashion, his theory instantly lost all plausibility. Moreover, Skinner had made no serious attempt to test his theory of language experimentally, relying instead on anecdote and speculation.

Chomsky did acknowledge that some operant-conditioning experiments on language had come up with surprising and non-trivial results. For instance, speakers can be—unknowingly—conditioned to produce more or fewer plural nouns, to use the personal pronoun more or less often, or to talk about one topic rather than another (Krasner 1958). The first of these is especially interesting, since it's defined in terms not of a single response, like uttering the word *I* or *we*, nor even of a phonemic pattern such as words ending in *-s*, but of a syntactic class. But Chomsky was sceptical: "Just what insight this gives into normal verbal behavior is not obvious" (Chomsky 1959*b*, n. 7).

This review, one-third as long as Chomsky's little book, attracted enormous attention. Its many readers were not only gripped by the argument, which was detailed and careful, but also amused and/or outraged by the rhetorical style. For this essay was very different from the dry and symbol-ridden *Syntactic Structures*.

Chomsky's pen had been dipped in vitriol, as well as in ink. Words such as *pointless*, *confused*, *gross*, *absurd*, *delusion*, *dogmatic*, *arbitrary*, *useless*, and *empty* leapt out from the pages. Skinner's fundamental concept (reinforcement) was scornfully said to have "totally lost whatever objective meaning it may ever have had", and Skinner himself to be "play-acting at science". There was delectable ridicule, too. According to Skinner, said Chomsky, the best way to show one's appreciation of a prized work of art owned by a friend is to shriek "Beautiful!" repeatedly in a loud and high-pitched voice, without ever pausing for breath.

This was welcome light relief from the daily academic grind. Small wonder that the piece was so widely read. Only Skinner avoided it—or so he claimed: "I have never actually read more than half a dozen pages of Chomsky's famous review of *Verbal Behavior*. (A quotation from it which I have used I got from I. A. Richards)" (Skinner 1967: 408).

Rhetoric apart, however, Chomsky's second point was unassailable. Skinner was guilty as charged: he was misusing his own terminology and betraying his own scientific method, by tacitly relying on everyday mentalistic intuitions. In that skirmish, Chomsky won hands down.

c. Nativist notions

Chomsky's third salvo caused even more sound and fury—and is still echoing today (see 7.vi, 12.vi.e, and 14.ix.c–d). More intellectually daring than the first two points, and arguably less well grounded, it was a combination of a negative claim and a positive one.

Negatively: that the child's language acquisition can't be fully explained by experience, even if this is conceptualized in non-Markovian terms. Positively: that babies must therefore have innate knowledge of the structure of language—which is to say, of the rules of some universal grammar.

The positive corollary, that there's some inborn "language acquisition device" specially apt for learning grammatical patterns, was even more obviously heretical than its negative ground. It had been "too audacious" to be included in *Syntactic Structures* (see above), and was merely hinted at in the notorious review, where Chomsky was more concerned to criticize Skinner's theory than to present his own. But it was made fully explicit—and defensively related to his august "Cartesian" precursors—in Chomsky's publications of the 1960s (Chomsky 1964, 1965, 1966, 1968). Indeed, he now built it into the very definition of linguistics as such:

> A theory of linguistic structure that aims for explanatory adequacy incorporates an account of linguistic universals, and it attributes tacit knowledge of these universals to the child. (Chomsky 1965: 27)

> Given a variety of descriptively adequate grammars for natural languages [which can be achieved if one sees the goal of linguistics as I defined it in *Syntactic Structures*], we are interested in determining to what extent they are unique and to what extent there are deep underlying similarities among them that are attributable to the form of language as such. *Real progress in linguistics* consists in the discovery that certain features of given languages can be reduced to universal properties of language, and explained in terms of these deeper aspects of linguistic form. Thus *the major endeavor of the linguist* must be to enrich the theory of linguistic form by formulating more specific constraints and conditions on the notion "generative grammar". (Chomsky 1965:35; italics added)

The negative nativist claim rested partly on Chomsky's anti-Markovian arguments, and partly on his assertion that the language the infant hears is inadequate to allow grammar to be induced so quickly. The child, he said, may hear a certain syntactic construction only very rarely. Moreover, everyday speech abounds with unfinished sentences, restarts, and grammatical errors. (People may get lost inside subordinate clauses, for instance.) So the child very often hears utterances that are grammatically flawed. Given such patchy and contaminated data, rapid learning of a perfect grammar simply couldn't happen.

This negative assertion was immediately criticized by those of an empiricist cast of mind, as being an argument from ignorance. The fact that no one has yet come up with an explanation of how children learn language, they said, doesn't mean that it's impossible to do so. Induction (as remarked above) needn't be conceptualized in the simple terms of traditional empiricism (Quine 1969: 96–7; Harman 1967, 1969). Moreover, the notion of "experience" was said to be richer than Chomsky allowed. It was no longer thought of as passive or atomistic, and with further psychological research might become richer still (M. Black 1970: 458).

This neo-empiricist shaft hit its mark, but left Chomsky standing. Facing only promissory notes from his opponents, his argument from ignorance remained in play. (It was soon followed by a formal proof that 'nested' languages can't be learned on the basis of *positive* evidence only: Gold 1967.)

As for Chomsky's comment that the infant's heard corpus is largely degenerate, this seemed obviously true to many people. But—like Cordemoy's remarks (see Section iii.a) about nouns preceding adjectives, preceding verbs—it hadn't actually been tested.

When it was, psycholinguists reported that mothers typically use a restricted dialect ('motherese') when speaking to babies and young infants, employing sentences both syntactically simpler and more grammatically correct than those of adult speech (R. Brown 1973). On the other hand, they also found that the child's syntax develops in predictable stages, two-word utterances appearing first, and the complexity increasing with age (7.vi.a). This was generally interpreted (perhaps wrongly: see Chapter 12.viii.c–d) as evidence of an innately driven developmental process, supporting the spirit, if not the letter, of Chomsky's nativism.

The positive nativist claim caused huge controversy. Much of the initial scepticism concerned what Chomsky—or indeed anyone—could mean by "innate knowledge" or "innate ideas". All of the questions, and most of the answers, that had been hurled at Descartes, Locke, and Leibniz resurfaced.

Some philosophers objected that innate knowledge is impossible, because "knowledge" implies justification (Edgley 1970; Harman 1969). Chomsky replied that unconscious knowledge (competence), unlike what Gilbert Ryle (1946) had called "knowledge that", requires neither verbalization nor justification (Chomsky 1969; 1980*a*, ch. 3).

Even when the debate was put in terms of innate ideas instead of innate knowledge, or dispositions instead of ideas, the problems sketched in Section ii.c (above) arose—and were debated with passion. High-visibility symposia pitting Chomsky against leading epistemologists and philosophers of language occurred from the mid-1960s on, and innate ideas became a hot topic of scientific and philosophical debate (e.g. *Synthese* 1967; Hook 1969, pt. ii; Stich 1975).

The discussions were often ill-tempered, and many of Chomsky's adversaries rivalled his rhetorical armoury in their attack. One leading philosopher, Gilbert Harman, said that Chomsky "misrepresents [my comments] at almost every point", to which Chomsky testily retorted: "I see no reason to try to trace the various confusions Harman develops, none of which have any relation to the views I actually hold . . . " (Harman 1969: 143; Chomsky 1969: 154). His irritation was doubtless fuelled by the fact that, some years earlier, Harman had denied the need for transformations (Harman 1963). Another influential philosopher dubbed Chomsky's theory 'The Emperor's New Ideas', and remarked:

> The theory of innate ideas is by no means crude. It is of exquisite subtlety, like the gossamer golden cloth made for that ancient emperor. But the emperor needs to be told that his wise men, like his tailors, deceive him; that just as the body covered with the miraculous cloth has nothing on it, the mind packed with innate ideas has nothing in it. (Goodman 1969: 141–2)

Willard Quine, in characteristically gentlemanly fashion, mildly declared that the behaviourist is "knowingly and cheerfully up to his neck in innate mechanisms of learning-readiness" (Quine 1969: 95–6). Reinforcement, for example, depends on "prior inequalities in the subject's qualitative spacing . . . of stimulations". And "unquestionably much additional innate structure is needed, too, to account for language learning".

But there was a sting in the gentleman's tail. Quine ended his paper thus:

> Externalized empiricism [which makes no reference to "ideas"] or behaviorism sees nothing uncongenial in the appeal to innate dispositions to overt behavior, innate readiness for language-learning. What would be interesting and valuable to find out, rather, is just what these endowments are in fact like in detail. (Quine 1969: 98)

Chomsky could answer that question only by saying: universal grammar, whatever that turns out to be.

d. Universal grammar?

What universal grammar will turn out to be—if it exists at all—is still unclear. Indeed, the very notion of *innateness* is now interpreted in a much more sophisticated way (Chapters 7.vi.g and 14.ix.c). Added to that is the fact that Chomsky himself changed his mind more than once about just what the "universal" base of language is.

Chomsky's initial postulation of universal grammar had depended only on argument, the main premiss being the supposed inadequacy of learning theory to explain language acquisition. But, as Quine had said, it needed empirical evidence also.

From 1965 onwards, Chomsky suggested a number of abstract grammatical principles that appeared to apply to several languages and might apply to all. Some concerned aspects of the base component (for example, that all languages share the same syntactic categories: noun, verb, NP, etc.). Others concerned the transformational rules.

For instance, in *Aspects of the Theory of Syntax*, the first major publication in which he made his nativism fully explicit, Chomsky said:

> Although a language might form interrogatives, for example, by interchanging the order of certain categories (as in English), it could not form interrogatives by reflection [reversing the order of words in the sentence], or interchange of odd and even words, or insertion of a [syntactic] marker in the middle of the sentence. (Chomsky 1965: 56)

Later, he posited the *A-over-A* principle. This forbids any language-specific (English, French, or . . .) transformation from picking an embedded noun phrase out of the noun phrase in which it is embedded—or more generally, from extracting any phrase of category *A* out of some larger phrase of the same category. Another example was a "principle of cyclic application" which Chomsky supposed to govern the way in which noun phrases are deleted and replaced by pronouns.

However, each of Chomsky's suggestions was challenged as applying only (if at all) to English, or to a small set of languages. Thorough testing would require detailed descriptions of a multitude of tongues.

This, of course, is what the out-of-fashion structuralists had aimed to provide. But Chomskyans didn't. One critic has described them as being as obsessed with English as the Port-Royalists were with French (Itkonen 1996: 487; cf. Section iii.c). Indeed, Chomsky himself would defiantly declare that "I have not hesitated to propose a general principle of linguistic structure on the basis of a single language"—an admission dubbed "preposterous" by this critic (Itkonen 1996: 487).

Some of Chomsky's "universals" were criticized even with respect to English. The *A-over-A* principle, for instance, was soon decomposed into several independent "island constraints" (J. R. Ross 1968), which covered the data better (and some of which hadn't been covered by the original principle). In the years that followed, Chomsky tried to integrate the island constraints into a single rule. He came up with the "Subjacency Principle", but even this didn't work for all of the island constraints. Again, Chomskyans tried to find some even more abstract principle underlying the variety of surface forms.

But non-Chomskyan linguists weren't persuaded. Worse (for Chomsky), they would eventually accuse him of constantly redefining his position over the years so as to make it, in effect, unfalsifiable. Sometimes, this would be said in waspish exasperation, caused by frustration at the still-continuing dominance of Chomskyans in the linguistic profession (see Section viii.a):

> I [have become] convinced that there is nothing, absolutely *nothing*, that could make Chomskyans admit that there is or has ever been anything amiss with their theory. In the game of linguistics, truth is a secondary consideration: not to lose face has top priority. (Itkonen 1996: 497; cf. pp. 490–4)

On this view, the passage from A-over-A to island constraints wasn't a step forward in a healthy Popperian progression of conjectures and refutations, but one of many intellectual retreats in a degenerating research programme (K. R. Popper 1935; Lakatos 1970). Today, half a century after Chomsky came on the scene, linguists are divided between these two judgements (see Sections viii & ix).

In short, Chomsky's universal grammar of the 1960s was, at best, a promissory note. Twenty years later, and largely due to Chomsky's influence, nativism would be expressed more acceptably and backed by considerable *non-linguistic* evidence (see Chapter 16.iv.c–d and Fodor 1981*b*). Twenty years later still, 'straightforward' nativism would be replaced by epigenesis, and modular theories of the mind reinterpreted, or rejected, accordingly (Chapters 7.vi and 15.x.a–c). Meanwhile, Chomsky couldn't meet Quine's challenge.

It's important to recognize, however, that (whatever one might say about his later theories) Chomsky's early suggestions on this matter had the Popperian merit of being mistaken. They weren't vague claims unamenable to testing—which isn't to say that the Chomskyans were eager to test them.

Still less were they a priori arguments from first principles, with no obvious relation to reality. On the contrary, they were empirical, falsifiable, *scientific* claims. Chomsky was saying "Languages are like *this*—but they could, conceivably, have been like *that*." As he put it:

> There is no *a priori* consideration that lends plausibility to a theory of [universal] syntactic operations [such as mine] that precludes the formation of interrogatives by left–right inversion of the corresponding declarative, or innumerable other perfectly conceivable, and quite efficient operations. It is just this fact that constitutes the scientific interest of such a theory. (Chomsky 1970: 468)

In other words, Chomsky wasn't a rationalist in the strict sense. The Cartesian philosophers of the seventeenth century took language to be close to reason, being informed by principles of necessary truth. And even Kant held that no alternative mental structure (system of intuitions) is conceivable by us (see Section ii.c). But Chomsky shared neither of these beliefs. Whether a language-universal grammar exists, and if so what it is like, are empirical questions.

On those points, Chomsky's behaviourist opponents were willing to agree. But even if he had identified a grammar suited, as a matter of fact, to all natural languages, they wouldn't have conceded defeat. For universality doesn't guarantee innateness. Max Black put the point thus:

> In science in general, reference to dispositions and powers is usually a promissory note, to be cashed by some identifiable internal structure . . . But there is no serious question at present of finding such internal configurations in human organisms: we hardly know what we should be looking for, or where to look. So far as stimulation of research goes, 'nativism' looks to me like a dead end . . . (M. Black 1970: 458)

Black's charge that no interesting research can be prompted by nativism fell wide of the mark. Cognitive scientists excited (whether for or against) by Chomsky's nativism did much interesting work as a result, the many studies of the development of grammar in children being just one example (see Chapter 7.vi). But Black was right to insist that, strictly, the nativist requires some independent evidence of internal mechanisms. Not until the 1990s would the debate on innate ideas, and inborn "dispositions", be illuminated by reference to specific neurological processes (see Chapter 14.vi.b and ix.c).

Not even Chomsky, of course, suggested that babies are born knowing French—or perhaps English, if destined for a life of persuasion, emotionalism, and deceit? (see iii.d, above). But their knowledge of universal grammar, he said, acts as a framework which guides them to attend to certain features, certain distinctions, in the language spoken around them. In syntax as in phonetics, the specific value of those distinctive features varies across natural languages: hence their apparent diversity.

On this view, the child is, in effect, like a scientist. Instead of collecting data by pure induction (which is impossible), the scientist formulates theories and hypotheses which suggest what to look for, and where to look for it. This, essentially, is what the baby has to do in learning its mother tongue.

The view of the speaker/hearer as a hypothesis tester had always been implied by Chomsky's work, quite irrespective of nativism. It was implicit in *Syntactic Structures*, in his claim that the speaker/hearer must ascribe unobservable, many-levelled, grammatical structure to utterances in order to understand them, and was made explicit in the first review (Lees 1957: 406–7). As he himself admitted, his theory that language users generate grammatical hypotheses owed much to Jerome Bruner's "New Look" in perception (6.ii), which swept cognitive psychology in the late 1940s and 1950s (Bruner 1980: 81).

What Chomsky's nativism added (in 1965) to the widely current idea of hypothesis testing was the claim that babies may not have to create their linguistic hypotheses out of thin air, nor even out of infantile dreams and visions. Rather, he said, they can produce them on the basis of a powerful theoretical framework—a "Language Acquisition Device"—already present in their minds.

9.viii. Aftermath

Chomsky's intellectual footsteps were greatly amplified in the two decades after his appearance, and are still clearly audible today.

His early papers had a huge, and lasting, influence on pure computer science. Virtually every introduction to compiler design cites Chomsky's hierarchical classification of languages (Edmund Robinson, personal communication). And a leading textbook on theoretical computer science remarks that "[Chomsky's] notion of a context-free grammar and the corresponding push-down automaton has aided immensely the specification of programming languages" (Hopcroft and Ullman 1979: 9; cf. 217–32). That's why I said (in Section vi.a) that the 'Three Models' paper was his most original and most well-founded contribution.

For cognitive science in general, his crucial writings were the formalist *Syntactic Structures* and the nativist *Aspects*—plus, for light relief, his coruscating review of Skinner. These three publications had a huge impact on psychology and philosophy, and a significant influence on AI (see 6.i.e, 7.vi.a–d, 10.i.g, and 16.iii–iv). In effect, cognitive science in the 1960s had a love affair with Chomsky: one historian of linguistics has remarked on "the sweeping and unrequited optimism of the honeymoon years of cognitive psychology and transformational grammar" (R. A. Harris 1993: 257).

With respect to linguistics as such, the sequel was clear: all linguists came to situate themselves with respect to Chomsky (Lyons 1991: 9). This was true even of those whose specialism was in some other area, such as sociolinguistics, historical linguistics, descriptive linguistics, and anthropological linguistics—all of which, and especially the last two, declined in status post-Chomsky (Sampson 1980: 78 ff., 146–7). All linguists had to decide whether to adopt some version of his account in their own work. And all had to allow that he'd set new standards of clarity for theoretical linguistics.

Whether his many followers lived up to those standards is questionable. Trouble was brewing even in the 1960s. A champion of formalization in science—echoing Leibniz's "calculemus" as the only way of achieving objectivity (see Chapter 2.ix.a)—praised Chomsky for seeding "one of the healthiest and most important controversies in psychology in this century", but complained that his leading disciples couldn't *prove* their claim that behaviourism is in principle inadequate (Suppes 1968). Twenty years later, things were even worse. One critic pointed out that, of the "thousands" of papers published on Chomsky's grammar between 1970 and 1986, *only one* (in a relatively obscure journal) attempted a mathematical statement (Gazdar 1987: 125). Indeed, Chomsky himself would eventually be fiercely criticized for lack of rigour (see Section ix.e).

In addition, he was accused of being mistaken in various ways even about *syntax*, never mind arcane philosophical topics such as innate ideas. This fact should be no great surprise. It's obviously possible to be influential without being right. What is surprising is the extraordinary emotional tone of many of those criticisms. Disinterested discussion of Chomsky's views was—and still is—hard to come by, as we'll now see.

a. Polarized passions

For linguists who specialized in syntax, semantics, or phonetics, Chomsky soon became a constant presence. Too much so, perhaps: the sociology of science hindered the science itself, as Chomskyan cliques came to dominate various conferences and journals.

Bruner, who as a non-linguist has no professional need to 'take sides' (and who'd influenced Chomsky deeply in the late 1950s: see vii.d, above), is one of many who've complained about this. He recently remarked:

> [Chomsky is a] systems builder, and ruthless systems at that. And it's so funny, because the fact of the matter is he claims to be a deep believer in democratic values. But the ways in which he goes about it, and the ways in which they, the Chomskians, have gone about it by a kind of taking over of the apparatus of linguistic scholarship in America is nothing less than just a hostile takeover of the whole damned system. Everybody else is out. (Shore 2004: 51)

Naturally, this phenomenon didn't go unchallenged within the profession. Eventually, linguists' complaints about Chomsky's hegemony in linguistics became even more heated than philosophers' debates about his theory of innate ideas (see Section vii.c). One observer recently remarked that "the level of enmity is truly stunning". As he put it, "there are people who genuinely wish that Chomsky would die, or retire, or move exclusively into political or philosophical domains, and just leave poor little linguists alone" (R. A. Harris 1993: 256).

This "blood-boiling animosity" arose from professional frustration. An English linguist surveying the international scene in 1980 saw it like this:

> The discipline of linguistics seems to be peopled by intellectual Brahmanists, who evaluate ideas in terms of ancestry rather than intrinsic worth; and, nowadays, the proper caste to belong to is American. The most half-baked idea from MIT is taken seriously, even if it has been anticipated by far more solid work done in the "wrong" places; the latter is not rejected, just ignored . . . To the young English scholar of today, the dignified print and decent bindings of the *Transactions of the Philological Society* smack of genteel, leather-elbow-patched poverty and nostalgia for vanished glories on the North-West Frontier, while blurred stencils hot from the presses of the Indiana University Linguistics Club are invested with all the authority of the Apollo Programme and the billion-dollar economy. Against such powerful magic, mere common sense . . . and meticulous scholarship (in which the London School [e.g. Firth, Halliday, Hudson] compares favourably, to say the least, with the movement that has eclipsed it) are considerations that seem to count for disappointingly little. (Sampson 1980: 235)

Comparable complaints have been made more recently, if less colourfully. A Finnish linguist in the mid-1990s remarked that Chomskyans often identify "linguistics" with "generative linguistics", and that "there are nowadays relatively few publications which subject generative linguistics to explicit critical scrutiny". The lack of publication was attributed to sociological factors:

> A lengthy discussion [on the Internet] concerning "mainstream linguistics" . . . was started by the observation that people who had to criticize the generative paradigm preferred to stay anonymous. This was interpreted as reflecting the opinion that open criticism of "mainstream linguistics" might jeopardize a person's career prospects and possibilities for publication. Although such a view was heatedly denied by representatives of the generative paradigm, at least my own experience confirms it. In several informal meetings I have found out that linguists agree with some or even all aspects of the criticism [that I am about to present]; they just do not want to say it publicly. (Itkonen 1996: 471)

In a nutshell, the "representatives of other schools seem anxious to maintain what might, depending on one's point of view, be called either 'peaceful coexistence' or a 'balance of terror' " (p. 471).

Similarly warlike language was occasionally heard on Chomsky's side of the Atlantic, too. One leading American linguist, a colleague of Chomsky's at MIT in the late 1950s and 1960s (see Section x.b), had lost patience with him by the 1980s. Victor Yngve (1920–) , arguing that linguistics should focus on people (communicators) rather than language, referred to "puffery, empty polemics, intellectual bullying, and the great-man syndrome", and bemoaned "unscientific" a priori reasoning "in support of philosophical preconceptions and maintained by early ecstatic reviews, political acceptance, largely unquestioned personal authority, and the abuse and contempt . . . poured on other positions" (Yngve 1986: 7, 43, 108). Chomsky's name wasn't mentioned. Clearly, however, the cap fitted and he was being asked to wear it.

In short, Chomsky was the high priest of a new orthodoxy, almost a new paradigm (Kuhn 1962). Most young linguists aspired to be his devoted acolytes. Heretics (if noticed at all) weren't treated lightly—a situation due largely to Chomsky's personality, which thrives on embattlement.

For example, the MIT linguist who defined island constraints (see Section vii.d) was systematically reviled and excluded by his so-called colleagues (R. A. Harris 1993: 245–6). And George Lakoff, a pioneer of generative semantics (see below), was venomously accused by Chomsky himself of (among other things) having "discussed views that do not exist on issues that have not been raised, confused beyond recognition the issues that have been raised and severely distorted the contents of virtually every source he cites" (quoted in R. A. Harris 1993: 157). Skinner, then, wasn't the last to be lacerated by Chomsky's pen.

Nevertheless, heretics there were. Not everyone agreed with him. Indeed, even Chomsky disagreed with Chomsky. In other words, he had second thoughts—and third, and fourth . . . and more. Before considering his linguistic critics (in Section ix), let's look briefly at his own revisions of his ground-breaking theory of 1957.

b. Revisions, revisions . . .

Revisions weren't slow to come, for the grammar of *Aspects* differed significantly from Chomsky's earlier approach. The main differences were its terminology of deep and surface structure, its new treatment of transformations, and its inclusion of semantics. (Besides the new treatment of *grammar*, there was also a new emphasis on nativism, as we've seen.)

The 1965 account—known as "standard" theory—lasted, with relatively minor revisions, for fifteen years, while Chomsky devoted much of his time to politics. These revisions reduced the number of transformations in his theory: in 1970, for example, he suggested that lexical rules should replace morphological transformations (Bresnan 1978: 5). But the revised standard theory wasn't his last gasp. In the early 1980s, he provided a new approach, the theory of "government-binding" (GB) (Chomsky 1980*a*,*b*, 1982).

GB theory was even more unlike *Aspects* than that had been unlike *Syntactic Structures*. Chomsky had already suggested, in the late 1970s, that what had previously been thought of as a number of different transformations (involving wh-questions and various other matters) could be seen as special cases of a more general one. By the early 1980s, only a single, highly general, transformation remained. (Known as "move alpha", its pivotal notion was subjacency.) Much of the explanatory weight was now carried by innate "principles" and "parameters" (Chomsky and Lasnik 1977).

Principles are unvarying linguistic universals. Parameters are like variables: each has a limited number of possible values, which are partly interdependent. The diversity of actual languages is explained by differences between their sets of parameter values. So infants, in acquiring language, must discover which values characterize their mother tongue, and set their internal parameters accordingly (Lyons 1991: 184–8). As Chomsky put it:

> If the system of universal grammar is sufficiently rich, then limited evidence will suffice for the development of rich and complex systems in the mind, and a small change in parameters may lead to what appears to be a radical change in the resulting system. (Chomsky 1980*a*: 66)

This theory is reminiscent of claims made by biologists who see the range of possible adult forms as highly constrained, yet different, with no "great chain of being" filling the gaps. Conrad Waddington, and his student Brian Goodwin, have argued that small changes in epigenesis can tip the embryo into one developmental pathway rather than another: see Chapters 14.ix.c and 15.ix.a–c.

Eventually, in the early 1990s, GB theory gave place to "minimalism" (Chomsky 1993, 1995). "Deep structure" was abandoned. Still more shockingly, the previously pivotal notion of grammaticality was dismissed as being "without characterization or known empirical justification" (Chomsky 1993: 44). And several new principles—including Greed, Procrastinate, and Last Resort—were introduced (Chomsky 1995: 200–12).

But minimalism was too much for some of Chomsky's critics. One reviewer remarked that it was so very different from its GB predecessor that Chomsky was here "contriving to reclaim the role of the lone revolutionary", without acknowledging that his "new" position owed much to the generative semanticists of twenty years before (see below)—Pullum 1996: 137. As for the core theoretical ideas of minimalism, these were "sketched so hesitantly as to be hard to describe", and afforded "a level of explication [that] is risible" (Pullum 1996: 140–1). He mocked the new principles of "Greed" and the like, and complained about "anthropomorphic" accounts of "phrases moving in a bid to get their needs satisfied, and abstract nodes yearning to discharge their feature burdens" (Pullum 1996: 142). Minimalism, this critic declared, involved "a complete collapse in standards of scientific talk about natural language syntax".

These heretical judgements cut no ice with Chomsky's followers. The mantle of intellectual guru had settled firmly on his shoulders in the 1960s, and sits there comfortably still. Late in 1998, for example, his theory of minimalism was reported as a quarter-page news item—not a book review, nor even a feature article—in a British national newspaper. This largely explains his reviewer's ill-tempered use of vocabulary such as "risible".

In truth, the rot had set in long before minimalism. Chomsky's own standards of rigour had always been much lower than was generally believed. This fact, which became clear (to those with eyes to see) in the mid-1970s, is important for cognitive science in general, and is discussed in Section ix.e.

The many theoretical changes in Chomsky's later writings, whether rigorous or not, don't concern us. The huge impact of Chomsky's work on cognitive science resulted from his first three books, and was already integral to the field by the 1980s.

However, two of Chomsky's post-1957 theory changes do have a wider relevance, so merit mention here. The first was his increasing stress on "modularity". His style of linguistic nativism always implied that the language "faculty"—or, as he also says, "organ"—is a self-contained part of the mind, functioning in independence of others. Over the years, he made this more explicit (now, he analyses the language faculty itself into distinct sub-modules: Chomsky 1986). Moreover, he speculatively generalized it to other mental "faculties" (Chomsky 1980*a*).

The idea of modularity didn't originate with Chomsky. Conceptualized as "hierarchy", it had already been stressed by Simon, in his widely read discussion of the design (and evolution) of complex computational systems (Simon 1962/9). Chomsky's championing of it influenced (and drew support from) research in the psychology of vision and developmental psychology (Chapter 7.v and vi); ran in parallel with the incorporation of expertise (as opposed to general reasoning methods) in AI (see Chapter 10.iv.b–c); and inspired an influential philosophy of mind developed by Fodor (16.iv.c–d).

Nevertheless, the general relevance of modularity is controversial (see Chapter 7.vi.d–i). It has been challenged, for instance, by wide-ranging research in developmental psychology (Karmiloff-Smith 1992). And Simon's arguments about the evolutionary necessity of modularity have recently been questioned by the philosopher Tadeusz Zawidzki (1998). He argues that Simon's desideratum that evolution pass through "stable intermediate stages" is met by Stuart Kauffman's models of genetic regulation (see 15.ix.b), which are *not* decomposable into separate, hierarchical, parts.

The second broadly influential change was that Chomsky included semantics in *Aspects*, having previously ignored it.

c. Semantics enters the equation

By introducing semantics into the discussion, Chomsky was trying (for example) to account clearly for the deviance of sentences like "Colourless green ideas sleep furiously", and to decide whether they should be regarded as truly grammatical.

This word string *seems* to be grammatical: two adjectives and a noun, making a nice NP, and a verb and adverb providing the necessary VP. From the syntactic point of view, then, it appears to be fine. (Some people even claimed to be able to give it a meaningful interpretation.) Nevertheless, there does seem to be something wrong with it—but what, exactly?

To deal with that question, among others, Chomsky now (in *Aspects*) said that any generative grammar must have both a syntactic and a semantic component. And he regarded the latter as "purely interpretive". That is, each deep-structure (syntactic) expression is sent to a semantic rule system, which gives it a formal description in terms of (universal) semantic primitives. These include the distinction between mass and count nouns, and between human, animate, inanimate, or abstract things.

Because the semantic component was supposed to be only interpretative, "it follows that all [this] information . . . must be presented in the syntactic component of the grammar" (Chomsky 1965: 75). But there is no a priori answer, he said, to the question whether—and if so, to what extent—it should influence the generation of syntactic structures. He did include various semantics-based "selectional rules" restricting the structures that his syntax could generate. But he pointed out that these could be transferred to the semantic component; conversely, all of the semantic interpretative rules might be applied before some of the syntactic generative rules, "so that the distinction between the two components is, in effect, obliterated" (Chomsky 1965: 158–9). In short, Chomsky regarded questions about the relation of syntax to semantics as a matter of judgement, not principle.

His close MIT colleagues Jerrold Katz and Fodor (1963) agreed that syntax and semantics are to some extent intertwined. Nonetheless, they insisted that we need a specifically semantic theory to describe and explain the role of meaning in language. Moreover, they said, such a theory would contribute to (and draw on) the psychology of language. Our ability to understand the meanings of novel sentences must depend on the meanings of the constituent words (morphemes).

This implies, they argued, that we need a semantic "dictionary" of morphemes, and also an understanding of the principles by which meanings can be sensibly combined. A dictionary and grammar alone cannot explain understanding. The prime reason for this is that many words are multiply ambiguous (*bank*, *bill*, *seal*, *bachelor*), and the dictionary must include all the possible senses. Syntax cannot always do the disambiguating for us: *He banked the bill* isn't ambiguous, but *They were approaching the bank* is. In addition, then, we need "projection rules" for using the dictionary and grammar, which take account of the meaning relations between morphemes in generating semantic interpretations of sentences.

The burgeoning interest in semantics took many forms. One of these was "generative semantics". This label, originated by Lakoff in 1963 (before the publication of *Aspects*), was attached to a number of theories in the late 1960s to mid-1970s (Dean Fodor 1977; R. A. Harris 1993, chs. 5 and 6; McCawley 1995). The single label obscured various differences, both theoretical and political.

Some generative semanticists criticized Chomsky's reliance on mathematical models, as opposed to detailed linguistic data. Some even saw Chomskyans as "scientific Calvinists" out of touch with the anti-elitist counter-culture of the 1960s (McCawley 1995: 344–5). Politics aside, the charge of mathematical affluence combined with data poverty would be levelled by many other critics too. But those generative semanticists who did focus on mathematical models argued that the basic generative system was semantically driven and interpreted by syntactic rules, rather than the other way around (see above).

It was as though they were saying "Meaning first, syntax afterwards"—which, at least on first hearing, is intuitively plausible. However, after "some years of involved and at times acrimonious argument", it was eventually agreed that there was less difference between the two positions than had been supposed (Lyons 1991: 93). Both "generative" and "interpretative" semantics concerned (timeless) linguistic competence, not performance. Giving the generativity to the syntax or to the semantics made little difference in the types of sentence that could be (non-procedurally) accounted for.

In general, a semantic theory should give us a principled account of synonymy, and explain the (non-syntactic) ambiguity of *The bill is large*, and its resolution in *The bill is large but need not be paid*. Katz and Fodor assumed that concepts are analysable into distinct semantic components, and posited semantic primitives universal to all natural languages—and all human minds. They explained synonymy in terms of shared entries in a semantic dictionary, and paraphrase in terms of shared semantic structure between sentences that might be lexically and syntactically very different (*The oculist examined me/I was inspected by the eye-doctor*). Different senses of one word (such as *bachelor*) falling in the same syntactic category (*noun*) were identified by differing "semantic markers" (primitives such as *Human*, *Animal*, *male*) and "distinguishers" (for instance, *never married*, *young knight*, *first academic degree*, *young fur seal*).

The componential analysis of meaning is a tenet of classical empiricism—found in Locke, for instance. But there are difficulties about how to identify semantic primitives (for example, how to differentiate Katz–Fodor markers from distinguishers—Dean Fodor 1977: 144–55). Worse, it's not obvious that the componential assumption is true. Even in apparently simple cases, such as the concept of *cousin*, it's not clear what the analysis should be. If one uses the 'obvious' constituents *mother*, *father*, *son*, *daughter* (and perhaps *brother* and *sister*), one can't represent someone's being a cousin without committing oneself on the sex of the various people involved (unless disjunctions are included). This problem disappears if one analyses in terms of *parent*, *child*, *sibling*, *male*, *female*; but then one can accept only blood cousins, not cousins by marriage (which would require *spouse*). Moreover, suppose the word *cousin* actually was used in this restrictive way: so what? Someone who creatively used it (in an appropriate context) to include cousins by marriage might raise some eyebrows, but would nevertheless be understood. This example illustrates what the philosopher Friedrich Waismann (1955) had called the "open texture" of language.

Scepticism about componential analysis and the identification of semantic primitives arose in linguistics, philosophy, and AI (e.g. Fodor 1970; Kempson 1977; Wittgenstein 1953: 31 ff.; Wilks 1978; Sampson 1979*b*). It was one reason why generative semantics went out of fashion. Fodor, for instance, abandoned his "primitivist" account, and later suggested—following the psychologist Eleanor Rosch (1978—see Chapter 8.i.b)—that certain psychologically salient concepts may be the semantic focus of language (Fodor 1983).

Besides these criticisms, however, a very different account of semantics—an account based in mathematical logic—began to gain ground by the mid-1970s (Dowty *et al.* 1981). Componential analysis, in the most general sense, was retained. But the components were now conceptualized as abstract ontological entities and mathematical relations, not as familiar notions such as *Human*, *Animate*, *parent*, *child*, or *male*. The assumption that the distinction between syntax and semantics is a matter of judgement and emphasis was roundly rejected. Moreover, the psychologism of Chomsky and his followers was rejected too.

In short, the Chomskyan paradigm was being challenged. This approach to semantics is outlined in the next section, along with some other important anti-Chomskyan work. Some of this would cast doubt not only on his *grammar*, but on his position on cognitive science in general.

9.ix. Challenging the Master

Some of the new theories challenging Chomsky would undermine only his linguistics, strictly so called. People not especially enamoured of NPs and VPs might not be much interested. Indeed, they might be downright bored.

But other challenges would also reject his close juxtaposition of language and mind, with its supposed implications for both philosophy and psychology. And one would drive a coach and horses through his claim that phrase-structure grammars are inadequate—so hugely improving the power of computer models of language.

In this section, I'll highlight these points of general interest, rather than concentrating on strictly linguistic nit-picking. Some nit-picking, however, there will have to be.

a. Linguistic wars

The grammarians who disagreed with Chomsky did so for a variety of reasons. Accordingly, the "linguistic wars" of the 1970s and early 1980s involved many different skirmishes (R. A. Harris 1993). One linguist in the mid-1980s identified forty-two competing grammars by name, ominously adding "among others" at the end of the list (Yngve 1986: 108).

Broadly speaking, the non-Chomskyan combatants fell into two armies. On the one hand, there were people who, perhaps even before Chomsky's first publications, took a fundamentally different approach to syntax. They might be spurred to greater clarity by his highly formal example. But they didn't accept his theory—or even his goal.

The London School, for example, didn't aim at a generative grammar from which all possible sentences could be derived (Sampson 1980: 212–35). Founded in the 1920s by J. R. Firth (1890–1960), these linguists focused on the functional properties of language as communication.

Where grammar was concerned, they asked how syntactically different types of sentence relate to different communicative contexts. They were much less interested in (Chomskyan) questions concerning word-by-word derivations of individual sentences, or decisions about whether a particular word string is grammatical. Like the Port-Royalists, they saw grammar as a communicative system that happens to be formalizable. Chomsky, by contrast, saw grammar as a formal system that happens to be used for communication. (His prioritizing of formal syntax was criticized also by philosophers working on speech acts, whose research dealt with issues—of meaning and speaker's intention—he left untouched; Searle 1972.)

Michael Halliday (1925–) was one of the London group. A specialist in Chinese, he had an early interest in machine translation, and was associated with Richard Richens and the Cambridge Language Research Unit, or CLRU (see Preface, ii). But his main contribution to linguistics was "systemic" grammar—later used in one of the first impressive automatic parsers (see Section xi.b).

Halliday classified sentences into a limited set of syntactic types. The speaker has to choose one of these, in deciding how to communicate what they want to say. Each type was defined in terms of the syntactic "features" included in it, and each feature was chosen from a feature set of mutually exclusive alternatives. The features were largely interdependent (hence the name, systemic grammar): choice of a feature not only ruled out its immediate alternatives, but also constrained the allowable feature choices from other feature sets.

Whereas the speaker must make the choices, the hearer uses them to aid interpretation. (An example of how syntax can help understanding is discussed in Section xi.c.) And both speaker and hearer are guided by context: if one syntactic type is more appropriate in context than another, then it will (or should) be chosen. Some linguists applied Halliday's approach to practical problems of language teaching, rhetoric, and literary criticism—hardly grist to the Chomskyan mill.

On the other hand, there were those who followed Chomsky onto his own ground and challenged him there. Unlike the London School, whose general approach pre-dated Chomsky, these linguists couldn't have done their work without him. They agreed that the goal of linguistics is to find some generative grammar—but not necessarily his.

Many merely quibbled about this transformation or that one. Someone could do that, yet still be playing on Chomsky's team. Others risked the charge of heresy by offering a very different transformational grammar.

But some went much further. They excommunicated themselves, by rejecting transformations altogether. And some of these, in turn, criticized Chomskyans' mathematical claims—and even Chomsky's formal skills—in uncompromising, often contemptuous, terms.

b. Who needs transformations?

Chomsky had argued that context-free phrase-structure (CFPS) grammars can't generate natural language, so must be supplemented by transformations. This strategy was theoretically inelegant, especially when—as in the early versions of Chomsky's linguistics—a long list of transformations was involved. But, for a quarter of a century, it was widely assumed to be unavoidable.

Some murmurings, however, were afoot. Even before *Aspects*, Harman had outlined a generative grammar without transformations (see Section vii.c). And the suspicious fact—first pointed out in 1961 (see Section vi.d)—that the class of transformational grammars is unconstrained, being equivalent to a Turing machine, was clarified a decade later by the Stanford logician Stanley Peters (1941–)—Peters and Ritchie (1973). Chomsky himself started to rely less on transformations (see Section viii.b). Eventually, then, some theorists stopped asking which transformations are needed, and asked instead whether they're needed at all.

One person who posed this question was Joan Bresnan (1945–), initially a student of Chomsky's at MIT. Bresnan's linguistics was clearly an example of cognitive science, and was highly influential at Stanford's (early 1980s) interdisciplinary Center for the Study of Language and Information (CSLI). Together with the computational linguist Ronald Kaplan (1946–) she developed "lexical-functional grammar", or LFG—so named because it stressed lexical rules for word morphology, and functional rules governing relationships such as *agent* and *passive* (Bresnan 1978; Bresnan and Kaplan 1982).

These two types of rule replaced some of the transformations defined in *Aspects*. Bresnan still felt some structural transformations to be necessary, and sometimes described her grammar as "transformational" accordingly (Bresnan 1978: 36–40). But she conceptualized them procedurally as online operations on a single string, not as changes in one (deep) string to turn it into another (surface) one. That is, transformations in the literal sense had disappeared.

This was no accident, for the main motivation underlying LFG was to avoid altering already built structural descriptions during the derivation or parsing of a sentence. Bresnan and Kaplan saw this strategy as computationally clumsy and psychologically implausible. Instead of (sequentially ordered) transformations, LFG provided several levels of description that apply simultaneously. In generating a sentence structure, each partial description could only add to, not cause changes in, the partial descriptions already accepted on other levels.

Bresnan took Chomsky's psychologism even further than he had done. In accordance with his structuralist training, Chomsky had developed his grammar first and only then suggested that it had psychological implications. Moreover, when psycholinguistic evidence went against his theory, he retreated into grammar as such: "[my] generative grammar does not, in itself, prescribe the character or functioning of a perceptual model or a model of speech-production" (Chomsky 1965: 9). But LFG was conceptualized from the beginning as the ground of detailed psycholinguistic hypotheses. As Bresnan put it:

> [A realistic grammar requires that we] define for it explicit realization mappings to psychological models of language use. These realizations should map distinct grammatical rules and units into distinct processing operations and informational units in such a way that different rule types of the grammar are associated with different processing functions. (Bresnan 1978: 3)

A realistic grammar, she also said, should be tested by computational methods, as well as by psychological experiments. A prime reason for her collaboration with Kaplan, who also saw grammars as "mental representations of language", was that he'd written computer programs—using ATNs (see Section xi.b)—intended to simulate what goes on in people's minds when they understand a sentence (R. M. Kaplan 1972, 1975). Similar models, and a related programme of psycholinguistic studies, were developed by two of Kaplan's co-researchers—who showed, for instance, how relative clauses could be parsed without backtracking (Wanner and Maratsos 1978; Boden 1988: 100 ff.).

Some years later, the computational linguist Martin Kay (a one-time member of CLRU: see Preface, ii) located Bresnan's work—and that of others discussed below—within the overall space of possible human parsing strategies (M. Kay 1980). He argued that the psycholinguist needn't be strictly constrained by considerations of computational elegance, still less by a devotion to step-by-step (sequential) algorithms. There are many reasons for thinking that people process language in parallel, in the sense that a number of different parsing strategies are at work at the same time, and are somehow integrated in generating the completed parse tree. And there are reasons for thinking that both top-down and bottom-up methods may be involved.

Each point in Kay's abstract space defined an *algorithm schema*, identifying a function to be computed but not saying just how this should be done. Actual algorithms could be generated in a principled way by combining a schema with an *agenda* listing various psychological strategies. These could direct both depth-first and breadth-first search (see Chapter 10.i.d). In addition, Kay discussed in general terms how parsers (using *charts*) could be constructed to avoid unnecessary duplication of effort, such as repeating prior computations during backtracking. When computer parsers proliferated in the 1980s and 1990s, this seminal attempt to see the wood as well as the trees was to be highly influential.

c. Montagovian meanings

Whereas Bresnan had insisted that grammar and psychology should be very closely related, some others who challenged Chomsky's account of language made a point of avoiding that claim. Richard Montague (1930–71) at UCLA was an important example (Dowty *et al.* 1981).

Montague was a logician and philosopher of language, working in the strongly anti-psychologist tradition of Frege (see Chapter 2.ix.b), and inspired by Carnap rather than Chomsky. His work wasn't an instance of cognitive science. He had no interest in the mind or mental processes, nor in computational linguistics. And his theoretical concepts were drawn not from computer science, but from mathematical logic. However, his approach appealed to some linguists whose views were highly relevant to cognitive science, as we'll see. In addition, it was taken up by a highly influential psychologist working on language and reasoning (Chapter 7.iv.e).

His influential paper on "universal grammar" had nothing to say about the relation between language and mind, still less innate ideas (Montague 1970). By invoking a universal grammar he meant, rather, that all languages, both natural and artificial, should be described in the same way. In this belief, he was echoing Carnap (see Section v.a). What was novel about his approach, and what commended it to some linguists, was the unusually close theoretical relation between syntax and semantics.

His system was called "intensional logic", because it dealt with meanings as well as logical formulae. It contradicted Chomskyan assumptions about the autonomy of syntax. It also rejected all semantic theories that merely offered translation into a more basic language—whether this was some form of logic, or a language of semantic primitives. For Montague, to translate one language into another isn't to address the central problem. The crucial questions of semantics concern the relation between linguistic expressions (of whatever kind) and entities in the world.

Logicians had long defined semantics in terms of truth conditions: what would have to be the case in the world for the expression to be true. And they had long assumed compositionality, so that the truth conditions of a complex expression depend recursively on those of its component parts. So far, so good—but just how were truth conditions to be conceptualized?

The logical positivists had answered this question in terms of (pure) observation protocols, but by mid-century it had become clear that this approach had intractable problems. In the 1960s (though not published until after his death in 1970), Montague developed a more abstract answer, and applied it to various types of sentence in English (Montague 1970, 1973).

Criticizing Chomskyan grammar as mathematically imprecise and unsystematic, he went back to Russell's set theory. With this as his ground, he defined "the set of all possible worlds" in terms of an abstract ontology of individuals, properties, *n*-ary relations etc. The semantics of a linguistic expression was then defined as its denotation (its model) in some possible world. That model was found by applying recursive mapping rules, such that each syntactic form (NP, VP, Det, etc.) had a corresponding semantic form. A proper name, for instance, always maps onto some individual.

This 'rule-for-rule' approach coupled syntax and semantics much more closely, precisely, and systematically than Chomsky and his followers had done. The theory of semantic markers, for example, had posited "primitives" such as *Animate* and *Human* (see Section viii.c). But to capitalize a familiar term of English isn't to display its meaning. Moreover, it wasn't clear just how to interpret the bracketed lists of semantic primitives offered by such theories: their syntax didn't establish their meaning. They were understood intuitively, much as the higher-level words (such as *bachelor*) were. In Montague's hands, by contrast, some specific—albeit highly abstract—meaning was definitively implied by the syntax.

Montague's theory was mathematically elegant, but couldn't solve most of the problems about meaning that had long plagued linguistics. He held, for example, that if a rule-for-rule mapping of a given English sentence succeeds in all possible worlds, it is necessarily true—and semantically equivalent to every other such sentence. Yet the component words of these distinct sentences may arouse very different associations in the hearer's mind. If, with Humboldt (or even Bloomfield), one regards these as part of the "meaning" of language, then Montague's theory is inadequate.

Again, semantic relationships such as synonymy, implication, sub/superordinacy, and consistency/inconsistency were rigorously defined by Montague. Synonymy, for instance, requires that both expressions be true in the same set of possible worlds. However, this abstract definition may not help us to decide whether two actual words are synonymous. (Montague agreed with Humboldt that accurate translation from one natural language to another is impossible—not because of different mental associations, but because it's unlikely that any two sentences in the different languages will share the same mapping in model theory.) Nor did his approach help in defining *cousin*: all the problems noted in Section viii.c, including the open texture of natural language, remained.

The new semantics offered clear definitions for everyday words such as *the*, *every*, *some*, *all*, *any*, *never*, *ever*, all of which are notoriously difficult to define in dictionary style. They were interpreted by Montague as set-theoretic expressions defined over possible worlds, whose logical–mathematical relations could be precisely stated.

However, most of the long-standing problems about assimilating natural language to logic had been brushed aside, rather than solved (see Chapter 4.iii.c). As for sentences about propositional attitudes, these were as intractable as ever (Dowty *et al.* 1981: 170–5). No semantic mapping for *Mary believed that the bracelet had been lost* could be completed, because the bracelet's fate still had to be left open (see Section v.a).

Model-theoretic semantics was a further step along the road of formalism pioneered by Frege, Russell, and the early Wittgenstein. This road had been followed also by Carnap and Reichenbach. But whereas in the 1930s–1940s few linguists had paid much attention to Carnap's work on language, by the 1970s linguistics itself was largely formalized, and had been strongly linked (by Chomsky) with the analysis of artificial languages. In other words, the ground had been prepared for Montague.

Even so, persuading linguists to read Montague took some time and effort. His work was so fiercely abstract, and so tersely written, as to border on unintelligibility. Most linguists tackled it by way of some introductory account, preferably one relating set theory (more familiar to philosophers of language than to linguists) to transformational grammar. Barbara Partee (1975) was especially influential in introducing Montague's ideas to other linguists. But even four years later, professional journals were still prepared to publish explanatory notes on Montague rather than original research, defending their decision by reference to his forbidding writing style (Halvorsen and Ladusaw 1979: 185).

After the mid-1970s, when (thanks to Partee) linguists in general became aware of him, Montague's work soon came to dominate theoretical semantics, and various theories of syntax too. (Ironically, Chomsky himself now puts much less stress on formalization than he used to, and even queries the relevance of "most of the results of mathematical linguistics"—Lyons 1991: 199.)

Psycholinguists, however, virtually ignored Montague. (Although some would take up his theories in the early 1980s: see 7.iv.e.) Being ignored by the psycholinguists wouldn't have bothered him at all. As remarked above, his theory was a purely logical–ontological one, making no claims about psychology.

d. Transformations trounced

Among the linguists inspired by Montague in the mid-1970s was Gerald Gazdar (1950–), of the University of Sussex. Together with Ewan Klein, Geoffrey Pullum, and Ivan Sag, Gazdar developed a new syntactic theory: generalized phrase-structure grammar, or GPSG (Gazdar 1979*a*, 1981*a*, 1982; Gazdar *et al.* 1985). GPSG adopted Montague's model-theoretic approach, and applied it to a much wider range of English syntax than he had done.

Like Montague, Gazdar treated language as a purely formal system, not as a psychological phenomenon. Nevertheless, he was inspired also by the psychologically oriented Bresnan. What drew him to Bresnan's work wasn't her use of grammar to relate language and mind, but her rejection of syntactic transformations.

Another formative influence was Peters—who in 1973 had expanded Putnam's insight about the Turing equivalence of transformations in general (see above). Gazdar took this mathematical discussion still further. Besides co-developing GPSG, he directly challenged some of Chomsky's formal claims about the power of various types of grammar. In so doing, he implicitly cast doubt on arguments that had influenced cognitive science in general (see Section ix.e–f).

In formulating GPSG, the prime aim of Gazdar and his associates was to develop a single-level (non-transformational) tree-based grammar that would describe English elegantly. That task could be, and often is, approached without any concern for the abstract mathematical properties of language. But two of the authors (Gazdar and Pullum) had already claimed that the widely accepted arguments against CFPS grammars were either formally invalid or based on false empirical premisses. Indeed, their work had revived interest in this long-neglected question, and led to some improved arguments for the opposing view (G. Gazdar, personal communication).

Partly because of this interest (which not all syntacticians shared), and partly as a matter of methodological discipline, the GPSG group took pains to prove that their new account of syntax was mathematically equivalent to a CFPS grammar.

The GPSG authors were careful not to claim that GPSG can describe every natural language, although they pointed out that it seemed to fit about twenty diverse tongues (Gazdar *et al.* 1985: 15). But they did assume that it could describe the whole of English.

Soon afterwards, it became clear that this was almost true, but not quite (see below). Nevertheless, it was evident by the early 1980s that GPSG could successfully describe a host of English sentences which Chomsky had said require appeal to transformations. Moreover, no extra rules had to be added to link each syntactic form with its semantic interpretation. These relations were transparent, by virtue of a model-theoretic account of the mapping between syntax and semantics.

GPSG captured the grammatical intuition noted in Section vi.d, that *The man reads the book*, *The book is read by the man*, and so on, are closely related. Nevertheless, it considered only the surface string, so didn't need transformations (which were introduced to derive surface-level from deep-level structures). It specified only a set of phrase-structure rules. GPSG was more flexible than the phrase-structure grammars considered by Chomsky, because in addition to the familiar syntactic categories (S, NP, VP, Det, and so on) it contained three other formal devices: slash-categories, metarules, and linking rules.

A slash-category marks a constituent with a syntactic gap: so S/NP, for instance, represents a sentence with a noun phrase missing. As this example shows, the nature of the gap was defined in terms of the basic (and familiar) phrase-structure rules. Given slash-categories, the problem arose of defining how the grammar should deal with them. Consider S/NP: this must denote either *an NP-missing-an-NP followed by a VP*, or *an NP followed by a VP-missing-an-NP*. These are the only possible interpretations, given that the basic grammar prescribes that S $\rightarrow$ NP + VP. Other slash-categories may allow more than two possibilities.

One might have expected, therefore, that GPSG would include an inelegant profusion of extra rules, several for each slash-category. Not so: instead, Gazdar's grammar contained a small number of metarules. Each of these was a general rule schema, from which (together with the basic rules) the new rules required for a given slash-category could be derived. Finally, the (more numerous) schematic linking rules dealt with the housekeeping: they defined how to introduce and delete slash-categories, and enabled the gap to be carried down to lower levels of the parsing tree as necessary.

This new grammar not only dealt with all the sentences generated by transformational grammar, but also explained some things simply which Chomsky could explain only in tortuous ways. For instance, consider these two sentences: (1) I think *Fido destroyed the kennel*, and (2) The kennel, I think *Fido destroyed*. Most linguists (including Chomsky) had assumed that the two italicized expressions fall into the same syntactic category—namely, a (subordinated) sentence. But because the second expression isn't a complete sentence (it cries out for a noun phrase telling us what Fido destroyed), they had to give a complicated account of (2) in order to show this. GPSG, by explicitly labelling the gap, placed the italicized strings in different syntactic categories. But it enabled one to give equally simple descriptions of how to generate the two whole sentences (Gazdar 1981*a*).

GPSG had a significant influence on linguistics, being taken up by a number of syntacticians in the early 1980s. Clearly, it could achieve a great deal—but how much? It seemed to encompass English, and at least twenty other languages too. But there were growing doubts. It was already known that some European languages contain phrase-structure trees lying outside GPSG, and in 1984 two further languages—one from West Africa, another a Swiss dialect of German—were found to contain recalcitrant sentences (Gazdar *et al.* 1985:16; Gazdar 1998). Indeed, English itself, as Klein had discovered, contains some sentences—involving nested comparatives—that couldn't be described by GPSG.

Accordingly, Sag went on to develop an improved, but no longer context-free, version: "head-driven" phrase-structure grammar, or HPSG (Pollard and Sag 1994). And Gazdar showed that both GPSG and HPSG were special cases of "indexed" grammar. This is a general type located just above context-free grammar in the abstract Chomskyan hierarchy, but still appreciably below context-sensitive grammar (Hopcroft and Ullman 1979: 389–92).

Even within linguistics, however, GPSG didn't attract attention as widely as it deserved. This was largely due to the sociological factors cited in Section viii.a—including the *not invented here* syndrome that still pervades MIT. Stanford, to be sure, was open to Gazdar's theory—but CSLI didn't rule the world.

Today, much of the best work on non-Chomskyan syntax (including LFG, GPSG, and HPSG) is done in departments of computational linguistics, not of linguistics as such. This is only partly because the theorists concerned are interested in computer modelling. More to the point, they are "locked out" of the many departments of linguistics still dominated by the Chomskyan paradigm. One indication of this dominance is the fact that the term "generative grammar" is sometimes used as if it referred only to Chomsky's work—the current version of which is arguably not generative at all (Gazdar *et al*. 1985: 6).

e. Why GPSG matters

In cognitive science *as a whole*, GPSG was even less well known than in linguistics. In a sense, it was its own worst enemy. For it was yet more fiercely abstract than Chomsky's early publications had been. Non-specialists would need plenty of black coffee.

Moreover, the GPSG linguists, unlike Chomsky, had no desire to discuss broad issues of "language and mind". And, unlike Bresnan and Kaplan, they had no interest in psycholinguistics. They drew no psychological morals from their grammar, and no psychological discovery could either support or falsify their (purely formal) work. Some psycholinguists, including Janet Fodor and Rosemary Stevenson, did experimental research inspired by GPSG (G. Gazdar, personal communication). But because of the GPSG group's explicitly anti-mentalist stance (and their limited visibility in professional linguistics), most psychologists were unaware of their work.

Nonetheless, it was—and is—potentially important for cognitive science in five different ways. (Four are discussed in this subsection, and the fifth in the next.)

First, GPSG questioned Chomsky's fundamental critique of CFPS grammars. He had admitted that he couldn't strictly prove that English lies beyond any conceivable phrase-structure grammar. Now, his claim (his hunch) that it could be so analysed, if at all, "only clumsily" had been challenged. GPSG, which covers most if not quite all English sentences, is a formally elegant theory, not a clumsy one. It didn't deserve Chomsky's contemptuous accusation that it was "simply a needlessly complex variant" of a transformational theory, supplemented by "a class of superfluous devices" (Gazdar 1981*a*: 280–1).

Moreover, if natural languages can indeed be described by some context-free phrase-structure grammar (a CFPSG, for short), then a huge mathematical literature becomes potentially relevant. This literature was developed (partly as a result of Chomsky's papers of the early 1950s) in theoretical computer science, to deal with the parsability and learnability of programming languages.

Compiler programs, for instance, have to parse context-free programming languages in order to translate them into machine code. This is "perhaps the most researched and best understood area of formal language theory" (Gazdar 1981*a*: 275). In other words, and as Montague had already shown, the mathematical analysis of language—and, significantly, the comparison of natural and artificial languages—can be taken much further than Chomsky took it.

Third (a corollary of the previous point), the standards of rigour set by Harris and raised by Chomsky had been raised further still. Indeed, Chomsky's own standards had already been criticized by people outside the GPSG group. The Stanford-based logician Montague didn't even deign to criticize them, but contemptuously ignored "the attempts emanating from the Massachusetts Institute of Technology" (Montague 1973: 247). He was presumably offended by (among other things) Chomsky's basing his core claim about the necessity of transformations not on a proof, but on a hunch.

Others agreed, and accused Chomskyans in general of not having done their mathematical homework. As remarked in Section vi.a, the technical evidence behind *Syntactic Structures* wasn't published until 1975. Meanwhile, the famous mimeograph ("three thick volumes of faint purple typewriting with handwritten annotations") had probably been read only by a handful of those who had bothered to get hold of it (Sampson 1979*c*: 356).

One reviewer of the published version confessed that he himself had for years taken on trust Chomsky's hints that the theoretical lacunae in *Syntactic Structures* had already been filled in by the work presented in those purple pages. He confessed also that when he did finally get hold of a copy (while waiting to receive the published book for review), he "rapidly laid it aside" because of its near-illegibility. And he acidly commented:

> The publication of this book [in 1975] symbolizes a remarkable situation in the recent history of linguistics, a situation which can hardly have many parallels in other disciplines . . . [For the best part of two decades], during the heyday of Transformational Grammar, its hundreds of thousands of advocates in the universities of the world not only didn't but *couldn't* really know what they were talking about. (Sampson 1979*c*: 355–6)

Part of the explanation for this unhealthy state of affairs was that most linguists couldn't have understood Chomsky's manuscript even if they had read it. Realizing this, they'd been content to leave its arcane mathematics to the "high priests" at MIT, assuming that it answered all their theoretical questions. But they were mistaken. Those with mathematical eyes to see would find, when they did actually read it, that Chomsky's argument was often "maladroit" and "perverse", and sometimes "just plain wrong" (Sampson 1979*c*: 366).

(I must confess that I can't justify this claim at first hand, for I *don't* have mathematical eyes to see. Several specific examples have been pointed out to me, but I have to take them on trust. However, I'm a colleague of two of Chomsky's major 'mathematical' critics, Gazdar and Sampson, and I've pressed both of them hard on whether they really mean what they say in the passages I've quoted. They do.)

Many of Chomsky's mistakes, according to Sampson (1979*c*), were "elementary", and some were crucial. For example, he'd repeatedly claimed that the syntactic phenomenon of "Affix Hopping" can be stated as a single transformation (though he never showed just how to do so). That claim had persuaded many linguists into his camp. Yet it now appeared, Sampson complained, that one of his own formal definitions rendered it impossible. In other words, Affix Hopping had arguably been "a standing refutation of [his] theory throughout its history". Yet Chomsky had never discussed this problem (Sampson 1979*c*: 365).

In light of such points, the professional adulation of Chomsky was—or anyway, should have been—distinctly embarrassing. To be sure, most linguists weren't mathematical animals. But their misplaced admiration for Chomsky's formal work was due to more than mere ignorance. It was encouraged by sociological factors: his by-then-established grip on departments of linguistics, and his towering status as a worldwide guru.

Similarly negative judgements of Chomsky's work and influence have been expressed for twenty years by Gazdar and his colleagues. Indeed, it was Pullum who recently dismissed Chomsky's theory of minimalism as "risible" (see Section viii.b). Admittedly, Gazdar had earlier declared—and still does (personal communication)—that *if* any work in syntactics is to be called "foundational", as Frege's work was foundational for mathematical logic, then it is Chomsky's—especially his hierarchy of grammars (Gazdar 1979*b*: 197). But that's not to say that he saw all of Chomsky's "proofs" (in Chomsky 1975) as equalling Frege's in respect of precision and accuracy. To the contrary, he didn't.

Commenting on Chomskyan linguists in general, Gazdar's group pointed out that "mathematically respectable" proofs of general claims about grammars are very difficult to obtain. For instance, new versions of transformational grammar had been proposed (by Chomsky and others) which were held to be formally equivalent to, though more elegant than, some previous version; and linguists often claimed that such-and-such a structure cannot be generated by this or that grammar. But proving it was a different matter. Since most Chomskyans didn't even try to do so, the GPSG theorists were less than impressed:

> Such claims [about the formal equivalence and computational power of different grammars] are made in many contemporary linguistic works without there being any known way of demonstrating their truth. In some works we even find purported "theorems" being stated without any proof being suggested, or theorems that are given "proofs" that involve no definition of the underlying class of grammars and are thus empty. (Gazdar *et al.* 1985: 14)

Rigour was necessary, they said, not just for clarity and a good mathematical conscience, but for providing explanation. Thus any proposed "universal" should be a necessary consequence of some formal definition of the class of natural grammars. Otherwise, it is (at best) a description of some feature found, as a matter of fact, to be universal—without any understanding of why this is so (pp. 2 ff.). This point was similar to Marr's claim that we can explain visual processing only if we can relate it to (ideally, derive it from) the abstract constraints on vision as such: Chapter 7.iii.b.

A fourth point of interest for cognitive science as a whole is that GPSG implied that Chomsky's nativist 'argument from ignorance' against behaviourism must be reconsidered. Even if the simple learning theories popular at mid-century are inadequate to explain language acquisition, because they cannot parse phrase structure, some more sophisticated—but still purely surface-level—version might succeed. In other words, GPSG might be learnable (without the need for any dedicated acquisition device), even if transformational grammar isn't.

To be sure, the framework for this more powerful type of 'induction' might be innate: even Quine had allowed that "unquestionably much additional innate structure is needed . . . to account for language learning" (see Section vii.c). Even so, and although linguistic nativism hadn't strictly been disproved by Gazdar's work, the empirical evidence demanded years before by Black had become even more necessary.

One reason why GPSG didn't spread widely throughout cognitive science was that evidence for some sort of language-specific nativism was by then more abundant. Besides the developmental studies mentioned in Section vii.c, it included analysis of the many detailed features of human anatomy and physiology seemingly evolved specifically (*sic*) for language (Lenneberg 1967), and the apparent impossibility of teaching even simple grammatical structure to non-human primates (Chapter 7.vi.c).

In addition, neuroscientific evidence for *visual* nativism had—seemingly—been accumulating since the 'Frog's Eye' discoveries in the late 1950s (Chapter 14.iv & vi). And the new popularity of modular theories of mind (see above) discouraged reliance on general learning mechanisms to explain language acquisition.

That's not to say that nativism had been conclusively proved, for it hadn't. Despite the best efforts of Chomskyans to marshal their evidence (Pinker 1994), the facts didn't—and still don't—show that language, even linguistic universals such as coordination and hierarchical structure, can't be learned and/or evolved (Sampson 1997, esp. chs. 3 and 4). Furthermore, even if the nativists were right, Black's point still stood: the nature of the innate dispositions supposedly concerned, and of their neurophysiological implementation, were—and still are—unclear.

Indeed, the notion of nativism is itself unclear. Recent research in both psychology and neuroscience has shown how over-simple it is to label behavioural or anatomical features as 'innate' (see Chapters 7.vi.g and 12.viii.c).

The fifth important aspect of GPSG is perhaps the most significant of all. It has deep implications for both psychology and software technology, and especially for NLP.

f. Computational tractability

Because of its equivalence to a CFPS grammar, GPSG is less computationally demanding than transformational grammar. And that means, in a nutshell, that more can be achieved by relatively simple systems—whether minds or machines—than Chomsky believed.

With respect to psychology, GPSG suggested (for instance) an explanation of why people can parse natural language so rapidly (Gazdar 1981*a*: 276). And it lessened the difficulty of seeing language as the product of evolution, since context-free parsing algorithms are less evolutionarily improbable than transformational ones. More generally, if natural languages are indeed CFPS structures, then psycholinguists don't need—so shouldn't propose—psychological theories with the power of a context-sensitive grammar.

This is a special case of Lloyd Morgan's canon, framed in the late nineteenth century to promote intellectual hygiene in comparative psychology (see Chapter 2.x.b). Conwy Lloyd Morgan had forbidden explanation in terms of "a higher psychical faculty" if the behaviour in question could also be explained in terms of "one which stands lower in the psychological scale" (Lloyd Morgan 1894: 53).

For sixty years, this hugely influential advice could be followed only by relying on some intuitive sense of what counts as a "higher" psychical faculty. But Chomsky's hierarchical classification of grammars had, in effect, defined "higher" in terms of distinct levels of computational power, requiring processing mechanisms of specific types—with or without push-down, for example. (This enabled 'extra' levels to be inserted in the hierarchy: indexed grammar, for instance, lying above CFPSGs but still significantly below context-sensitive phrase-structure grammars, or CSPSGs.)

In short, the linguistic hypothesis that natural languages can be described by CFPSGs implied a psychological hypothesis about the minimum computational power required by the human mind/brain. Humboldt's "creative spirit of humanity" would have to rise as far as CFPSGs to generate language, but—if the GPSG theory was correct—need rise no further.

As for the software implications of GPSG, these were quickly recognized and highly influential. By the early 1980s, this grammar had already been adopted in various NLP projects in Europe, Japan, and the USA.

One of these was the "Alvey Tools Grammar", developed for the UK's Alvey Project (11.v.c) and still in use in various places today. And Sag's HPSG, which shares many features with GPSG although it isn't context-free (it permits unbounded recursion in certain circumstances), is now probably the most widely employed grammar formalism in NLP, especially in Europe (G. Gazdar, personal communication). (Very recently, Gazdar (1998) has defined a class of grammars, which includes GPSG, that is mathematically equivalent to a type—known as "linear tree adjoining grammars"—now gaining popularity in NLP circles.)

These NLP projects of the 1980s fell clearly into "technological" rather than "psychological" AI (see Chapter 1.ii.b). GPSG's psychological plausibility was irrelevant. So was the 'pure' linguist's question whether it could describe *every* syntactic construction in *every* human language. From the point of view of NLP research, that question is comparable to theological disputes about angels dancing on pinheads. What was important about GPSG, besides its success in describing almost the entire range of English (and some other languages), was its computational tractability.

Even devotees of GPSG, however, don't deny Chomsky's seminal importance. His work, besides affecting computer science as such, profoundly influenced NLP. Or rather, his classification of different types of grammar did so. His ever-changing syntactic theories have had less influence here. Partly because of his own lack of interest in, even hostility to, computer modelling, very few Chomskyan programs have been written (but see Chapter 7.ii.b). And, for reasons remarked above, many computational linguists are explicitly anti-Chomskyan. However, to be anti-Chomskyan isn't to be uninfluenced by Chomsky: LFG, GPSG, and HPSG were all developed in reaction to his pioneering efforts.

Moreover, Chomsky's formal–generative approach to linguistics in the late 1950s (and his linking of different grammars to different automata) buoyed the spirits of people already doing computer modelling of language, and encouraged others to start trying. (Section x will recount the early development of this field.)

g. Linguistics eclipsed

The last item within the tenfold Chomsky myth (i.a, above) was that linguistics is as prominent in cognitive science today as it was in the 1950s–1960s. That's not true. Ironically, its eclipse was largely due to Chomsky himself.

Part of the problem was his continual recasting of his own theory of syntax, described above. But that alone wouldn't have sufficed: if the price of doing good cognitive science had been keeping up with Chomsky, then that price would have had to be paid. More important than his inconvenient changes of mind was his mid-1960s distinction between *competence* and *performance* (Chapter 7.iii.a). This put a theoretical firewall between the psychology of language and linguistics as such. The firewall prevented commerce in both directions. Linguists, in effect, had been instructed to take no heed of psychology. But by the same token, psychologists were tempted to regard abstract linguistics as largely irrelevant to their concerns.

Researchers in non-linguistic disciplines didn't stop working on language, of course. But they stopped looking to Chomskyan theory for their key inspiration. As for the people who weren't working on language, they no longer gave linguistics a second thought. The general lessons (about formalization and generativity) that Chomsky had taught had been well learnt, and were no longer thought of, at least by the youngsters entering the field, as having any special connection with linguistics. By the 1990s, the discipline was "arguably far on the periphery of the action in cognitive science" (Jackendoff 2003: 651).

Recently, things have started to change. For instance, Pinker has put linguistics in the context of cognitive psychology, and in particular of evolutionary psychology (Pinker 1994, 1997, 2002). Moreover, the late 1990s saw a new field being named: cognitive linguistics. Its adherents are *linguists*, but linguists who take special note of the cognitive processes involved in using language (see 7.ii, preamble, and 12.x.g).

Similarly, the Brandeis (now Tufts) linguist Ray Jackendoff (1945–) has recently tried to integrate linguistics with psychology, neuroscience, and evolutionary theory (Jackendoff 2002, 2003). An ex-pupil of Chomsky, Jackendoff is still wedded to the idea of a universal grammar. But he claims to have shown (what Chomsky questioned) that it's possible for this to have evolved *incrementally*. He also argues that semantics doesn't need to be "paralysed" by a failure to solve the philosophical problem of intentionality (2002: 280). And he couches his argument in computational terms, describing a parallelist architecture (blackboard-based: see 10.v.e, and xi.g below) in which the rules of grammar are directly involved in language processing.

Jackendoff's book is described as a "masterpiece" by one *BBS* commentator, namely the philosopher Daniel Dennett (2003*b*: 673). Others disagree. For instance, another philosopher (F. Adams 2003) complains of its cavalier attitude to semantics and intentionality (although Dennett defends Jackendoff on this point—2003*b*: 673–4). And a neuroscientist objects to its assumption that the brain has evolved specialized components of language, as opposed to a small number of general capacities (he names seven) that make it possible to discover such "components" over the course of many millennia and/or to learn them within a few years (Arbib 2003: 668). Right or wrong, however, Jackendoff's theory is a far cry from Chomsky's chastely firewalled linguistics.

Or perhaps not? Employing the revolutionary analogy cited in Section i.b, Edelman has described Jackendoff's book as

> one of several recent manifestations in linguistics of the Prague Spring of 1968, when calls for putting a human face on Soviet-style "socialism" began to be heard (cf. the longing for "linguistics with a human face" expressed by Werth 1999: 18). (S. Edelman 2003: 675)

Edelman's prognosis for Chomskyan theory isn't hopeful:

> In a totalitarian political system, this may only work if the prime mover behind the change is at the very top of the power pyramid: Czechoslovakia's Dubcek in 1968 merely brought the Russian tanks to the streets of Prague, whereas Russia's Gorbachev in 1987 succeeded in dismantling the tyranny that had sent in the tanks. In generative linguistics, it may be too late for any further attempts to change the system from within, seeing that previous rounds of management-initiated reforms did little more than lead the field in circles. . . . *If so, transformational generative grammar, whose foundations Jackendoff ventures to repair, may have to follow the fate of the Communist bloc to clear the way for real progress in understanding language and the brain.* (S. Edelman 2003: 676; italics added)

To be sure, Edelman isn't a disinterested critic. For he argues, contra the Chomskyans, that the inductive learning of hierarchical structure is in principle possible. He and his colleagues have devised a pioneering "ADIOS" algorithm (Automatic DIstillation Of Structure), which can induce both context-dependent and context-free grammars (Solan *et al.* forthcoming). It uses a combination of statistics and structured generalization to extract hierarchical regularities from input sequences without knowing what patterns to expect, and without being given clues as labels on sequence parts (e.g. part-of-speech tags on the input words). Its only presupposition is that the sequence contains "partially overlapping strings at multiple levels of organization" (p. 3).

ADIOS has been applied to large sets of 'grammatical' sentences (e.g. the Bible and the *Wall Street Journal*), and also to the CHILDES corpus of spontaneous—and largely 'ungrammatical'—speech. (Those scare quotes are needed because Edelman rejects Chomsky's view that only strings fitting some set of abstract syntactic rules can be termed grammatical: language, for Edelman, is what people actually speak.) The system works even better with CHILDES than with the *WSJ* corpus, which Edelman (personal communication) explains in terms of Jeffrey Elman's stress on the importance of "starting small" (see Chapter 12.viii.c): whereas CHILDES contains many short word strings, the *WSJ* contains many longer ones.

In addition, ADIOS has coped well with six natural languages, including English, French, and Chinese. It has also been tested on artificial context-free languages, some having thousands of rules. And it has even been applied to DNA and protein sequences from several species. In each case, it has learnt to recognize the hierarchical patterns involved, and to reject anomalous sequences.

As its success with CHILDES illustrates, it has done this *despite* having ungrammatical strings in the input (which Chomsky saw as a huge barrier to inductive learning). Moreover, simulating the "creativity" of language, it has generated new (and legal) structures too—the first learning algorithm to do so. (The patterns it classifies in the learning phase are transformed into rewrite rules, which are then used for the generation phase.)

It doesn't follow that this is what infants do. Indeed, the authors point out that ADIOS has no access to conceptual knowledge, nor any "grounding" of the verbal input in current action and/or environment. (They also remark, however, that babies appear to use both statistics and rule-learning in acquiring language: Saffran *et al.* 1996; Seidenberg 1997; G. F. Marcus *et al.* 1999.) But ADIOS's success does dispatch arguments for the *necessity* of linguistic nativism.

In sum, linguistics may regain its place in cognitive science—if not as a *root* of theorizing in the other disciplines, at least as an equal, and collaborative, partner. If that happens, however, linguistics will be a very different enterprise from that engaged in by Chomsky.

9.x. The Genesis of Natural Language Processing

At mid-century, Harris wasn't the only person hoping to apply computer technology to linguistic problems. On both sides of the Atlantic, there were people "enchanted by the idea that [machine-based experiments to test linguistic hypotheses] could now be performed in the flesh rather than only in the spirit" (Bar-Hillel 1970: 293). And some were already doing so.

The late 1940s and early 1950s saw the first computer models for NLP. These attempted machine translation (MT), automatic classification, information retrieval (IR), text generation, and (for a while) the induction of grammars.

As it happened, these topics weren't the focus of the early research in any of the three pioneering AI labs at MIT, Stanford, and Carnegie Mellon (see 10.ii.a). Consequently, when people talk about the origins of AI they often ignore work in MT which pre-dated their foundation. By the same token, however, when MT suddenly fell into disfavour in the mid-1960s (a story told in Section x.e–f, below) AI as a whole—although it suffered—wasn't so badly affected as one might have expected.

a. Ploughman crooked ground plough plough

Hopes for automatic translation long pre-dated the computer age. It had been envisaged in some of the "universal language" projects of the seventeenth century (see Section iii.d, above). And the first (purely mechanical) MT patents had been taken out, in Russia and France, in 1933.

Whereas the French patent merely described a simple paper-tape dictionary that stored equivalent words in several languages, Petr Smirnov-Troyanskii's (1894–1950) machine could transform a grammatical analysis of a sentence in one language into its equivalent in another. Moreover, he claimed that it should also be possible to automate the (two-way) conversion between these analyses and actual sentences. But he wasn't able to demonstrate this, for lack of funds.

As late as 1944, the Institute of Automation and Telematics refused his bid for support; and his 1948 design for an electromechanical machine similar to the Harvard Mark I (see 3.v.b) was similarly ignored. (He died two years later, so had no chance to develop it.) This lack of appreciation was perhaps inevitable, for his ideas about MT were far ahead of his time:

> In retrospect, there seems to be no doubt that Troyanskii would have been the father of machine translation if the electronic digital calculator had been available and the necessary computer facilities had been ready. History, however, has reserved for Troyanskii the fate of being an unrecognised precursor; his proposal was neglected in Russia and his ideas had no direct influence on later developments; it's only in hindsight that his vision has been recognised. (W. J. Hutchins 1986: 24)

With the development of computers in the UK and USA, their use for various language-related purposes soon followed. In 1946, for instance, Andrew Booth (1918–) of Birkbeck College, London, suggested using computers for translation—and had designed and built a computer by 1947 (Lavington 1975: 5). In the late 1940s he started work with Richard Richens on what would become a lifelong project (Richens and Booth 1955; Richens 1958; Booth 1967, p. vi), and in the early 1950s he co-edited the first volume on MT (W. N. Locke and Booth 1955).

Richens, who was later attached to Margaret Masterman's Cambridge Language Research Unit (CLRU), has been credited with "the first serious contribution" to MT (Bar-Hillel 1959: 303; 1970: 303). His work (initially carried out by hand simulation using punched cards) was interesting not least because it considered grammar as well as vocabulary: the input verbs were split into stem and affix, so that *start*, *started*, and *starting* would have only one vocabulary entry, while -*ed* and -*ing* could be attached to indefinitely many different verbs. Booth and Richens's first experiments on MT were mentioned at a conference (on 'Science Abstracting') in Paris in June 1949, and featured briefly in the *Scientific American* six months later. But they became more widely known thanks to Warren Weaver's memo on MT, written after Booth's visit to the USA in 1947.

Booth's discussions in 1947 with the information theorist Warren Weaver, then Director of the Natural Sciences Division of the Rockefeller Foundation, indirectly influenced the initiation of MT research in America. MT programmes were set up at several US universities in the early 1950s, as a result of an influential memorandum written by Weaver two years after Booth's visit (Weaver 1949; W. J. Hutchins 1986: 24–30).

Financially, the burgeoning of NLP was largely driven by Cold War funding. This was especially true of machine translation. For military purposes, the prime languages were English and Russian, but in the USA they were soon joined by Vietnamese (Newquist 1994: 115).

The first public demonstration of an MT system, in January of 1954, translated forty-nine Russian sentences into English (using a 250-word vocabulary and six rules of grammar). Despite the six rules of grammar, this was basically a word-for-word translator. As such, it was hugely fallible. Nevertheless, it was used right up to the mid-1960s by the USA's Atomic Energy Commission (which was monitoring nuclear sites in Italy, as well as America), and by other governmental agencies too. In other words, it was *still* state-of-the-art: nothing better (for practical use) than word-for-word translation had been achieved.

That program, developed with CIA funding at Georgetown University, was one of many MT projects initiated in the USA in the 1950s. Even in the UK, much of the money for MT and other types of NLP came from various parts of the American defence programme. For instance, CLRU (founded in 1954) was partly supported by the US Office of Naval Research.

Predictably, NLP was developed in the Soviet Union too. Following the visit of a Russian computer scientist to the USA in 1954, for whom the Georgetown demonstration was proudly repeated, several research programmes were set up in the USSR (Hutchins 1986: 37, 133–45). By 1959, there were many hundreds of Soviet researchers in MT alone.

Intellectually, the field was driven by linguistics, by various logical and philosophical theories of language, and by the recent discussions of communication in cybernetics and information theory (see Chapter 4.v). Much early NLP research made a serious attempt to think about language systematically and/or to work towards practically useful ends.

But to say it was serious isn't to say it was successful. The well-known story—or rather, the group of mutually inconsistent stories—about "The spirit is willing but the flesh is weak" being rendered (after translation and retranslation) as "The whisky's fine, but the meat is rotten" is almost certainly apocryphal (W. J. Hutchins 1986: 16–17). However, there were certainly many such early-MT howlers. In the mid-1950s, for instance, Virgil's precisely inflected Latin sentence *agricola incurvo terram dimovit aratro* was once translated at CLRU as *ploughman crooked ground plough plough* (Masterman *et al*. 1957/1986). As remarked in the Preface, ii, this happened because CLRU used a thesaurus, not a dictionary.

With no computable theory of syntax—as opposed to a few simple programmed grammatical rules—available at the time, that's hardly amazing. (A fairly modest dose of syntax would have sufficed to deliver *A/The ploughman ploughs the ground with the crooked plough*.) And with no adequate theory of semantics either, it's not surprising that the CLRU's computational work on meaning didn't prevent this infelicity. Not until the early 1970s would there be powerful computational models of syntax, or of syntax allied with semantics (see xi.b, below).

b. Shannon's shadow

Shannon's theory of information was an important early influence on NLP. He himself used it in the late 1940s to produce "approximations" to English, as we saw in Section vi.c.

With hindsight, it's easy to see that these were word strings rather than sentences, because neither grammatical structure nor meaning were represented in information theory. At the time, this wasn't so obvious. Karl Lashley's (1951*a*) arguments against Markovian theories (see Chapter 5.iv.a) were unknown to most linguists and logicians, and Chomsky's more formal critique hadn't yet been published. Accordingly, many people were enthused by Shannon's ideas about the automatic generation of language.

Besides attempts to generate English text probabilistically, a number of people in the 1950s tried to induce grammars by statistical programs. In effect, they were aiming at the automatic discovery procedure sought by structuralism (see Section v). They focused on artificial rather than natural languages, not least because natural syntax was both complex and unclear.

Chomsky, for instance, cooperated with the psychologist George Miller (then at Stanford University, but soon at Harvard) in describing the formal properties of finite state grammars (Chomsky and Miller 1958). They also did experiments to find out what strategies people use, consciously or otherwise, to discover them (G. A. Miller 1967). Letter strings were labelled by the experimenters as grammatical or ungrammatical, and the subjects tried to learn to distinguish the two classes. If they could actually state the regularities involved, so much the better.

Their Cambridge colleague Bruner had investigated the information strategies used in concept formation (see Chapter 6.ii.b). Moreover, when Miller and Chomsky first described their experimental work (at the University of Michigan in 1957), they alerted the information scientist Ray Solomonoff, a participant at the Dartmouth meeting in the previous year, to define ways of inducing (artificial) grammars automatically (Solomonoff 1958, 1964). (These methods underlie the 1990s work on statistical grammar induction mentioned in Section v.d.)

That there should be linguists, psycholinguists, and information theorists at one and the same venue in 1957 was no accident. Communication between these groups had been growing throughout the 1950s. The Dartmouth meeting had caused an extra upsurge of interest, especially in AI models of learning and problem solving (see Chapter 6.iv.b).

But even earlier, in 1951, the information-theoretic approach to language was already predominant at Harvard and MIT—where Chomsky and his wife, Carol, respectively, were then working:

> Few linguists in Greater Boston at that time dared not use freely "message" and "code", "information" and "bits" in their shop talk, and nobody was "in" if he did not master, or at least professed to master, a good amount of probability theory and statistics. Everybody who was somebody in the field would sooner or later show up at MIT . . . All of us were enormously impressed by Shannon's well known [probability-based] experiments in what he called "approximations to ordinary English" and were convinced that speech, in English or any other language, was a Markov process. From this to the conviction that the English language, i.e. the set of all English sentences, can be generated by a Markov source, was only a small step, and I am not sure that we noticed at the time that this was a step at all. (Bar-Hillel 1970: 294–5)

The person reminiscing here, and quoted above as initially "enchanted" by the possibilities of the new technology, was the logician Yehoshua Bar-Hillel (1915–75). He'd encountered the logical positivists' approach to language in 1935, while studying at the Hebrew University in Jerusalem. He experienced their work as "nothing short of a revelation", later describing Carnap's volume on the logical analysis of language as "the most influential book I read in my life" (Bar-Hillel 1964: 1). And Bloomfield's linguistics "kept the fire of my interest burning".

So Bar-Hillel shared the structuralists' dream of an inductive discovery procedure for grammars. His hope was intensified when Harris, on a visit to Palestine in 1947, remarked that he was considering using the new electronic computers for this purpose. And, at close of the 1940s, he became "the first prophet in Israel" of cybernetics, including its potential relevance to the mind–body problem (Bar-Hillel 1964: 5).

With these views firmly in place, in 1950 Bar-Hillel accepted a fellowship at the University of Chicago, where he got to know Carnap personally. In the following year he visited MIT, well primed to be drawn into the enthusiastic information-theoretic community described above. Indeed, he was very soon appointed the first director of MIT's MT research. His remit was to survey the nascent field, to assess its prospects, and to plan MIT's research accordingly. His initial survey—of which, more below—was submitted in 1951, and in 1952 he organized the first MT conference (Hutchins 1986: 34–6).

Bar-Hillel didn't stay at MIT very long. On his return to Jerusalem in 1953, he was succeeded by Yngve, also from the University of Chicago. Yngve, now in his eighties, has been described by one NLP expert as "still the most original of all computational linguists" (Y. Wilks, personal communication).

In the 1950s, Yngve used both cybernetic and symbolic approaches. Besides doing early work on MT (Yngve 1955), he studied statistical methods for generating English (Yngve 1961) and developed the COMIT programming language, the first to be designed specifically for NLP (Yngve 1958). In the 1960s, he wrote a *Scientific American* article describing MT to the general public, and a technical survey defending MT against a damaging attack from an official committee (see below) (Yngve 1962*c*, 1967).

Yngve also made an early, and influential, attempt to relate syntactic theory to specific hypotheses about how people process language (Yngve 1960). That was an indication of the difference between his interests and those of the more austere grammarians, whether Chomsky or Gazdar. Over the years, his interests moved more towards "people" and away from "language" (Yngve 1986). And as remarked in Section viii.a, he attacked Chomsky on sociological as well as intellectual grounds.

During Bar-Hillel's brief stay in Massachusetts in the early 1950s, he met Chomsky. The two men interacted regularly, and Chomsky expressed his gratitude to Bar-Hillel several times in his early manuscript on the logical properties of language (e.g. Chomsky 1975: 31).

By 1955, Chomsky had accepted an appointment at MIT's Research Laboratory of Electronics (jointly with Modern Languages). In 1957, as remarked above, he and Miller were doing psychological experiments that immediately prompted information-theoretic work aimed in part at the development of NLP. And his hierarchy of grammars, as we've seen, helped to suggest what type of computational device would be needed to compute natural language.

Nevertheless, Chomsky never shared Bar-Hillel's enchantment with NLP. On the contrary, he was deeply sceptical from the outset:

> My personal reaction to this particular complex of beliefs, interests, and expectations [that information-theoretic computer modelling, combined with empiricist psychology, would make NLP possible, and further the positivist unification of science] was almost wholly negative. The behaviorist framework seemed to me a dead end, if not an intellectual scandal . . . I had no personal interest in the experimental studies and technological advances. [I felt the latter to be] in some respects harmful, [leading people to focus on] problems suggested by the available technology, though of little interest and importance in themselves. As for machine translation and related enterprises, this seemed to me pointless, as well as probably quite hopeless. As a graduate student interested in linguistics, logic, and philosophy I could not fail to be aware of the ferment and excitement [in the early 1950s]. But I felt myself no part of it and gave these matters little serious thought. (Chomsky 1975: 40)

When he did become interested in them, soon after completing his *magnum opus*, he wrote his fierce review of Skinner (see Section vii.b). His only foray into NLP work was a program schema that used transformational grammar (rewrite rules), not statistics, in its processing (G. A. Miller and Chomsky 1963: 464–82).

Nor was he a fan of AI in general. Quite apart from any intellectual reasons for scepticism, there appears to have been some unpleasant rivalry between him and Marvin Minsky—both young, brilliant, famous, and ambitious. In the mid-1960s, when the *Alchemy* scandal was at its height (see 11.ii.b), he went over to the MIT AI Lab, to sit in on one of the heated debates between Weizenbaum and Hubert Dreyfus on one side and Minsky and Seymour Papert on the other. David Waltz, an AI student at the time, now recalls:

> I remember once Noam Chomsky came over [to these debates] also. There was personal animosity between him and Minsky, and people were very rude to Chomsky when he came to the AI lab. They booed when he talked, and he was very miffed. (interview in Crevier 1993: 123–4)

As for Bar-Hillel, and partly because of his discussions with Chomsky, his love affair with—though not his interest in—NLP was to be short-lived, as we shall see.

c. Love letters and haikus

Despite the "ferment and excitement" aroused at mid-century by information theory and statistical cybernetics, some NLP pioneers chose instead to use the newly available digital computers as symbol manipulators.

Even very early on, these machines could be programmed with rules (written in binary notation) for selecting words from distinct grammatical classes. But with the introduction of list-processing languages, such as IPL and COMIT, in the mid-1950s (see 10.v.b), hierarchical syntactic structures could be represented more easily.

Probably the earliest computerized text generator was Turing's playful love-letter program. This was written in the late 1940s for MADM, the world's first electronic stored-program digital computer (Chapter 3.v.b). Using random numbers to choose the words, MADM produced this (Lavington 1975: 20):

Darling Sweetheart,

You are my avid fellow feeling. My affection curiously clings to your passionate wish. My liking yearns to your heart. You are my wistful sympathy: my tender liking.

Yours beautifully

M.U.C. [Manchester University Computer]

The program hasn't survived. But presumably the last two lines, apart perhaps from the adverb *beautifully*, were provided in 'canned' form, while some overall template instructed the machine to select an adverb here, a noun there, and perhaps to pick two endearments (*Darling*, *Dearest*, *Sweetheart*, *Love* . . .) for the first line. Template fillings of this type were soon to be used at CLRU to generate texts fitting the strict constraints of a Japanese haiku (Masterman and McKinnon-Wood 1968; Masterman 1971). Two examples are:

All green in the leaves
I smell dark pools in the trees
Crash the moon has fled

All white in the buds
I flash snow peaks in the spring
Bang the sun has fogged

As the CLRU team realized, the apparent meaningfulness of these mechanically generated haikus was due primarily to the human reader, whose "effort after meaning" (Bartlett 1932: see 5.ii.b) projected significance onto the initially puzzling lines. But if the CLRU realized this, many others didn't. The program was exhibited at the first international exhibition of computer art, and impressed the visiting public more than it should have done (J. Reichardt 1968).

d. Wittgenstein and CLRU

Some of the intellectual sources of early NLP were surprising, being very far in spirit from structuralist linguistics and information theory, and from symbolic computing too. This applies, for instance, to the pioneering research done at CLRU under the direction of Masterman (1910–86).

Their computational work was informed by the anti-positivist philosophy of the later Ludwig Wittgenstein. In scientific circles, this approach to language was nothing if not unorthodox. Wittgenstein himself had likened it to Goethe's views on morphology—see Chapter 2.vi.e–f (Monk 1990: 303–4, 501).

Although these new ideas weren't published until after his death, Wittgenstein had lectured on them—and his lecture notes had circulated—in Cambridge for many years. Moreover, he'd dictated some of them to Masterman herself (see Preface, ii). The lectures provided a radical critique of his own earlier (logicist) position (Wittgenstein 1953), and a very different view of language. Whereas he had once declared that "Everything that can be said at all can be said clearly" (Wittgenstein 1922: 27, 79), he now saw an indefinable penumbra of meaning surrounding every statement.

One of his main points was that words don't have cut-and-dried (essentialist) definitions, in terms of necessary and sufficient conditions. Rather, the items falling under a given word (*game*, for example) are linked by a network of "family resemblances" (Wittgenstein 1953: 31 ff.). Analogously, relatives may share the family nose, the family eyebrows, the family fingernails . . . without any one person having all these characteristics. Admittedly, some items are better examples of the concept concerned, so can function as "paradigm cases". (These have Rosch-typicality ratings close to 1: see Chapter 8.i.b.) But there's no clear cut-off point.

Masterman was convinced by the doctrine of family resemblances, and disturbed by the all too evident weakness of word-for-word dictionary translations (first attempted by Richens and Booth 1955 and Yngve 1955). Her critique of the early word-for-word efforts (Masterman 1967) was backed up by her own thesaurus-based approach to NLP (Masterman 1957, 1962). One of the early AI workers influenced by her research was M. Ross Quillian, who visited King's College, Cambridge, in 1961 to give a talk about his early ideas on semantic networks (Chapter 10.iii.a).

The thesaurus was envisaged as a semantic interlingua, through which the translator could pass from one natural language to another. (For an early application of CLRU's interlingua approach, see the discussion of "catataxis" in Parker-Rhodes 1956.) Many examples of semantic ambiguity, which would defeat a word-for-word MT procedure, could be resolved by this means. The semantically ambiguous word would fall under at least two different heads in the thesaurus, and the one actually selected by the program would be the one under which some other words in the sentence fell also. So *I put my money in the bank* and *I know a bank whereon the wild thyme blows* would be recognized as dealing with financial and rural matters, respectively. (Syntactic analysis would be needed to deal with *I drew some money from the bank to buy some wild thyme*.)

Initially, Masterman used a thesaurus derived by human intuition, namely, a cut-down version of Peter Roget's pioneering work published 100 years earlier. But her group also developed procedures for constructing thesauri automatically. Formal clustering methods were used to identify word classes, in terms of the co-occurrence of various words. Those word classes determined lexical substitutability for the purposes of machine translation. In addition, they aided in the automatic classification, indexing, and retrieval of documents (Sparck Jones 1988). (The pioneering librarian Gabriel Naudé would have been intrigued: see Preface.)

CLRU's thesaurus approach offered flexibility of various kinds. It could be applied to texts of any length, not just to isolated sentences. It was a source of semantic primitives that could be used to define meaningful messages (or "gists"), based on *actor–action–object* patterns, that were helpful for single-language interpretation and précis as well as for translation. And, not least, it enabled Masterman's group to take some account of the unpredictable extensibility of language. This had been stressed by Wittgenstein and Waismann—and, as Masterman often remarked, by the philosopher Ernst Cassirer (1874–1945), who followed Humboldt in highlighting the unceasing development of language (Cassirer 1944).

The extension of language was modelled, for instance, by Yorick Wilks (1939–). Like Masterman, he was especially interested in the interpretation of metaphysical texts—dismissed as "meaningless" by the logical positivists fashionable at mid-century. He wrote a program in the 1960s to assign semantically coherent representations to textual snippets drawn from the writings of a wide range of philosophers.

Wilks's program could do this successfully for some test paragraphs from Descartes, Leibniz, and David Hume, *without* any need to extend word meanings. But other test paragraphs did require that. Accordingly, his program would locate the semantically problematic word in a chunk of text (not a trivial matter), and then use thesaurus-driven methods to identify its meaning with some other term or phrase within the passage (Wilks 1972: 166–72). If the problematic word was an extension of sense relative to some interpretation already stored in the (thesaurus-based) dictionary, Wilks's EXPAND algorithm could develop the rules for 'literal' interpretation so as to make sense of the anomalous word in context.

This procedure wasn't foolproof. What if there were no other suitable word in the text, and/or some major sense of the problematic word had been omitted from the dictionary? And what is to count as a "major" sense, anyway? (Bar-Hillel, clearly, was hovering in the background.) Nevertheless, it gave acceptable readings of some notoriously tricky texts, including passages drawn from Wittgenstein's *Tractatus Logico-Philosophicus* and Spinoza's *Ethics*. Short of torturing it with extracts from *Finnegans Wake*, one can hardly imagine a more taxing test. In short, this early NLP model tried to escape the logicist straitjacket by acknowledging the creative extensibility of language. (Some of Wilks's later work did so too: his "preference semantics" was used in automatic parsing, interpretation, and inference—Wilks 1975.)

The methods developed by CLRU weren't intended as theories of how human beings translate, interpret, or classify. As remarked in Chapter 1.ii.b, this NLP group was doing technological, not psychological, AI. (Quillian, by contrast, was at least as interested in simulating human psychology as in producing useful technology: 10.iii.a.) But the practical demands were taken seriously from the start. One of the strengths of CLRU was its emphasis on careful testing and experiment to compare IR (information-retrieval) systems (Sparck Jones 1988: 14–17, 25–6). Mere plausibility wasn't good enough—and was sometimes found to be misleading. In short, Drew McDermott's well-deserved criticisms of much early AI didn't apply to Masterman's group (see 11.iii).

Whether Wittgenstein himself would have approved of any computerized approach to translation or classification, even one (such as Masterman's) part-inspired by his own ideas, is another matter. He didn't explicitly discuss any of the budding AI work. Nor do we know of any informal meeting between him and Turing, still less any discussion between them about the possibility of AI. (For an entertaining account of an imaginary meeting, see Casti 1998.) But when Turing attended some of Wittgenstein's lectures on the philosophy of mathematics, the two men repeatedly disagreed on fundamental points (Monk 1990: 417–22). The whole tenor of Wittgenstein's mature philosophy suggests that he wouldn't have welcomed computational work on language.

In part, this follows from his view of language as a network of social conventions rooted in the human "form of life", a notion that can hardly be applied to a computer. But in part, it follows from his stress on the flexibility and open texture of language (and concepts), which seems to lie beyond any scientific explanation.

This flexibility may be approximated by some computational methods, such as CLRU's thesaurus. But even if anti-logicist NLP is in practice less inadequate than logicist approaches, the philosophical problems remain. For Wittgenstein, presumably (S. Shanker 1998)—and for his follower Dreyfus, certainly (H. L. Dreyfus 1972; Dreyfus and Dreyfus 1988)—neither type of NLP can escape the fundamental limitations of AI in general. The Wittgensteinian position agreed with what Descartes had argued long ago (2.iii.c): machine mimicry of language use can only be highly imperfect.

e. Is perfect translation possible?

However, in this area of AI as in others, the question whether some task could in principle be performed (or mimicked) *perfectly* by a machine must be distinguished from the question whether it could in practice be performed (or mimicked) *well enough to be useful*.

The first question is especially difficult where translation is concerned, because it's not even clear what—if anything—would count as perfection in the human case.

Humboldt had claimed that no human translator can achieve perfect equivalence (see Section iv.b, above). In scientific texts, he allowed, translators could hope to do their work with no important loss of meaning. But in literary contexts, they couldn't (Novak 1972). In discussing the (many) possible criteria for a good literary translation, he suggested that the translator should choose words and sentence structures that somehow indicate the original language. The more one does this, however, the less acceptable (in the derivative language) the new sentence will become. Mark Twain's spoof of the German habit of delaying the verb illustrates the point:

> I am indeed the truest friend of the German language—not not [*sic*] only now, but from long since—yes, before twenty years already. And never have I the desire had the noble language to hurt; to the contrary, only wished she to improve—I would her only reform. It is the dream of my life been . . . I would only some changes effect. I would only the language method—the luxurious, elaborate construction compress, the eternal parenthesis suppress, do away with, annihilate; the introduction of more than thirteen subjects in one sentence forbid; the verb so far to the front pull that one it without a telescope discover can. With one word, my gentlemen, I would your beloved language simplify, so that, my gentlemen, when you her for prayer need, One her yonder-up understands . . .
>
> After all these reforms established be will, will the German language the noblest and the prettiest on the world be. (Twain 1897)

Moreover, it's not even clear that this particular criterion of fidelity to the source language should be considered at all. Humboldt himself was well aware that there are conflicting views on what counts *in principle* as a high-quality translation. In short, "perfect" translation is a chimaera. And even high-quality translation is often very difficult.

If this is true for expert human translators, the difficulties facing machines (or rather, their programmers) are so much the greater. Both meaning and grammar stand in their way.

Meaning is notoriously elusive: if it can be pinned down in principle, which even empiricists often deny (Quine 1960), it doesn't follow that it can be captured in practice. Semantic ambiguity and vagueness, not to mention metaphor, are obvious problems. Moreover, a translation machine (eschewing word-for-word translation and statistical methods) needs a universal grammar or an artificial interlingua to mediate between languages, and/or full descriptions of the relevant pair of natural-language grammars. But defining a grammar to cover a single language, never mind a universal grammar, is no mean feat, as we saw in Section ix. And to define a grammar isn't to program it: successful parsers weren't available until the early 1970s (and true computational effectiveness had to await GPSG, more than a decade later).

Most of these theoretical points were made in the 1950s by an increasingly disillusioned Bar-Hillel (1951, 1953, 1960). Despite having been enchanted by the linguistic promise of the new technology in the 1940s, he later expressed deep pessimism—not only about machine translation, but about computational linguistics in general (Bar-Hillel 1970: 298 ff.). He even apologized for his own early influence on the field:

> I myself was led into confusion by the strong surrounding currents [of cybernetics] and was responsible to a degree for increasing this confusion and leading others into blind alleys from which some never returned. (Bar-Hillel 1970: 289)

In recanting his earlier beliefs, Bar-Hillel didn't merely lament the unexpectedly slow progress. He claimed to prove that certain things, their feasibility largely taken for granted by the young NLP community, simply cannot be done by automatic means.

For example, Harris's suggestion that texts made up of syntactically complex sentences can be represented as transformations of several much simpler kernel sentences had led many to hope that NLP could proceed by normalizing texts to some simpler form. Bar-Hillel proved, by contrast, that "Not everything that can be said at all can be said, in a given language, by using syntactically simple sentences exclusively" (Bar-Hillel 1963: 31).

As for semantic ambiguity, this often calls for world knowledge as well as knowledge of language as such. The simple sentence *The box is in the pen*, for instance, can be understood only if one knows that playpens are large enough to contain boxes whereas writing pens are not (Bar-Hillel 1960, app. iii). An MT program concerned specifically with such topics could of course be given this information beforehand. But, Bar-Hillel insisted, such priming can't be done in the general case.

In short, what he called FAHQT (fully automatic high-quality translation) is impossible.

f. Is adequate translation achievable?

Bar-Hillel was complaining about the "science-fictional claims" made by some proponents of NLP, namely those who were promising FAHQT. He wasn't arguing that computer models of language are wholly useless.

In his first report to MIT, he'd said that whereas high-accuracy, fully automatic, MT isn't feasible, "there appear to be less ambitious aims the achievement of which is still theoretically and practically valuable" (Bar-Hillel 1951: 154). Almost a decade later, when he surveyed the progress of MT in the USA and UK (and USSR), Bar-Hillel again stressed the impossibility of human-quality MT (Bar-Hillel 1960). Instead of aiming for this unattainable goal, he said, AI should aim for imperfect-but-intelligible automatic translations, possibly supplemented by human post-editing.

Alongside this potentially hopeful advice, however, he argued that even this less ambitious project should be undertaken in a different way. In his second report, after MT had been under way for some years, he rejected approaches based on statistics and information theory, which he'd previously thought so promising. Nor did he have much time for the then-fashionable methods of GOFAI (Good Old-Fashioned AI: see Chapter 10). He criticized virtually all the major MT groups by name, and attacked the most common methodologies.

The immediate effect of Bar-Hillel's second report was almost to destroy faith in MT on the part of the general public, including scientists not working in the field. A best-selling popular book on *Computers and Common Sense* by Mortimer Taube (1910–65), for instance, took up Bar-Hillel's negative remarks while ignoring his positive ones (Taube 1961).

Taube's book announced in highly emotional terms that MT was impossible and MT research—indeed, AI in general—a waste of taxpayers' money. The nation's monetary resources, Taube said acerbically, would be better devoted to "research looking toward the Second Coming" than to research aiming to build machines with "intelligence, knowledge, and knowledgeability". His dismissive conclusion was that "One may wonder why reputable scientific journals publish material of this sort and why it should have an audience beyond the readers of the Sunday supplements."

Taube, an internationally famous librarian, knew some of his computer onions: he'd pioneered powerful new forms of indexing and information retrieval. Even so, his attack cut little ice with the AI community. Being called "ignorant" and "jejune", and being accused (repeatedly) of "scientific aberration", led to so much offence that they hardly bothered to marshal a defence. Moreover, the book was—as one early cognitive scientist put it—"neither reliable nor responsible", being largely comprised of "allegations presented as facts, of misunderstanding, of debaters' tricks identical to those he decries in others, and of statements about the work of others which are simply untrue" (Reitman 1962: 718).

However, Taube's attack encouraged an attitude of scepticism, not to say ridicule, in the general reader. And the "general readers", in this case, included not only counter-culturalists such as Theodore Roszak (1969: 295) but also people who might at some point have to make decisions about funding policies for MT.

As if that weren't damaging enough, Bar-Hillel's critique also dampened the optimism of MT researchers who were aiming to approach human-quality results. Those who were willing to tolerate many linguistic infelicities, provided that these didn't compromise understanding, were less set back. Even so, and even though significant (military-led) funding continued for over a decade, many MT workers lost heart. At Harvard, for example, almost all research in this field had ceased by 1964.

In that year, the scepticism deepened further. A survey had just been conducted by the US government's Automatic Language Processing Advisory Committee (ALPAC), chaired by the Bell Telephones executive John Pierce (Hutchins 1986: 164–7). Their report was thirty-four pages of dynamite—plus ninety pages of smouldering appendices. It concluded that previous MT research had been a failure, that fully automatic MT was impossible, and—even worse—that there was "no immediate or predictable prospect of useful machine translation".

This negative judgement wasn't a purely scientific one. It was partly based on ALPAC's view that not enough customers would want to use MT to make it commercially viable. Machine aids for human translators, however, might well be feasible.

The effect on informed (or rather, "semi-informed") public opinion was predictable. Yngve described it like this:

> The [general] public is now where the pioneers were many years ago, vaguely aware of the difficulties but allowing that the thing is possible, while the workers, with more than a little knowledge now, see quite clearly that there is very much yet to be done. And the semi-informed public gives indications of swinging back to [their] original position, that the task is virtually impossible, therefore no proper area for research. In this they may be echoing a few despairing noises coming from some of the workers who had expected a quick victory. (Yngve 1967: 453)

Despite criticisms that the ALPAC report was "narrow, biased, and shortsighted", and imbued with a "hostile and vindictive attitude", it had a devastating effect on MT research in the USA. The activity was virtually halted for over a decade. And the devastation was near-instantaneous: "Funding for [MT] projects dried up within weeks of the ALPAC report, and several research labs shut down soon thereafter" (Newquist 1994: 116).

It had a negative influence in the UK and Soviet Union, too. Work on translation at CLRU, for instance, ceased in the late 1960s—even though the Unit itself continued, and did valuable work on various aspects of information retrieval (Sparck Jones 1988; Wilks forthcoming).

If the negative aura of ALPAC could cross oceans, it could cross some intellectual boundaries too. It threw a shadow over AI in general, and NLP in particular. Although the three main AI departments weren't much affected (they weren't doing NLP anyway), several US universities that had been planning to start departments of AI decided not to do so (Newquist 1994: 117).

Nevertheless, ALPAC didn't end NLP as a whole. On the contrary, it favoured increased support of *basic* research in computational linguistics. If the embarrassingly prolific MT cousins of *ploughman crooked ground plough plough* were ever to be avoided, this would require a better understanding—and effective computational implementation—of theoretical linguistics.

9.xi. NLP Comes of Age

NLP would come of age (in the 1970s) with the acceptance of more realistic goals for MT, with the appearance of effective computer parsers, and with the recognition that, besides processing single sentences, one must also model the rhetorical coherence of entire texts. The last two lines of development—which required advances in general AI—are still under way.

Moreover, it's now clear that improved statistical methods, let loose on huge samples of real-life linguistic data, can go much further than Shannon's early critics imagined. Even as late as 1987, that was less clear. At a statistics/linguistics conference in Oxford, sponsored by the Engineering and Physics Research Council (EPSRC), the atmosphere was civilized but there was no real communication (G. Gazdar, personal communication). At a similar meeting held at the Royal Society in 1999, there was (I witnessed that for myself). What's more, while the EPSRC may have expected all the benefit to go from the statisticians to the linguists, in fact it went in both directions.

For instance, Markovian three-word chunks can now be used to make word predictions almost as well as people can, although they can't generate grammatical sentences as reliably (Pereira 2000). These probabilistic models typically induce "hidden" rules from huge databases, and some rely on symbolic rules or tree structures derived from theoretical linguistics (Gazdar *et al.* 2000). Many are trained on collections made up of several hundreds of millions of words. Even so, not all sentences (or newly coined words) can be included. Chomsky's complaint that Markovian theories can't allow for the creative use of language is countered by "smoothing algorithms" that avoid assigning zero probabilities to unencountered events.

Some advances, whether symbolic or statistical, are of primarily technological interest. They will affect our daily lives, but not necessarily our understanding of the mind. Others are of interest only to highly specialized linguists. Below, I outline a few post-1960 landmarks, selected for their relevance to cognitive science *in general*.

Strictly, a certificate of maturity for NLP would also require advances in speech processing—both understanding and synthesis. The former involves the analysis of continuous speech into individual words, and the classification of acoustically different sounds as one and the same phoneme. It had to await the development of vocal-acoustic technology, and also required an integration of several theoretical levels: acoustic, phonetic, morphemic, syntactic, and semantic.

Since I've virtually ignored phonetics throughout this chapter, I'll ignore speech systems too. An early machine for learning to recognize phonemes was mentioned in Chapter 6.ii.c, but this was merely an exploratory toy. The pioneering HEARSAY program of the early 1970s will be mentioned in Chapter 10.v.e, because of its innovative "blackboard" architecture, which integrated processes based in syntax and semantics with processes grounded in phonetics. And the 1980s connectionist system NETtalk is discussed in Chapter 12.vi.f, because of its methodological interest. But speech processing as such isn't crucial for our purposes. Apart from a few remarks at the close of the chapter (subsection g), it won't feature here.

a. MT resurrected

As for machine translation, the ALPAC authors were shown in the long term to have been over-pessimistic. But their suggestion that basic research in computational linguistics would be helpful was justified. Bar-Hillel, by contrast, turned out to be correct on both main counts: although human-quality MT isn't achievable in the general case, practically useful MT is. Masterman's key problem of word-sense disambiguation, for example, can be approached today by methods fundamentally similar to hers but much more powerful (Ide and Veronis 1998).

There are now many MT systems in daily use (W. J. Hutchins 1994, 1995). Most, as Bar-Hillel had predicted, deal only with a highly restricted area, but some can cope with a fairly wide range of topics. They are used, for example, to scan foreign-language news media or scientific publications, or to express political debates and legislation.

Their translations are typically good enough to be understood by informed human readers, and/or to show whether it would be worthwhile to ask a human translator to improve them. Sometimes, only minimal post-editing is required. But sometimes (for instance, when Japanese is involved), significant pre-editing and post-editing are needed.

The SYSTRAN program is one well-known example (P. J. Wheeler 1987). A descendant of an early Russian–English system funded by the US Air Force, it was written in the late 1960s by a team led by Peter Toma, a participant in the famous Georgetown project. The European Community adopted it for English and French in 1976, since when it has been broadened to include all the official languages of the European Union. It is used also by NATO and the International Atomic Energy Authority, and by commercial companies such as Xerox and General Motors.

Some of these applications appear at first glance to evade Bar-Hillel's criticism. For many documents in English can be translated by SYSTRAN into other European languages with near-perfection. The trick is to use only a small subset of English for the original document. A number of other commercially available programs can also provide near-perfect translation of technical manuals and the like, by using highly restricted vocabulary and syntax.

What might well surprise Bar-Hillel is that some very recent MT systems rely heavily on statistical pattern matching (W. J. Hutchins 1995: 440–1; 1994). Rule-based approaches have given way to corpus-based ones, and the emphasis has shifted from syntax to the lexicon (Hutchins 1994, sect. 9). Instead of careful syntactic analysis, these systems rely on probabilistic methods based in some huge corpus of paired source/translation texts. Shannon, one might say, has returned with a vengeance.

IBM's CANDIDE system, for instance, draws on the voluminous (and multi-topic) French–English 'Hansard' of the Canadian Parliament. In the mid-1990s this program contained no grammatical rules whatsoever, relying purely on statistical matching of words and phrases. Its performance amazed most observers, given that the analytical approach had been largely taken for granted in NLP for thirty years. Even ALPAC had seen detailed computational linguistics as essential for automatic translation. Almost 50 per cent of phrases were acceptably translated, being matched either with their database originals or with equivalent expressions.

However, one must not forget the other 50 per cent. To deal with those, future versions of CANDIDE will include some morphological and syntactic information, as well as more sophisticated statistical methods. But grammar and morphology will be used as sparingly as possible. In other words, the team expect the statistics to trump the syntax. It remains to be seen whether they are right, and to what extent MT can afford to ignore theoretical linguistics.

Certainly, grammatical correctness is MT's Achilles' heel. The general assumption, *pace* CANDIDE, still is that this demands painstaking attention to syntax (gender, number, tense, and so on). Measures of intelligibility and correctness have shown a dramatic improvement in the former, but much less progress on the latter.

For instance, in the two-year period 1976 to 1978, the intelligibility of translations generated by SYSTRAN rose from 45 to 78 per cent for the raw text, and from 95 to 98 per cent for the post-edited text (W. J. Hutchins 1986: 261). Human translation, it's worth noting, gave not 100 per cent but only 98–9 per cent.

SYSTRAN's correctness, where every tiny error counts, scored much lower. In 1976 the measure for grammatical agreement was only 61–80 per cent; and in 1978, only 64 per cent of words were left untouched by the human post-editors. Even so, human post-editing of a page of SYSTRAN output took only twenty minutes in the mid-1980s, whereas normal (fully human) translation would have taken an hour. However, over half of the amendments involved substituting a different word, not simply altering a word ending. In both cases, the main source of errors seemed to be the dictionary, which in 1978 contained only 45,000 entries. Accordingly, the computerized dictionaries have been gradually enlarged.

The more ambitious EUROTRA system was multilingual as opposed to severally bilingual, dealing with forty-two different language pairs (King and Perschke 1987). It held 140,000 words in its prototype stage, in 1988: 20,000 words for each of seven languages, with the expectation of at least two more to come. But no operational version was ever built. EUROTRA was influential, nevertheless. It was recently recognized in an official report as providing much of the basis for later work in MT across Europe (*Euromap Report* 1998: 59).

More ambitious still would be an MT system that could switch between languages in very different language groups, such as English, Japanese, and Hindi. In Japanese, for example, the words aren't segmented as they are in English (no equivalent of our *post-ed*, for the past tense of *post*), and the phrase orderings in the parse trees are reversed (Powell 2002: 112). Although there are MT programs combining Japanese and English, and also for switching between several of the many languages of India, they aren't as efficient as the best European-language systems.

(I've mentioned corpus-based work only in MT, but similarly statistical approaches are increasingly being used in other areas of computational linguistics too: see Sampson and McCarthy 2004.)

b. Automatic parsing

If computerized dictionaries are relatively easy to improve, automatic parsing is a different matter.

NLP in the 1950s and early 1960s, as we've seen, was hampered by the lack of a powerful theory of syntax. Computational work on syntax started to show results in the early 1970s. This development was largely due to the post-Chomsky interest in formal grammars. (This is *not* the same thing as an interest in Chomsky's formal grammar: as remarked in Section ix.f, very few people used Chomsky's theory as the basis of their NLP programs.)

The first impressive computer parser was written by William Woods at Bolt, Beranek & Newman (BBN), using "augmented transition networks", or ATNs (W. A. Woods 1970, 1973). The computational advantage of ATNs, which soon became widely used in NLP, was that they considered the input sentence in one pass, working left to right on one word or syntactic constituent at a time. Nevertheless, they could deal—recursively—with nested sentences or noun phrases (such as *the judgement of the court*), and with nouns embellished by several adjectives.

In effect, they worked top-down, continually making syntax-driven predictions about grammatical form and testing the input word to see whether it fitted the current prediction (see Boden 1988: 91–102). On encountering the word *the*, for example, the ATN would predict either an adjective or a noun, in that order; but if an adjective were found, it would immediately predict another one before looking for a noun. This simple procedure would enable it to recognize, for instance, *The girl*, *The clever girl*, and *The clever French girl*.

Woods used his parser in his LUNAR program, which answered questions about lunar geology, drawing on NASA's data about the mineral samples collected from the moon on the Apollo 10 mission. Its answers were in 'written' English (though Woods soon worked also on the HEARSAY speech system). This early version couldn't handle relative clauses in a theoretically elegant manner. Nevertheless, LUNAR was a huge advance on the question-answerers of the 1960s (surveyed in Simmons 1965, 1970).

These included the BASEBALL program, co-authored by Carol Chomsky (B. F. Green *et al.* 1961). This could search and reason about its database so as to answer questions such as "How many games did the Yankees play in July?" (see Chapter 10.iii.a). To deal with the English-language input, BASEBALL and its contemporaries had relied on a menu of ELIZA-like pattern matches (10.iii.a), not on parsing. Simple rules enabled ELIZA to switch pronouns so that (for example) the input *I ***** you* would elicit *WHY DO YOU ***** ME*. Woods's program, by contrast, modelled syntactic structure in some detail.

ATNs had originally been defined by a research group in Scotland (Thorne *et al.* 1968). Their NLP program was successfully tested on Lewis Carroll's 'Jabberwocky', among other things. It was based on transformational grammar, and built syntactic representations of the input sentence on both deep and surface levels. Woods adopted the general approach of ATNs, but eschewed Chomsky's grammar.

Nor was he the only one to do so. Because of the computational load involved in mapping between two entire syntactic structures, and thanks to the development of non-Chomskyan grammars (see Section ix), very few NLP projects have implemented transformations. The most interesting exception is Mitch Marcus's PARSIFAL, a program inspired not only by Chomsky's syntax but also by his views on language-and-mind (see Chapter 7.ii.b and M. P. Marcus 1980).

The Scottish team, like Marcus, had been doing psychological AI. They saw Chomsky's theory, and their left-to-right ATNs, as modelling some of the mental processes involved in language. Woods wasn't much interested in whether his parser was psychologically plausible. But some people used his approach to formulate specific psychological hypotheses about how people use language—and memory, too (see Chapter 7.ii.d and iv.d–e). The first to do so was Kaplan (1972), who later went on to co-author a psychologically motivated grammar, LFG, that has been borrowed extensively for technological purposes (see Section ix.b). Another was David Rumelhart, who used ATN models of grammar to study the psychology of reading errors (Stevens and Rumelhart 1975).

This swinging pendulum of intellectual influence illustrates a general point made in Chapter 1.ii.b. The boundary between psychological and technological AI, and therefore between what is cognitive science and what is not, is often fuzzy. ATNs were first devised (in Scotland) with psycholinguistic intent, soon adapted (by Woods) for technical purposes, and then picked up (by Kaplan and others) for use in psycholinguistics, memory research, and theoretical syntax. The new formal grammar that eventually resulted was later adopted by a variety of technological NLP researchers—some of whom would hardly recognize a mental process if it punched them on the nose.

A further illustration of this point was provided by the most famous achievement of early NLP: Terry Winograd's program SHRDLU (Winograd 1972; Boden 1977, ch. 6). The name, by the way, wasn't an acronym but a slyly subversive joke. In the publishing technology of the 1960s, one row of the keyboard of a standard linotype typesetting machine consisted of these letters. Typesetters would often signal a mistake by inserting them in a faulty line, so that the proof-readers would easily spot that a mistake had been made. Bad proof-reading might result in this deliberate gibberish being printed in the final text. That fact was made much of in the counter-cultural *MAD* magazine, which was deliberately peppered with multiple occurrences of this nonsense word. As a devotee of *MAD*, Winograd (1946–) decided to use it as the name of his program (T. Winograd, personal communication).

Winograd's program was written, at MIT's AI Laboratory, in the heyday of NewFAI. In 1961 George Ernst (with Minsky and Shannon on his thesis committee) had started to build a robot that would pick up blocks from a table, build them into towers, and put them into and out of a box. And by the late 1960s, the MIT robotics team were trying to integrate computer vision with their robot's problem solving, as well as working on a better robot hand (L. J. Fogel and McCulloch 1970: 256–65). It was in this intellectual environment that Winograd's doctoral research began.

The overall aim was to enable the team to give instructions to the robot in English, and for the robot to respond in real time. But for that to be possible, AI needed to advance in English as well as robotics.

Winograd's thesis title, 'Procedures as a Representation for Data in a Computer Program for Understanding Natural Language', showed that he wasn't interested only in English, or even NLP. To the contrary, he was largely concerned with questions about virtual architectures for AI. Indeed, his concepts of "heterarchy" and "procedural programming" had a huge impact on GOFAI (see Chapter 10.iv.a).

What's more, he wasn't doing psychological AI—except, perhaps, in the most general sense. Although his opening sentence was "When a person sees or hears a sentence, he makes full use of his knowledge and intelligence to understand it," he was careful to stress that SHRDLU wasn't intended as a detailed model of what goes on in people's minds.

Nevertheless, his work was considered so significant by the cognitive science community that an entire issue of the journal *Cognitive Psychology* was devoted to it—and it was simultaneously published in book form by a leading university press. In explaining their unprecedented decision, the psychologist editors said:

> Winograd's system is not a "simulation", but it incorporates important ideas about human syntactic, semantic, and problem-solving abilities, and, in particular, about their interactions in understanding natural language. Human intelligence goes far beyond this system, and human

language comprehension may turn out to differ in major respects from the means employed here. At the very least, however, Winograd's system should prove an invaluable tool for thinking about what we do when we understand and respond to natural language.

(Winograd 1972, p. vii)

Their reference to "interactions" between syntactic, semantic, and problem-solving abilities, and Winograd's opening sentence (quoted above), indicated the sense in which SHRDLU went far beyond previous NLP work.

Like Woods, Winograd provided a powerful computer parser (based on Halliday's systemic grammar: see Section ix.a). It could even parse long sentences with a syntactic sting in the tail, such as: *How many eggs would you have been going to use in the cake if you hadn't learned your mother's recipe was wrong?* But the most important NLP novelty was that the parser was closely integrated with semantic analysis, problem solving, and world knowledge (see 10.iv.a).

SHRDLU's world knowledge—of a table, a box, and various coloured objects that could be moved by a robot arm—had to be laboriously programmed in. So Bar-Hillel's argument citing *The box is in the pen* still held. As Winograd himself pointed out, many obvious implications of English sentences about boxes, blocks, and tables were invisible to the program.

This was a prime reason why, in the 1980s, he rejected the approach underlying his early work (Winograd 1980*a*; Winograd and Flores 1986). He'd been persuaded, by Fernando Flores and Dreyfus, that various neo-Kantian thinkers were better guides to language than the logicist philosophers whose views imbued GOFAI—and most computational linguistics, too (see Chapter 16.vi–viii). So his 1986 book, co-authored with Flores, highlighted the hermeneutic philosophers Martin Heidegger (sixteen index entries) and Hans-Georg Gadamer (nine), the autopoietic theorist Humberto Maturana (fifteen), and the later Wittgenstein (two). None of these had figured in his earlier work—and many of his erstwhile admirers were aghast. (For more on Winograd's volte-face, see Chapter 11.ii.g.)

The cognitive integration, in SHRDLU, went beyond the fact that syntax, semantics, reasoning, and world knowledge were all included in the one *program*. The crucial point was that they could interact in the processing of a single *sentence*. When interpreting the syntactically ambiguous instruction *Put the blue pyramid on the block in the box*, for instance, the program checked to see whether there was one and only one blue pyramid already sitting on the block, or (if not) whether there was one and only one block already inside the box. (Compare this with Quillian's semantic network, which used *purely lexical* information to disambiguate sentences like *I threw the man in the ring*: 10.iii.a.) And in obeying the instruction, and deciding how to reply to it, SHRDLU might have to do some problem solving, to enable the robot arm (which could pick up only one thing at a time) to get at the blue pyramid concerned.

With the help of predicate logic ("one and only one thing such that ..."), also included in the program, Russell's theory of definite descriptions (see 4.iii.c) had been implemented in computational form. However, Winograd defined the program's lexicon, and its syntactic and semantic terms, not as formulae of the predicate calculus but as mini-programs. So his definition of *the* didn't simply assert (as Russell had done) that this word was apt only if there was exactly one thing that fitted the description given. Rather, it instructed the system to go and look for such a thing on encountering the word. As a result, when SHRDLU was asked to "Grasp the pyramid", there being *three* pyramids in the scene, it sensibly replied, "I don't understand which pyramid you mean." However, if a specific pyramid had already been mentioned in the conversation, it assumed that "the pyramid" referred to *that* one. Similarly for *pyramid*, whose definition was a procedure telling SHRDLU to consider—and, if appropriate, to look for—a thing of a certain sort. The syntactic definitions, of *noun*, *adjective*, and the like, were also expressed as procedures rather than assertions.

An ATN too, of course, would look for a 'suitable' word on encountering *the*, or *pyramid*, and would employ some specific procedure to find a noun, or a verb. But these syntactic categories weren't *defined* procedurally. This gave ATNs a generality that SHRDLU lacked, for *some* ATN could be devised for *any* grammar.

SHRDLU thus constituted a pioneering example of "procedural semantics", an approach to meaning and understanding that became widely influential in cognitive science. As we saw in Chapter 7.ii.d and iv.d–e, procedural semantics was taken up in studies of language and problem solving. It also affected work in practical AI (10.iv.a). With respect to technological NLP, the specifics of Winograd's parser were less influential than Woods's ATNs. But Winograd's work had shown the possibility of integrating syntax and semantics (and world knowledge, when available) in understanding language.

This general lesson could be applied by those who came after him, even if they used a different grammar and method of processing. For example, the author of the speech program whose many errors betrayed underlying anxieties (7.ii.c) singled out Winograd, for having enabled him "for the first time . . . to get a handle on how to think about and represent cognitive and linguistic processes" (Clippinger 1977, p. xv). That stuttering program was very different from the grammatically perfect SHRDLU. Nevertheless, SHRDLU was crucial in its ancestry.

Awesome though SHRDLU was, it wasn't quite so impressive as my description, above, has suggested. For one thing, the "conversation" wasn't really a conversation:

> I believe that in the process of debugging I got it to go through the whole dialog in a single run, but I wouldn't want to state that for sure. Most of the work was done in a more fragmentary manner. (T. Winograd, personal communication)

(Each fragment was relatively speedy: the system, when running compiled, was "fast enough to carry on a real-time discourse", with between 5 and 20 seconds needed for the combined interpretation and response; and the display was designed to move at the speed of a real arm: personal communication.)

For another, the system couldn't be used 'off the shelf':

> The program was at times run effectively by others (including Stu Card who was there for a summer and helped get out many of the bugs), but only after they took some trouble to become familiar with it. It wasn't at a level of robustness for walk-up users. (T. Winograd, personal communication)

In this, SHRDLU was far from unique. It was a common failing of early GOFAI programs that they couldn't be run—or developed—by others, because of bugs. Indeed, this was one of the complaints made by McDermott in his famous criticism of AI work in the mid-1970s (11.iii.a).

c. 'What I did on my holiday'

The growing interest in automatic parsing throughout the 1970s and 1980s, which fed and was fed by the theoretical work on grammars outlined in Section ix, wasn't matched in respect of automatic composition. Even a young child can write an essay on 'What I did on my holiday'. But a computer, even if it could go on holiday in the first place, would be very hard put to do so.

The output of many (non-MT) NLP programs, including SHRDLU, was based on pre-assigned templates. Although these could be syntactically varied to some extent, they were essentially similar to the 'canned' responses of very early programs such as ELIZA. Indeed, the problem of how to enable computers to generate syntactically elegant text was rarely addressed. The reasons were part-practical, part-theoretical.

For most practical purposes, the automatic composition of elegant syntax was, and still is, a stylistic luxury. Many NLP programs, such as the human–computer interfaces used by the general public, can get by with very simple syntax. Even MT programs can acceptably simplify the original syntax, provided that intelligibility is retained. Often, the original text (of technical manuals, for instance) can be written in a deliberately simple fashion without loss of meaning. And even programs that generate the text—not just the plot—of stories can 'succeed' by emulating the syntax of stories written for 5-year-olds, not of Marcel Proust.

The theoretical obstacle was a version of Bar-Hillel's problem: subtle use of syntax requires subtle world knowledge. When people use language, they typically generate syntactic structures that are both apt (with respect to the situation being described) and helpful (with respect to intelligibility). In doing so, many questions arise. For instance:

* When is clause subordination appropriate,
* and which event should be referred to by the subordinate clause?
* When are two separate sentences preferable to a single sentence in which two clauses are conjoined?
* When should one employ subjunctives and conditionals?
* How should one do so?
* And how can one choose *appropriate* referring expressions, determiners, modifiers, tense, aspect, and modal verbs?

These intriguing theoretical questions were investigated in the late 1970s by Antony Davey, who related them to various psycholinguistic phenomena. Here, the relevant point is not the psychology, but the fact that he implemented some of his answers in an NLP program (Davey 1978). He showed how syntax could be used, in a principled way, to signal the strategy driving individual games of noughts and crosses (tic-tac-toe).

Thus his program produced the following description of the game depicted in Figure 9.5:

I started the game by taking the middle of an edge, and you took an end of the opposite one. I threatened you by taking the square opposite the one I had just taken, but you blocked my line and threatened me. However, I blocked your diagonal and threatened you. If you had blocked my edge, you would have forked me, but you took the middle of the one opposite the corner I had just taken and adjacent to mine and so I won by completing the edge.

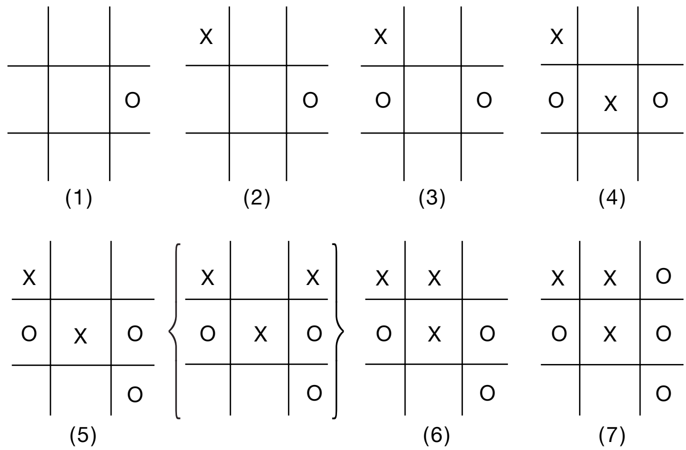

FIG. 9.5. Game of noughts and crosses (tic-tac-toe). The move in brackets wasn't actually made, but was 'imagined' by the program when it described the game depicted in the numbered frames. (Based on an example in A. C. Davey 1978: 18.)

No post-editing was done by Davey: this passage was the actual output of his program. The syntax nicely reflected the structure of attack, defence, and counter-attack informing the game, and helped distinguish strategy from tactics. For instance, it would have been less apt to reverse the last two ideas in the second sentence (as in "...but you threatened me and blocked my line"), or to start it by saying "I took the square opposite the one I had just taken and so threatened you." Many other superficially equivalent alternatives were specifically—and rightly—avoided by the program.

This was the first automatic system capable of generating complex, semantically appropriate, syntactic structures. As such, it was a significant achievement. But, for the reasons outlined above, it had little influence on the development of NLP. Its unprecedented power was unnecessary in most practical contexts, and wasn't achievable in the general case—for composing essays on holidays, for example. Davey had chosen noughts and crosses as his domain precisely because the rules are easily statable, the games are fully describable, and the strategy is readily intelligible. It would be difficult, and often impossible, to adapt his program to other domains.

Different approaches to grammar generation were sometimes considered. For instance, rules governing shifts in the focus of attention (see below) were used in choosing the active or passive voice, pronouns or definite descriptions, and the sequence of sentences within a paragraph (McKeown 1985). But these didn't match the detail of Davey's work. Consequently, today's NLP programs are asymmetrical in their grammatical abilities, much as SHRDLU was. Their expertise in parsing belies their simple-mindedness in sentence construction.

d. Semantic coherence

The maturation of NLP also required a concern for the semantic coherence and rhetorical structure of entire texts. Let's consider the first topic here (the second is discussed in subsection f).

Impressive though it was, Winograd's program had virtually no ability to deal with a text, as opposed to a series of single sentences. Admittedly, anaphora and definite descriptions were interpreted by noting what objects had been referred to in previous sentences. But there was no representation of the theme of the conversation as a whole, nor any prediction as to what the human might say next.

Still less was there any systematic representation of conversational etiquette, or of the distinction between the words uttered and the communicative intention behind them. Eventually, and partly driven by the problems involved in designing usable natural-language interfaces, these issues of semantics and pragmatics came to the fore (see 13.v).

One early attempt to identify the theme of an entire text had been made by Wilks, using CLRU's thesaurus approach (see Section x.d, above). Another, even earlier, was the "General Inquirer", initiated around 1960 by Philip Stone (1936–) at Harvard and colleagues (Stone *et al.* 1962). The method relied on an "affective" classification of the English dictionary, based on tags drawn from Fred Bales's small-group psychology. For example, *union*, *cooperation*, and *love* were coded under the Affiliation tag, whereas *fight* and *war* were classed as Hostility. Counting tag words was straightforward. And hierarchical structures, wherein one semantic tag seemed to be providing the context for another, could be found by list processing—using Yngve's new list-processing language, COMIT.

This approach provided only a very broad analysis—and could be highly misleading, to boot. The computer analysis might show Affiliation governing Hostility, for instance, or Hostility governing Affiliation. However, to decide whether these seeming contradictions reflected actual incoherence, as opposed (for example) to a threat followed by conciliatory remarks, the researcher would have to go back to the text itself.

I myself used a prototype version of the Inquirer in October 1962, as a student in Stone's team analysing the ongoing Khrushchev–Kennedy interchange about the Cuban missile crisis. (The texts were being made immediately available to us in electronic form by an international press association.) The hope was that implicit affective changes in Nikita Khrushchev's attitude might be picked up even before they'd been made explicit by him.

The attempt wasn't successful. For example, Khrushchev's speeches, but not John Kennedy's, scored very high on Affiliation. On checking the original transcripts, the explanation was clear. What Kennedy called "the USA" and "the USSR", Khrushchev (or his translator) called "the *United* States" and "the Soviet *Union*". The problematic words, of course, were then removed. But they might not have been identified in the first place, if the initial scorings hadn't been so highly counter-intuitive. In addition, it was unclear what significance—if any—should be attached to the fact that frequency counts showed one tag hierarchically dominating another.

Both Wilks's system and Stone's suffered from a broad-brush approach. They could identify general themes and overall affective tone, respectively, but couldn't follow the development of a theme from one sentence to the next. And there was no question of their being able to predict the likely content of sentences yet to come.

The first people to model these matters were the social psychologist Robert Abelson and the computational linguist Roger Schank. One reason for their collaboration was that Schank was doing psychological, not technological, AI. He did eventually produce a number of commercially marketed NLP tools, but he saw them—correctly or not—as reflecting how humans process language (see Chapter 7.i.c and ii.d).

By the mid-1960s, Abelson (at Yale) had simulated individual systems of political beliefs, predicting responses to a variety of new inputs. Soon afterwards, he outlined a computational theory of social roles, of motivational–emotional "themes", such as betrayal and cooperation, and of "scripts", such as escape (see Chapter 7.i.c).

By this time, Schank—then at Stanford, though soon to join Abelson at Yale—was doing NLP based on his conceptual dependency (CD) theory. This combined a semantic primitive analysis of verbs with a version of case grammar (Schank 1973; Boden 1977, ch. 7). He aimed not only to represent the meaning of an input sentence, but also to predict the likely topic/s of the following sentences. The verb *hit*, for example, involves an "instrumental" case. If this case slot was left open (as in the sentence *John hit Mary*), the CD representation enabled/predicted the follow-up query "What with?"—where the default inference was that the instrument was the actor's hand.

In their collaborative work at Yale, Schank and Abelson (1977) used the term "scripts" in a new way, to represent the behaviour of various actors in culturally stereotyped situations. Their restaurant script, for example, codified the roles of customer, waitress, and cashier in a hamburger bar. From the early 1970s, Schank and his students combined CD theory with this new theory of scripts to generate questions and answers about (specially composed) texts.

For instance, Wendy Lehnert (1978) wrote a question-answering program that integrated CD representations of verbs with world knowledge about hamburger restaurants. Given a mini-story including the sentence "When the hamburger arrived, it was burnt", the program would infer—what was not made explicit in the input—that the customer who had ordered it neither ate it nor paid for it. These texts provided the story snippets that would soon be used as examples by John Searle (1980) in his discussion of the Chinese room (16.v.c).

Meanwhile, Jim Meehan at Yale, and researchers in other places, had been using Schankian ideas in modelling the generation and summary of simple story plots (Meehan 1975; Rumelhart 1975; Schank and Riesbeck 1981; Lehnert and Ringle 1982). And up to the early 1980s, Schank's student Michael Dyer (at Yale and then UCLA) considered more complex types of story, involving developments of Abelson's ideas about motivational and emotional structures (Dyer 1983).

These Yale-based programs combined linguistics with GOFAI, for they relied heavily on AI ideas about planning in interpreting texts. In this, though in little else, they resembled Winograd's SHRDLU.

For example, a key concept in Dyer's NLP approach was the TAU, or thematic abstraction unit. TAUs were used to organize memory, direct the process of understanding, support analogical reasoning, and enable one story to remind the system of a different one. They were defined as abstract patterns of planning and plan adjustment, involving aspects such as enablement conditions, cost and efficacy, risk, coordination, availability, legitimacy, affect, skill, vulnerability, and (legal) liability. Those dimensions were used by the system to recognize individual episodes in the text as examples of one TAU or another, and different story plots would involve different sets and/or sequences of TAUs. The TAUs included (for instance) *incompetent agent*, *a stitch in time saves nine*, *too many cooks spoil the broth*, *red-handed*, *hidden blessing*, and *hypocrisy*.

As these labels indicate, TAUs were abstractly defined as general planning schemata, applicable to a wide range of texts. As such, they could in principle be used to describe the internal coherence of many different stories. But because of the specific world knowledge needed to consider individual plans in actual texts, Dyer's approach in practice fell foul of the Bar-Hillel problem. (It also fell foul of the McDermott problem: using meaning-rich English phrases, such as 'a stitch in time . . .', as the names for semantically impoverished technical terms: see Chapter 11.iii.a.)

Dyer's program had to be specially designed to anticipate a specific range of stories—which in turn usually had to be specially written. So this approach to NLP, or to knowledge representation in general, was limited in terms of its potential applications. If the text domain wasn't laboriously mapped onto a specific set of planning problems, a Schankian program couldn't generate questions and answers appropriate to it.

In Winograd's (unpublished) invited lecture at the 1973 international AI conference in Los Angeles, when his own name was still on the lips of everyone interested in cognitive science, he pointed out—with characteristic modesty and intellectual generosity—that Schank was addressing important problems which he himself hadn't touched. As for Abelson's work, this was being recommended at much the same time in Minsky's influential, and widely pre-circulated, paper on "frames" (see 10.iii.a). A few years later, Minsky (1977, 1985) would draw on Abelson's ideas again, and on Schank's as well, in his "society" theory of mind (see 12.iii.c).

In short, Schank and Abelson's ideas eventually influenced AI in general (and psycholinguistics too: 7.ii.d), not just NLP.

e. The seductiveness of semantic networks

To say—as Winograd did—that Schank was addressing important problems isn't to say that he was doing so successfully. Quite apart from doubts about semantic primitives in general (see Section viii.c), his own list wasn't drawn up in a principled fashion. It dropped from fourteen to twelve, to eleven. Moreover, his work suffered from an ailment that was all too common in the AI of the time: a logical and linguistic vagueness, grounded in the intuitive use of semantic links.

This charge may seem strange, since computer models can't function by intuition. Any semantic link implemented in a computer program must have some specific definition. But the case is different when one considers the human researcher's interpretation of *diagrams* of semantic networks, such as Figures 9.6 to 9.8. Many such diagrams adorned the NLP literature of the mid-1970s. They flourished in the AI work on knowledge representation, too (see Chapter 10.iii.a). Indeed, they'd been originated by Quillian (1961, 1968) to model the structure of long-term memory.

Quillian's networks represented conceptual associations of *various* kinds, including class membership and similarity (see 10.iii.a). His general approach was taken up by many NLP researchers, including Schank, to represent the meaning of sentences. (It

$$\text{John} \Leftrightarrow \text{hit} \overset{o}{\leftarrow} \text{dog}$$

FIG. 9.6. Conceptual-dependency diagram representing "John hit the dog." Reprinted with permission from Schank and Colby (1973: 193.)

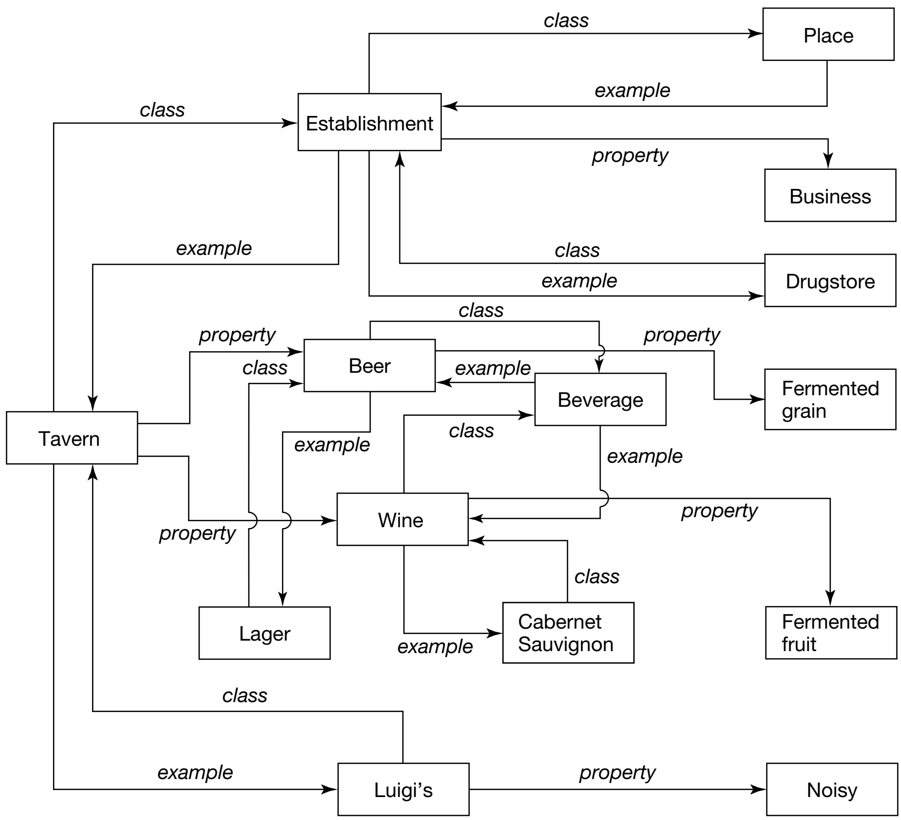

FIG. 9.7. Semantic network representing the concept of "tavern". Redrawn with permission from Lindsay and Norman (1972: 389)

was also applied to vision and to concept learning, by Patrick Winston and John R. Anderson, for instance: see 10.iii.d and 7.iv.c.) But, as semantic networks became increasingly popular, W. A. Woods (1975) rang a warning bell.

Woods pointed out that these representations were usually ambiguous, and—in their naive form—ill-suited to capture many crucial semantic distinctions. In particular, there were many problems concerning relative clauses and quantification:

> [When] one does extract a clear understanding of the semantics of the notation, most of the existing semantic network notations are found wanting in some major respects—notably the representation of propositions without commitment to asserting their truth and in representing various types of intensional descriptions of objects without commitment to their external existence, their external distinctness, or their completeness in covering all of the objects which are presumed to exist in the world. I have also pointed out the logical inadequacies of almost all current network notations for representing quantified information and some of the disadvantages of some logically adequate techniques. (W. A. Woods 1975: 79–80)

Additional problems that needed to be addressed, he said, included (among others) the representation of mass terms, probability, time, tense, and the use of adverbial modification.

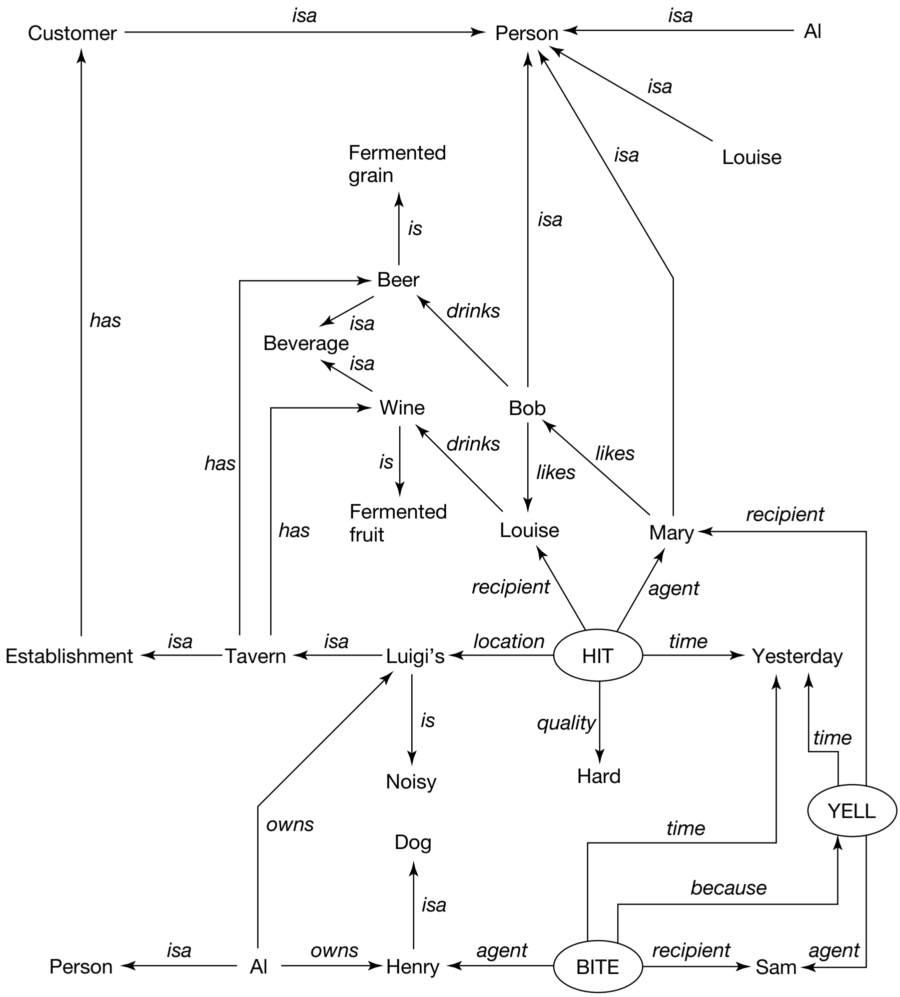

FIG. 9.8. Semantic network representing data about several people—and a dog. Redrawn with permission from Lindsay and Norman (1972: 400)

The justice of Woods's critique can be illustrated by the Schankian CD network shown in Figure 9.6. The link between "hit" and "dog" is labelled as being one of a specific kind, denoting the objective case. And Schank glosses this diagram accordingly, as being equivalent to "John hit the dog". But it could equally well be interpreted as "John hit *a* dog". If Winograd's SHRDLU had relied on this type of representation, it could not have queried the meaning of "Grasp the pyramid" when no specific pyramid had been indicated, nor interpreted that same phrase correctly later in the conversation. Moreover, if even this simple diagram is ambiguous, then one may expect more complex ones (such as Figure 9.7) to be highly problematic.

Where Schank's system was concerned, semantic ambiguity wasn't the only weakness. The system's syntactic power was grossly limited in comparison with that of SHRDLU, or of Woods's ATNs. It couldn't cope with complex auxiliary verbs and nested sentences, as in the query concerning the eggs in the cake (see above). And because it had no representation of logic, it couldn't interpret quantifiers such as *all* or *some*—nor, as we've just seen, definite versus indefinite articles.

Accordingly, those NLP workers who had a healthy respect for logical semantics and grammatical niceties weren't much influenced by Schank's work. Rather, they paid careful attention to syntax, and tried to fill the semantic gaps listed in Woods's critique.

Woods himself, for example, soon improved his LUNAR system in many ways, such as enabling it to interpret the natural-language quantifiers *all*, *every*, *some*, and *most* (W. A. Woods 1978). And others soon improved ATNs by defining a procedure (the "HOLD" function) that enabled them to parse relative clauses in an efficient, and theoretically elegant, way (Wanner and Maratsos 1978). From the early 1980s on, much NLP research implemented highly detailed syntactic theories such as GPSG and its computational cousins (see ix.d–f, above). And some developed theories of logical semantics broadly inspired by Montague, but better fitted to the constraints of actual language use (e.g. Barwise and Perry 1984).

f. Whatever will they say next?

The actual *use* of language involves not only syntax and semantics, but pragmatics too. This became increasingly clear through the 1970s, as computational linguists tried to design human–computer interfaces for interactive tasks (13.v). These included querying databases, planning (of travel itineraries, for instance), and giving/seeking advice on ongoing action (as in repairing a car).

Such applications require a rhetorically coherent conversation, as opposed to a string of isolated sentence pairs. Moreover, the human should be free to use language in everyday ways, and the computer should converse 'naturally' also.

Accordingly, NLP researchers in the mid-1970s began to model how people plan what they are going to say, how they use language to do things other than state facts, how they give or request information in a way appropriate to the interlocutor's state of mind at that moment, and how they keep track of shifts in conversational focus. In doing this, they drew on previous work in the philosophy of language.

Both Wittgenstein (1953) and John Austin (1962*a*) had shown that language can be used to do many different things, and Searle (1969, 1975) had recently formalized some of Austin's ideas. Austin (1911–60) had pointed out, for instance, that some verbs are used not to state facts but to promise, to command, to warn, or to marry. (Consider "I'm warning you . . .", "I promise", and the bride's "I do".) He believed that there's a finite number of speech acts, perhaps a few hundred, in terms of which language use could be systematically classified.

However, Austin (and Searle) didn't assume a one-to-one match between words and speech acts, because people often use language well suited to one speech act in order to perform another. For example, the verb "I suggest" can be used not to suggest, but to command. An indicative sentence can be used to ask a question ("I don't know the time"), or to make a request ("It's cold in here"). And a Yes–No question can convey a command ("Can you pass the salt?"), or elicit substantive information ("Can you tell me the way to the station?").

It follows that human speakers can—and interactive computers should—not only recognize different speech acts, but also distinguish the literal meaning of the sentence from the communicative intention of the speaker. Consequently, Searle's theory of speech acts was applied by Philip Cohen (a colleague of Woods at BBN) and Raymond Perrault to the computer modelling of conversation (P. R. Cohen and Perrault 1979).

The intellectual benefits flowed in both directions. This NLP work uncovered some flaws in Searle's theory, which was later developed—and axiomatized—as a result (Gazdar 1981*b*; Searle and Vanderveken 1985).

Cohen and Perrault drew on other philosophical work, too. H. Paul Grice (1913–88) had noted that people choose what to say—and what to leave unsaid—by considering not only truth, but also helpfulness, relevance, and brevity (Grice 1957, 1967/1975, 1969). These "conversational postulates" are involved also in understanding someone else's remarks. If they are clearly contravened, the remark will be interpreted differently: as a joke, or sarcasm, for instance. In order to apply them, both speaker and hearer must consider each other's beliefs and interests. What is helpful to one person may be unnecessary or confusing for another, or for the same person at a different time. (Grice's seminal work eventually led to a more elaborate theory of "relevance": see Chapter 7.iii.d.)

An interactive NLP system, then, should be able to adjust its output to suit different human individuals, or one individual at various points in the conversation. It should also include procedures for recognizing and correcting *misunderstandings*—on its own part, or the human's.

In implementing Grice's ideas, Cohen and Perrault used AI planning (Chapter 10.iii.c) to show how the multilevel plans of two conversationalists can be mutually adjusted so that communication takes place as intended. Each of these plans included a (continually updated) model of the other speaker's beliefs and intentions. (One might say that the programmers had sketched a "Theory of Mind" for AI systems: see 7.vi.f and 13.iii.e.) And each plan was used to guide the way in which different topics were 'naturally' introduced, or dropped, as the conversation progressed.

Similarly, Richard Power (1979) modelled the ongoing conversation between two robots cooperating in opening a door, where the relevant information wasn't directly available to both of them (see Chapter 13.iii.e). Only the robot on the bolt-side of the door could see whether the bolt was up or down. Opening the door involved problem-oriented planning concerned with how to get the door open (by moving the bolt). It also required *communicative* activities such as:

* attracting the other robot's attention;
* suggesting,
* and then negotiating, an agreed common goal
* (followed by a series of sub-goals);
* communicating or requesting information accessible to only one robot;
* confirming that this information has been duly noted;
* instructing another agent to do something which one cannot do oneself;
* . . . and so on.

From time to time, a robot would have to find out what the other robot knew and compare that with its own knowledge. This required processes that modelled each robot's changing knowledge and its (continually updated) model of the other's state of mind.

Related work was being done at much the same time by Barbara Grosz (1977; Grosz and Sidner 1979). For instance, she modelled the changing reference of terms such as "it" as the conversational focus shifted. As remarked above, focus shifting was later used in generating appropriate syntax, as well as in planning *when* to talk about *what* (McKeown 1985).

One of the general trends in 1980s NLP research was to integrate the two non-linguistic constraints of *attention* (related to focus of interest) and *intention* (related to plans and goals). An interdisciplinary workshop devoted to this aim was held in Monterey, California, in 1987 (Cohen *et al.* 1990). Many of the papers presented there showed how far NLP had moved on since Power's model of the conversation between the door-opening robots. Others were highly relevant, also, to AI work on planning—and on plan recognition, which involves the identification of the other agent's beliefs and intentions.

Several of the Monterey talks were devoted to one or another aspect of speech act theory. And Searle himself gave a paper in which he argued that "collective intentions" can be formally modelled, but can't be analysed in terms of "individual intentions" (Searle 1990*c*). Indeed, some of his remarks could be paraphrased as asserting the existence of something akin to a "group mind" (see 8.iii, preamble).

In addition, Grosz's (and other) work on the structure of conversations would eventually feed into AI teaching programs (Rickel *et al.* 2002). This enabled (for instance) virtual-reality agents to converse with human technicians about how to operate complex machinery (Chapter 13.vi.b).

None of these examples, of course, could match the human use of language. Never mind fancy syntax. The following interchange (Nicholas Negroponte's ultimate goal for NLP) could well make sense for two intimate friends, but would challenge all but the ideal example of "personalized" computers:

"Okay, where did you hide it?"

"Hide what?"

"You know."

"Where do you think?"

"Oh."

(quoted in Brand 1988: 153)

Evidently, sensible conversation depends on much more than *language*. It often needs knowledge of the situation, and of the speakers, too. Flawless computerized conversation is as unattainable as "perfect" translation and interpretation, at least in the general case.

Nevertheless, the attempt to attain these unattainable goals, and to enable computers to parse sentences of indefinite complexity, led to results of both theoretical and practical importance. Some relevant psychological research was discussed in Chapter 7.ii, and the benefits for philosophy included advances in speech act theory (see above). As for the practical aspects, applications burgeoned.

By the mid-1980s, a fairly superficial survey of "technological" NLP required no fewer than 460 large pages (T. Johnson 1985). Now, it would need many more. For significant progress has been made since then. Today's survey, in addition to updating the older topics, would need to add work on non-verbal pragmatics: not just inflection and tone of voice, but eye-gaze, hand gestures, eyebrow movements and other facial expressions, and general body posture.

All these things, which involve significant cultural variations (think of the Indian way of indicating "Yes": not by nodding, but by inclining the head from side to side), help us to interpret conversations in real life. And all are being mimicked, with more or less success, in applications of virtual reality (VR)—see below, and Chapter 13.vi.

g. A snippet on speech

Some of the technological advances in NLP over the past thirty years concern speech processing, which I've chosen not to discuss. Suffice it to say that a great deal was learnt from the blackboard-based early speech-understanding systems, such as the single-user HEARSAY and its speaker-independent successor HARPY (Reddy *et al.* 1973; Newell *et al.* 1973; Erman and Lesser 1980; Lowerre and Reddy 1980). Indeed, one of today's most successful commercial applications, a voice-recognition program marketed by Dragon Systems, was developed by Janet Baker—whose DRAGON program was a component of HARPY (J. K. Baker 1975).

Many of the difficulties that plagued these pioneering attempts (for an overview, see Barr and Feigenbaum 1981, ch. 5), and which led DARPA to halt their funding in 1976, have now been overcome. This has been done largely by combining rule-based knowledge (of phonological tree structures, for instance) with statistical pattern recognition, and by strengthening the phonetics (e.g. Rowden 1992, chs. 6–8; Stolcke 1997; P. A. Taylor 2000). Sometimes, speech recognizers have relied on self-organized "maps" of quasi-phonemes, rather than built-in phonetic representations (Kohonen 1988; see 14.ix.a).

The results have benefited both speech recognition and speech generation. They've also allowed for these to be combined in speech-to-speech translation.

Today's commercially available speech recognition systems can accept menu-based inputs from virtually any speaker on first hearing, so are widely used in telephone-answering services. HEARSAY's notorious slowness—which caused some disillusionment with AI in official circles at the end of the 1970s (Klatt 1977)—is a thing of the past. (Current speech recognizers rely on adaptive statistical models, not on GOFAI: cf. Chapter 12.) Some inexpensive desktop systems today allow one to dictate at normal speeds, without any unnatural pauses between the words. Their powers of recognition are flexible, to some degree: they can learn extra items to enlarge their vocabulary, and adapt to the individual accent of the user.

Visual speech recognition may be feasible in the near- to mid-future, since only sixteen lip-movements are needed to represent English—all of them used in saying *I thought you really meant it*. Already, "talking heads" can be generated (to 'read' input text) which pronounce words with highly effective lip-synchronization (Lucena *et al.* 2002). The sixteen lip positions may be generic, but can also be based on photographs of a specific person's face (cf. Chapter 13.vi.e).

Speech translation is progressing too. A program is being marketed which can deal with telephoned conference registrations, switching between English, German, and Japanese. Research throughout the 1990s sought to aid face-to-face commercial transactions in English between Germans and Japanese who don't speak the language fluently (W. J. Hutchins 1994, sect. 8). Indeed, the new century saw a detailed report on a speech-to-speech translation system which, unlike its few rivals, works in real time. It's not yet (as I write this page) commercially viable—although it may be, by the time you read this book. But a trustworthy reviewer has described it as highly promising, and refreshingly free of hype (Sampson 2001). This achievement is a long way from HEARSAY, whose linguistic knowledge often outweighed its phonetics—so much so, that it might emit a whole sentence in 'reply' to a cough.

Even virtual reality is taking speech on board (see Chapter 13.vi.d–e). In some computer games, one can ask a dragon, "Can you fly?"—and it does. This VR dragon is no SHRDLU: it's merely recognizing the word *fly*. But its future descendants may be able to use syntax and speech act theory to interpret the input sentence either as a question *or* as a request. Certainly, plausible interfaces for virtual reality will demand no less.

Meanwhile, a system called TESSA has been developed for the British Post Office to enable counter-clerks to communicate with deaf people whose native language is BSL, or British Sign Language (Cox *et al.* 2002). This program translates the clerk's speech into sign language, which is visibly expressed as the finger movements, bodily gestures, and facial expressions of a VR avatar. (The screen-creature even manages to look "rueful" when explaining that there are no more first-class stamps.)

By early 2003, about 400 pre-designed translations were being used by the avatar. The team had discovered early in their research that human clerks can't be trained to speak *only* the precise words defining these 400 phrases. So TESSA listens to what the clerk says and then offers a menu of up to half a dozen 'equivalent' English sentences, from which the person chooses one. It's that one which is displayed in BSL form via the avatar. In short, TESSA combines speech recognition with MT and advanced computer graphics.

Automatic speech synthesis is improving also. The first example, Ray Kurzweil's reading machine for the blind (gratefully used by Stevie Wonder, among others), was announced in January 1975. It was immediately featured on America's TV news: Walter Cronkite even allowed it to read the sign-off message for that evening's newscast. Given that it could read virtually any printed font, it was a superb achievement at the time. But it was most unnatural to listen to. The voice of its Xerox successor (Kurzweil sold the company to Xerox in 1980) is a good deal more pleasant.

In addition, there are now many more types of AI-voice application. For instance, a system recently developed for use in a hospital cardiac unit employs near-naturalistic intonation to reflect the relative importance of many different items of medical data. These are passed automatically from the operating theatre to the nurses busily preparing for the patient's arrival in intensive care (McKeown and Pan 2000). And in March 1999, my daily newspaper reported that the physicist Stephen Hawking (current holder of Isaac Newton's and Charles Babbage's Lucasian Chair) had acquired a new speech synthesizer, much more 'human'—and more British—than the one he'd been using for many years.

By the time this book is published, the choices available will doubtless be better still. I say that not just because technology inevitably moves on, but because of a specific research advance. By the close of the century, NLP researchers at Edinburgh University had produced a "dialect independent" lexicon suitable for synthesizing spoken English in many different local accents (Fitt and Isard 1999; Fitt 2003).

In this lexicon, which is based on the work of the descriptive linguist John Wells (1982), speech sounds are classified in terms of the words in which they occur—as in the "boot" vowel, the "foot" vowel, and so on. The description of a dialect for synthesis purposes specifies which of these vowel classes to merge (for instance, the vowels of "father" and "bother" for many American dialects) and which to keep separate (such as the vowels of "tide" and "tied" in most Scottish dialects). The sounds are 'glued together' to form the sentences required—"sentences", not "words", because the pronunciation of a particular word often varies according to the word that follows it.

The sounds themselves aren't synthesized from scratch, but are retrieved from the recorded speech of someone asked to read a script containing examples of each of the sounds required. In May 2004 the accents that had been synthesized in this way included received pronunciation of British English, several American dialects, and a Scottish, an Irish, and an Australian dialect (S. D. Isard, personal communication). More can, and will, be added. The collection of key phonetic symbols (in 2004, just thirty-four vowels and twenty-seven consonants) may turn out to be adequate. But it can be expanded if necessary.

Whether Hawking would be pleased to hear of this advance—which was licensed for commercial use in 2001, and is already being employed in the 'rVoice' product—is unclear. The newspapers in April 2004 reported that he was distressed about his familiar voice system wearing out, as he felt that its Dalek-like tones had become part of his identity. (As he put it when Intel Chairman Gordon Moore donated £7.5 million for a new science library in Cambridge, "I'm Intel inside myself"—Mialet 2003: 453.) In other words, he seems *not* to want his speech synthesizer to pass the Turing Test (16.ii.c).

The Edinburgh-based synthesizer doesn't quite do that, anyway. For example, dialects also vary in intonation and segment duration, but these features (which are independent of the lexicon) aren't represented in the system. So although it's clear that *this* synthesized voice is Scottish and *that* one Australian, neither sounds quite right to anyone with a good ear for accents. Nevertheless, speech processing has come a long way since HEARSAY.

From artificial voices to artificial intelligence: our next topic is the history of AI. *This* chapter, of course, has already begun that narrative. Language may be less central to intelligence than early AI took it to be (see 13.iii.b and 15.vii.a). Nevertheless, NLP is crucial to the AI project. And Chomsky's wider influence affected AI, as well as psychology and philosophy—as we'll now see.

MIND AS MACHINE

A History of Cognitive Science

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